

IDENTIFICATION

PRODUCT CODE: MAINDEC-08-DHRKC-H-D
PRODUCT NAME: RK8E/RK8L DATA RELIABILITY PROGRAM
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1.

ABSTRACT

THE RK0E/RK0L DATA RELIABILITY PROGRAM IS DESIGNED PRIMARILY AS AN ACCEPTANCE TEST TO VERIFY DISK DATA TRANSFERS WITHIN THE DISK SYSTEM.

THE "ACCEPT MODE" OF OPERATION VERIFIES THE CAPABILITY OF TRANSFERRING A TOTAL 3×10^9 BITS OF DATA TO AND FROM EACH INDIVIDUAL DISK DRIVE ON THE DISK SYSTEM.

THE "MANUAL INTERVENTION MODE" IS AVAILABLE AS A HARDWARE DEBUGGING AID TO ALLOW THE OPERATOR TO SELECT DATA PATTERNS, TRANSFER LENGTHS, AND ADDRESSING.

(NOTE: LOCATION 0 CONTAINS REVISION LEVEL (IN ASCII) OF PROGRAM ON PROGRAM LOAD).

2.

RESTRICTIONS

THE RK0L CONTROL, WHICH CAN CONTROL UP TO 8 DRIVES, WILL NOT RUN WITH THE DW0E BUS ADAPTER. THE REASON FOR THIS STATEMENT IS THAT THE RK0L CONTROL USES IOTS FOR EXTENDED DRIVES 4-7 WHICH IS NOT AVAILABLE ON THE DW0E.

2.1

HARDWARE

- A. PDP-8/A, 8/E, 8/F, OR 8/M COMPUTER OR OTHER FAMILY OF 8 COMPATIBLE COMPUTER WITH NECESSARY DW0E BUS ADAPTER.
- B. AT LEAST 4K OF READ/WRITE MEMORY. AT LEAST 8K OF MEMORY IS NECESSARY FOR OPERATION OF THE CONSOLE PACKAGE.
- C. ASR-33 TELETYPE OR EQUIVALENT
- D. RK0E OR RK0L DISK CONTROL
- E. RK05J OR RK05F DISK DRIVE(8)
- F. FORMATTED 2200 DPI-16 SECTOR PACK(8).

NOTE: THE RK05F DISK DRIVE IS CONSIDERED AS TWO SEPARATE UNITS. WHEN ANSWERING ALL QUESTIONS THE SEPARATE DRIVES MUST BE SPECIFIED. DSK07, DSK17, DSK27, ETC.

2.2

PROGRAM STORAGE

THE PROGRAM OCCUPIES OR UTILIZES LOCATION 0000 TO LOCATION 7577 OF FIELD 0. ALL EXTENDED MEMORY LOCATIONS, IF AVAILABLE, ARE UTILIZED FOR TESTING.

2.3

PRELIMINARY PROGRAMS

THIS PROGRAM REQUIRES A FORMATTED CARTRIDGE ON ALL DRIVES TO BE TESTED.

ALL BASIC AND EXTENDED MEMORY DIAGNOSTICS SHOULD BE RUN PRIOR TO RUNNING THIS PROGRAM.

RK8E CONTROL: RUN THE RK8E DISKLESS CONTROL TEST AND THE RK8E/RK8L DISK FORMATTER IF THIS DIAGNOSTIC FAILS TO OPERATE PROPERLY.

RK8L CONTROL: RUN THE RK8L INSTRUCTION TEST AND THE RK8E/RK8L FORMATTER IF THIS DIAGNOSTIC FAILS TO OPERATE PROPERLY.

2.4 EXECUTION TIME -----

THE PROGRAM EXECUTION TIME (I.E. PASSING 3 X 10⁹ BITS OF DATA ON A DISK DRIVE), IS APROX. 4 HOURS PER DISK DRIVE ON A 4K MEMORY SYSTEM OR APROX. 3.5 HOURS PER DISK DRIVE ON SYSTEMS WITH EXTENDED MEMORY.

3. SWITCH REGISTER SETTINGS -----

SWR0#1 LOOP ON WRITE SEQUENCE.
SWR1#1 LOOP ON READ SEQUENCE.
SWR2#1 INHIBIT ALL ERROR TIMEOUTS
SWR3#1 TYPE "STATUS-COMplete" REPORT.
SWR4#1 PROGRAM STOP ON HALT.
SWR5#1 DRIVE DISCONNECT AFTER PASS COMPLETION.
SWR6#1 PERFORM ONLY "OVERLAP SEEKS", DO NOT EXECUTE DATA BREAKS.

4. OPERATOR AND/OR PROGRAM ACTION -----

4.1 STANDARD TEST PROCEDURE -----

- A. START AS SPECIFIED THROUGH OUT THIS DOCUMENTATION IS KEY CLEAR AND THEN KEY CONTINUE ON PDP8/E, PDP8/M, AND PDP8/F COMPUTERS.
- B. LOAD THE PROGRAM INTO MEMORY FIELD 0 USING THE STANDARD BINARY LOADER TECHNIQUE.
- C. IF IT IS DESIRED TO CHANGE THE IOT CODES WITHIN THE PROGRAM, FOLLOW THE PROCEDURE IN SECTION 4.6.
- D. RUN THE ACCEPTANCE MODE OF DATA RELIABILITY WITH ALL DRIVES AND MEMORY AVAILABLE BY FOLLOWING THE PROCEDURE

IN SECTION 4.4.

E. THE MANUAL INTERVENTION MODE, SECTION 4.5, MAY BE USED FOR TROUBLE SHOOTING, IF DESIRED.

F. IF POSSIBLE SWR4#1 SHOULD ALWAYS BE USED TO STOP THE PROGRAM.

G. IF THE PROGRAM HAS BEEN STOPPED DUE TO SWR4#1, THE PROGRAM CAN BE RESTARTED, AND THE INITIAL STARTUP QUESTIONS BYPASSED, BY USING 0205 AS THE RESTART ADDRESS.

H. FOR THE ABSOLUTE LOCATIONS OF ALL KNOWN HALTS IN THIS PROGRAM, ACCESS PAGE 1-22 OF THE PROGRAM LISTING.

4.2

RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE

THE FOLLOWING IS THE CORRECT CARTRIDGE MOUNTING PROCEDURE FOR THE RK05J DISK DRIVE. ANY DEVIATION ENCOUNTERED DURING THIS PROCEDURE WILL BE CONSIDERED AN ERROR CONDITION.

A. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION.

B. TURN AC POWER TO DISK DRIVE ON.

C. VERIFY THAT THE LIGHT LABELED "PWR" IS ON.

D. WAIT FOR THE LIGHT LABELED "LOAD" TO COME ON.

E. VERIFY THAT THE LIGHTS LABELED "RDY", "ON CYL", "FAULT", "WT", AND "RD" ARE OFF.

F. OPEN ACCESS DOOR.

G. INSERT CARTRIDGE.

H. CLOSE ACCESS DOOR.

I. SET SWITCH LABELED "RUN/LOAD" TO THE "RUN" POSITION.

J. WAIT FOR THE LIGHTS LABELED "RDY" AND "ON CYL" TO COME ON.

K. TOGGLE SWITCH LABELED "WT PROT" AND VERIFY THAT THE LIGHT LABELED "WT PROT" GOES ON AND OFF.

L. TOGGLE SWITCH LABELED "WT PROT" UNTIL THE LIGHT LABELED "WT PROT" IS OFF.

M. VERIFY THAT LIGHTS LABELED "FAULT", "WT", "RD", AND "LOAD" ARE OFF.

4.3

RK05F DRIVE SETUP PROCEDURE

THE FOLLOWING IS THE CORRECT SETUP PROCEDURE

FOR THE RK05F DISK DRIVE. ANY DEVIATION ENCOUNTERED DURING THIS PROCEDURE WILL BE CONSIDERED AN ERROR CONDITION.

- A. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION.
- B. TURN AC POWER ON.
- C. VERIFY THAT LIGHT LABELED "PWR" IS ON.
- D. WAIT FOR LIGHT LABELED "LOAD" TO COME ON.
- E. VERIFY THAT LIGHTS LABELED "RDY", "ON CYL", "FAULT", "WT", AND "RD" ARE OFF.
- F. SET SWITCH LABELED "RUN/LOAD" TO THE "RUN" POSITION.
- G. WAIT FOR LIGHTS LABELED "RDY" AND "ON CYL" TO COME ON.
- H. TOGGLE SWITCH LABELED "WT PROT" AND VERIFY THAT THE LIGHT LABELED "WT PROT" GOES ON AND OFF.
- I. TOGGLE SWITCH LABELED "WT PROT" UNTIL LIGHT LABELED "WT PROT" GOES OFF.
- J. VERIFY THAT LIGHTS LABELED "FAULT", "WT", "RD", AND "LOAD" ARE OFF.

4.4

RK08E/RK08L DATA RELIABILITY (ACCEPT MODE)

- A. MAKE READY ALL DRIVES TO BE TESTED USING THE RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE SECTION 4.2 OR THE RK05F DRIVE PROCEDURE IN SECTION 4.3.
- B. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION ON ALL DRIVES NOT BEING TESTED.
- C. VERIFY THAT AC POWER IS ON, ON ALL DRIVES NOT BEING TESTED.
- D. SET THE SWITCH REGISTER TO 0200 AND PRESS LOAD ADDRESS.
- E. SET THE SWITCH REGISTER TO 0000 AND PRESS START.
- F. THE OPERATOR MAY SET SWRS=1 IF IT IS DESIRED TO HAVE THE PROGRAM AUTOMATICALLY DISCONNECT EACH DISK DRIVE AS EACH MAKE THEIR PASS COMPLETION. (NOTE: IF SWRS=0, ALL DISK DRIVES WILL CONTINUE TO RUN AFTER THEIR PASS COMPLETION)
- G. THE TTY WILL PRINT THE FOLLOWING PROGRAM NAME AND QUESTION.

RK08E/RK08L DATA RELIABILITY
EXTENDED R/W MEMORY (0-7)?

THE OPERATOR SHOULD THEN TYPE THE AMOUNT OF EXTENDED READ/ WRITE MEMORY BANKS NUMBERED SEQUENTIALLY FROM BANK 0, AS INDICATED BY THE TTY QUESTION.

- H. THE TTY WILL PRINT THE FOLLOWING QUESTION(S), ASKING THE

DESIRED DISK DRIVE(S) TO BE USED IN TESTING.

EXERCISE DISK07
EXERCISE DISK17
EXERCISE DISK27
EXERCISE DISK37
EXERCISE DISK47
EXERCISE DISK57
EXERCISE DISK67
EXERCISE DISK77

FOR THE QUESTION(S) ABOVE, TYPE Y FOR YES, IF IT IS DESIRED TO TEST THE DISK DRIVE IN QUESTION, OTHERWISE, TYPE N FOR NO.

I. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ACCEPT MODE?

THE OPERATOR SHOULD THEN TYPE Y FOR YES TO RUN THE ACCEPTANCE MODE OF OPERATION.

J. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ARE YOU SURE?

IF THE OPERATOR IS CERTAIN OF THE AMOUNT OF MEMORY, THE DISK DRIVE(S) SELECTED, AND THE MODE OF OPERATION, TYPE Y FOR YES. TYPING N FOR NO WILL RESULT IN A REPEAT OF ALL MESSAGES AND QUESTIONS ENCOUNTERED THUS FAR.

K. THE PROGRAM SHOULD START TESTING THE DISK DRIVE(S) AND MEMORY SELECTED.

L. THE "STATUS-COMPLETE" TYPEOUT SHOULD OCCUR UPON PASS COMPLETION OF EACH DISK DRIVE. ALL OTHER TYPEOUTS OR HALTS WILL BE CONSIDERED AS AN ERROR CONDITION. SEE SECTION 5.5 FOR "STATUS-COMPLETE" TYPEOUT.

M. A SUCCESSFUL PASS COMPLETE ON A DISK DRIVE WILL BE CONSIDERED AS NO "HARD" ERRORS AND NO MORE THAN ONE (1) "SOFT" ERROR PER PASS COMPLETE.

N. IF ANY ERRORS DO OCCUR, THE OPERATOR SHOULD ACCESS SECTION 5 IN THIS DOCUMENTATION.

4.5

RK8E/RK8L DATA RELIABILITY (MANUAL INTERVENTION MODE)

THE MANUAL INTERVENTION MODE IS AVAILABLE AS A TROUBLE SHOOTING AID AND SHOULD ONLY BE USED FOR SUCH PURPOSES, IF DESIRED.

A. MAKE READY ALL DISK DRIVES TO BE TESTED USING THE RK05J DRIVE CARTRIDGE MOUNTING PROCEDURE SETION 4.2. OR THE RK05F DRIVE PROCEDURE SECTION 4.3.

B. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION ON

ALL DRIVES NOT BEING TESTED.

- C. VERIFY THAT AC POWER IS ON, ON ALL DRIVES NOT BEING TESTED.
- D. SET THE SWITCH REGISTER TO 0200 AND PRESS LOAD ADDRESS.
- E. SET THE SWITCH REGISTER TO 0000 AND PRESS START.
- F. THE TTY WILL PRINT THE FOLLOWING PROGRAM NAME AND QUESTION.

RK0E/RK8L DATA RELIABILITY
EXTENDED R/W MEMORY (0-7)?

THE OPERATOR SHOULD THEN TYPE THE AMOUNT OF EXTENDED READ/ WRITE MEMORY BANKS NUMBERED SEQUENTIALLY FROM BANK 0, AS INDICATED BY THE TTY QUESTION.

- G. THE TTY WILL PRINT THE FOLLOWING QUESTION(S), ASKING THE DESIRED DISK DRIVE(S) TO BE USED IN TESTING.

EXERCISE DISK0?
EXERCISE DISK1?
EXERCISE DISK2?
EXERCISE DISK3?
EXERCISE DISK4?
EXERCISE DISK5?
EXERCISE DISK6?
EXERCISE DISK7?

FOR THE QUESTION(S) ABOVE, TYPE Y FOR YES, IF IT IS DESIRED TO TEST THE DISK DRIVE IN QUESTION, OTHERWISE, TYPE N FOR NO.

- H. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ACCEPT MODE?

THE OPERATOR SHOULD THEN TYPE N FOR NO TO RUN THE MANUAL INTERVENTION MODE OF OPERATION.

- I. THE TTY WILL THEN PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT A CONSTANT MEMORY FIELD, RATHER THAN THE NORMAL RANDOM FIELD SELECTION.

FIELD?

IF THE OPERATOR DESIRES TO SELECT A CONSTANT FIELD, TYPE Y FOR YES, OTHERWISE, TYPE N FOR NO. IF Y WAS TYPED THE TTY WILL SPACE OUT ONCE AND WAIT FOR THE OPERATOR TO TYPE THE DESIRED FIELD IN OCTAL (0-7).

- J. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT A CONSTANT TRACK, RATHER THAN THE NORMAL RANDOM TRACK SELECTION.

TRACK?

IF THE OPERATOR DESIRES TO SELECT A CONSTANT TRACK, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED, THE TTY WILL SPACE OUT ONCE AND WAIT FOR THE OPERATOR TO INPUT THE DESIRED TRACK ADDRESS (00000-14537).

- K. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT HALF BLOCK OR FULL BLOCK TRANSFERS, RATHER THAN THE NORMAL RANDOM SELECTION.

BLOCK LENGTH?

IF THE OPERATOR DESIRES TO CHANGE THE BLOCK LENGTH, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED THE TTY WILL SPACE OUT ONCE AND WAIT FOR THE OPERATOR TO TYPE THE BLOCK LENGTH DESIRED (0-256 WORD BLOCK OR 1-128 WORD BLOCK).

- L. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT A CONSTANT NUMBER OF SECTORS TO BE TRANSFERRED, RATHER THAN THE NORMAL RANDOM SECTOR SELECTION.

EXTRA SECTORS?

IF THE OPERATOR DESIRES TO SELECT A CONSTANT AMOUNT OF SECTORS, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED THE TTY WILL SPACE OUT ONCE, AND WAIT FOR THE OPERATOR TO TYPE IN THE EXTRA SECTORS DESIRED (00-17). (NOTE: IF THE FIELD AND THE BLOCK LENGTH PREVIOUSLY SELECTED WAS 0, THE AMOUNT OF EXTRA SECTORS WILL BE LIMITED TO 07. OTHERWISE THE MAXIMUM AMOUNT IS LIMITED TO 17.)

- M. THE TTY WILL PRINT THE FOLLOWING QUESTION, ASKING IF THE OPERATOR DESIRES TO SELECT A DATA PATTERN, RATHER THAN NORMAL RANDOM DATA SELECTION.

DATA?

IF THE OPERATOR DESIRES TO SELECT A DATA PATTERN, TYPE Y FOR YES, OTHERWISE, N FOR NO. IF Y WAS TYPED, THE TTY WILL DO A "CRLF" AND WAIT FOR THE OPERATOR TO TYPE IN 12 OCTAL DATA WORDS TO BE USED IN TESTING.

- N. THE TTY WILL PRINT THE FOLLOWING QUESTION.

ARE YOU SURE?

IF THE OPERATOR IS CERTAIN OF THE INFORMATION SELECTED, TYPE Y FOR YES. TYPING N FOR NO WILL RESULT IN A REPEAT OF ALL MESSAGES AND QUESTIONS ENCOUNTERED THUS FAR.

- P. THE PROGRAM SHOULD START EXECUTING THE OPERATIONS SELECTED.

- R. IF ERRORS ARE ENCOUNTERED, ACCESS SECTION 5 IN THIS DOCUMENTATION.

THE PROGRAM NORMALLY RECOGNIZES DEVICE IOT CODE X74X. TO CHANGE THE DEVICE IOT CODES WITHIN THE PROGRAM:

- A. SET THE SWITCH REGISTER TO 0204 AND PRESS LOAD ADDRESS.
- B. SET THE SWITCH REGISTER TO 0000, SET SWITCH REGISTER BITS 3-8 TO THE DESIRED DEVICE IOT CODE, AND PRESS START.
- C. THE PROGRAM WILL CHANGE THE DEVICE IOT CODES WITHIN THE PROGRAM AND THEN HALT.
- D. PRESSING KEY CONTINUE WILL START THE PROGRAM AT LOCATION 0200 (SEE SECTIONS 4.4 OR 4.5 FOR OPERATION INSTRUCTIONS).

5. ERRORS -----

5.1 USEFUL INFORMATION -----

ALL STATUS ERRORS WILL BE REPORTED AS STATUS ERRORS. ALL DATA ERRORS WILL BE REPORTED AS DISK DATA ERRORS.

WHEN DATA IS BEING READ OFF THE DISK AND A CRC ERROR OCCURS THE PROGRAM WILL REPORT THE ERROR AS A READ STATUS ERROR. THE PROGRAM WILL THEN CHECK THE DATA READ FOR DATA ERRORS. IF DATA ERRORS EXIST THEY WILL BE REPORTED AS DISK DATA ERRORS.

5.2 ERROR HALTS -----

ERROR HALTS FOR WHICH THERE ARE NO ERROR TYPEOUTS ARE LISTED AND DEFINED AS FOLLOWS.

BIGSTP	MASTER ERROR HALT FOR ALL OF THE FOLLOWING ERROR STOPS. WHEN THE COMPUTER HALTS THE AC REGISTER WILL INDICATE THE PC OF THE FAILING ERROR STOP.
INTER1	NO DISK INTERRUPT
ERHLT0	SKIP TNAP FOR IOT "DLSC"
ERHLT2	SKIP TNAP FOR IOT "DCLR"
ERHLT3	SKIP TNAP FOR IOT "DLAG"
ERHLT5	SKIP TNAP FOR IOT "DRST"
ERHLT6	SKIP TNAP FOR IOT "DLDC"
BADHLT	CHECKSUM FAILED BUT WORD-BY-WORD COMPARE WORKED

NODSKS

NO DISKS AVAILABLE TO RUN

FLOHLT

PROGRAM WILL ONLY RUN IN FIELD 0

FOR THE ABSOLUTE LOCATIONS OF THE HALTS LISTED ABOVE,
ACCESS PAGE 1-22 OF THE PROGRAM LISTING.

5.3

ERROR TYPEOUTS

WHEN AN ERROR OCCURRES THE PROGRAM WILL PRINT AN
"ERROR HEADER" WHICH WILL SPECIFY THE PARTICULAR TYPE
OF ERROR FOUND AT THE TIME OF THE FAILURE.

POSSIBLE "ERROR HEADERS" ARE AS FOLLOWS.

WRITE STATUS ERROR
READ STATUS ERROR
DISK DATA ERROR
RECALIBRATE STATUS ERROR

AFTER THE "ERROR HEADER" MENTIONED ABOVE IS TYPED, THE
PROGRAM WILL PRINT THE FOLLOWING ERROR INFORMATION
FOUND AT THE TIME OF THE FAILURE, PERTAINING TO THE
FAILURE. POSSIBLE TYPEOUTS ARE AS FOLLOWS.

PC: PROGRAM LOCATION OF THE ACTUAL FAILURE.

ST: CONTENTS OF THE STATUS REGISTER.

EX: EXTENDED DRIVE BIT

CM: SOFTWARE COMMAND REGISTER.

IA: INITIAL SOFTWARE DISK ADDRESS REGISTER OR THE
CYLINDER, SURFACE, AND SECTOR BITS.

DA: FINAL SOFTWARE DISK ADDRESS REGISTER OR THE
CYLINDER, SURFACE, AND SECTOR BITS.

CA: SOFTWARE INITIAL CURRENT ADDRESS

WC: SOFTWARE INITIAL WORD COUNT

FW: SOFTWARE FINAL WORD COUNT

AS: SECTOR IN ERROR ON THE PARTICULAR CYLINDER
AND SURFACE IN QUESTION.

WA: WORD ADDRESS WITHIN THE SECTOR IN ERROR

AD: BREAK ADDRESS OF DATA BREAK IN COMPUTER.

DG: EXPECTED DATA

DB: DATA FOUND DURING DATA BREAK.

ERROR RECOVERY AND ERROR DISCONNECT

WHEN A READ, WRITE, OR DISK DATA ERROR OCCURS (SEE SECTION 5.3), THE PROGRAM WILL TRY TO REPEAT THE FAILING SEQUENCE FOUR (4) TIMES. IF THE ERROR HAS OCCURRED FOUR (4) TIMES SIMULTANEOUSLY, THE ERROR WILL BE CONSIDERED AS A NON-RECOVERABLE ERROR, THE "ERROR HEADER" WILL BE CHANGED TO INDICATE "NON-RECOVERABLE" ERROR, ANOTHER DISK ADDRESS WILL BE SELECTED FOR TESTING. IF A "SOFT" ERROR SHOULD OCCUR ON A TRACK, THE PROGRAM WILL RETRY THE READ SEQUENCE (64) TIMES BEFORE SELECTING ANOTHER TRACK FOR TESTING. (NOTE: THIS 64 RETRYs ON "SOFT" ERRORS WILL BE TERMINATED ON A "HARD" ERROR).

POSSIBLE NON-RECOVERABLE ERROR HEADERS ARE AS FOLLOWS.

NON-RECOVERABLE READ STATUS ERROR
NON-RECOVERABLE WRITE STATUS ERROR
NON-RECOVERABLE DISK DATA ERROR

IF A NON-RECOVERABLE READ OR WRITE ERROR SHOULD OCCUR, THE DISK IN QUESTION WILL THEN BE RECALIBRATED (RESTORED TO CYLINDER 0). IF THE RECALIBRATE SEQUENCE FAILS, THE DISK DRIVE IN ERROR WILL BE DISCONNECTED BY THE PROGRAM AND NO LONGER BE TESTED.

THE FOLLOWING "DISCONNECT" AND "STATUS-COMPLETE" TYPEOUTS SHOULD OCCUR.

RECALIBRATE ERROR DISCONNECT!
DISK X DISCONNECTED!
DSK HARD SOFT COMP
X 0030 0010 0001
X 0240 5670 0001

IF ALL DISKS ON THE SYSTEM HAVE BEEN DISCONNECTED DO TO RECALIBRATE ERRORS THE FOLLOWING TYPEOUT WILL OCCUR AND THE PROGRAM WILL HALT.

DISK SYSTEM SHUT DOWN, NO DISKS TO RUN!

STATUS-COMPLETE TYPEOUT AND PASS COMPLETE DISCONNECT

ALL ERRORS AND PASS COMPLETES ARE TALLIED BY THE PROGRAM PER DISK DRIVE.

THE FOLLOWING IS AN EXAMPLE OF THE "STATUS-COMPLETE" TYPEOUT THAT WILL OCCUR WHEN SWR3=1 INDICATING TYPE THIS REPORT, A PASS COMPLETE OCCURS ON A DRIVE UNDER TEST, OR A DRIVE IS DISCONNECTED DO TO A RECALIBRATE ERROR.

DSK HARD SOFT COMP
X XXXX XXXX XXXX
X XXXX XXXX XXXX
X XXXX XXXX XXXX

X XXXX XXXX XXXX

THE TYPEOUT AS MENTIONED ABOVE IS DESCRIBED AS FOLLOWS.

DSK DISK DRIVE IN QUESTION.

HARD ALL ERRORS OTHER THAN THAT DEFINED AS A SOFT ERROR.

SOFT A READ CRC STATUS ERROR WITH BAD DATA PER TRANSFER WITH RECOVERY POSSIBLE WITHIN FOUR (4) RETRYs. (NOTE: FOUR (4) CONSECUTIVE RETRYs WILL BE CONSIDERED AS A NON-RECOVERABLE ERROR OR A "HARD" ERROR).

COMP PASS COMPLETES. <3 X 10(9) BITS>

IF SWR5=1 INDICATING "DISCONNECT ON PASS COMPLETION", AND A DISK DRIVE UNDER TEST MAKES A PASS COMPLETION, THE FOLLOWING TYPEOUT WILL OCCUR AND THE DRIVE WILL BE DISCONNECTED.

DISK X PASS COMPLETE!
DISK X DISCONNECTED!
DSK HARD SOFT COMP
X XXXX XXXX XXXX
X XXXX XXXX XXXX

IF SWR5=0 INDICATING DON'T "DISCONNECT ON PASS COMPLETION", AND A DISK DRIVE UNDER TEST MAKES A PASS COMPLETION, THE FOLLOWING TYPEOUT WILL OCCUR AND THE DRIVE WILL CONTINUE TO RUN.

DISK X PASS COMPLETE!
DSK HARD SOFT COMP
X XXXX XXXX XXXX
X XXXX XXXX XXXX

IF SWR5=1 AND ALL DRIVES HAVE MADE THEIR PASS COMPLETION AND HAVE BEEN DISCONNECTED, THE FOLLOWING TYPEOUT WILL OCCUR AND THE COMPUTER WILL HALT.

DISK SYSTEM SHUT DOWN, NO DISKS TO RUN!

5.6

TYPICAL ERROR TYPEOUTS

THE FOLLOWING IS AN EXAMPLE OF AN "ERROR HEADER" AND ERROR TYPEOUT THAT COULD HAVE OCCURRED ON A WRITE STATUS ERROR. (NOTE CRC IN THE STATUS INDICATOR "ST1")

WRITE STATUS ERROR
PC12371 ST14010 EX10001 CM14000 IA10001 DA10002
CA13600 WC17000 FW10000

THE FOLLOWING IS AN EXAMPLE OF AN ERROR TYPEOUT THAT COULD HAVE OCCURRED IF THE STATUS REGISTER FAILED ON A SEEK ONLY FUNCTION.

SEEK STATUS ERROR

PC:2076 ST:4002 EX:0001 CM:3000 DA:4007

THE FOLLOWING IS A TYPICAL EXAMPLE OF AN "ERROR HEADER"
AND ERROR TYPEOUT THAT COULD HAVE OCCURRED ON A DISK
DATA ERROR. (NOTE: ADDITION DATA ERRORS IN BUFFER)

DISK DATA ERROR

PC:1674 ST:4010 EX:0001 CM:1432 IA:1035 DA:1021

CA:0001 WC:5000 FW:7400

AS:0015 WA:0007 AD:0010 DG:0537 DB:0536

AS:0015 WA:0077 AD:0100 DG:7777 DB:7776

AS:0016 WA:0002 AD:0405 DG:6167 DB:6166

6. RESTRICTIONS

ALL DISK DRIVES SHOULD BE SET TO THE LOAD POSITION
THAT ARE NOT BEING TESTED.

7. TROUBLE SHOOTING INFORMATION

IOT

FUNCTION

6740 DLSC

LOAD SECTOR COUNTER AND EXTENDED
DRIVE BIT FOR RK8L.

AC

--

0-3

LOAD THE DESIRED AMOUNT OF SECTORS
TO BE TRANSFERRED WITH THE BINARY
VALUE IN AC BITS 0-3.

4

EXTENDED DRIVE BIT FOR DRIVES 4-7

6741 DSKP

"SKIP" SKIP IF TRANSFER DONE FLAG
OR ERROR FLAG IS SET.

6742 DCLR

"CLEAR" FUNCTION IS REGULATED BY
AC BITS 10 AND 11. THE AC IS THEN
CLEARED.

AC10

AC11

0

0

CLEAR THE AC AND STATUS REGISTER.

0

1

CLEAR THE AC, CONTROL, AND MAJOR
REGISTERS. THIS INSTRUCTION WILL
STOP THE CONTROL EVEN IF IT IS
WRITING A HEADER. THIS IS THE ONLY
INSTRUCTION THAT CLEARS MAINTENANCE
MODE.

1	0	CLEAR AC, RECALIBRATE DISK DRIVE, AND CLEAR STATUS REGISTER.
6743	DLA G	"LOAD DISK ADDRESS AND GO" LOAD THE DISK CYLINDER, SURFACE, AND SECTOR FROM THE AC, CLEAR THE AC, AND DO THE COMMAND IN THE COMMAND REGISTER.
AC	--	
0-6		CYLINDER
7		SURFACE (1=UPPER) (0=LOWER)
8-11		SECTOR
6744	DLCA	"LOAD CURRENT ADDRESS" LOAD THE CURRENT ADDRESS FROM AC. THE AC IS THEN CLEARED.
AC	--	
0-11		CURRENT ADDRESS
6745	DRST	"READ STATUS" CLEAR THE AC AND READ THE CONTENTS OF THE STATUS REGISTER INTO THE AC.
AC	--	
0		TRANSFER DONE
1		READY TO SEEK, READ, OR WRITE.
2		NOT USED
3		SEEK FAIL
4		DISK FILE READY
5		CONTROL BUSY ERROR
6		TIME OUT ERROR
7		WRITE LOCK ERROR
8		CRC ERROR
9		DATA RATE ERROR
10		DRIVE STATUS ERROR
11		CYLINDER ADDRESS ERROR
6746	DLDC	"LOAD COMMAND" LOAD THE COMMAND REGISTER FROM AC, CLEAR THE AC, AND CLEAR THE STATUS REGISTER.
AC	--	
0-2#0		READ DATA
0-2#1		READ ALL
0-2#2		WRITE LOCK
0-2#3		SEEK ONLY
0-2#4		WRITE DATA

0-205	WRITE ALL
0-206	NOT USED
0-207	NOT USED
3	ENABLE INTERRUPT
4	ENABLE SET TRANSFER DONE ON SEEK DONE
5	HALF BLOCK 128 WORDS
6	EXTENDED MEMORY ADDRESS
7	EXTENDED MEMORY ADDRESS
8	EXTENDED MEMORY ADDRESS
9	UNIT SELECT
10	UNIT SELECT
11	EXTENDED CYLINDER ADDRESS

6747 DMAN

"MAINTENANCE IOT" LOAD THE MAINTENANCE REGISTER FROM THE AC. THE FUNCTION IS REGULATED BY THE AC BITS. MAINTENANCE MODE CAN ONLY BE CLEARED BY DCLK "CLEAR CONTROL".

AC
--

0	ENTER MAINTENANCE MODE
1	ENABLE SHIFT TO LOWER BUFFER
2	AC BIT 10, CRC REGISTER, AND THE LOWER DATA BUFFER ARE CONNECTED AS A SHIFT REGISTER. AC BIT 10 DATA SHIFTS TO THE CRC, THE CRC SHIFTS TO THE LOWER DATA BUFFER.
3	SHIFT COMMAND REGISTER TO THE LOWER DATA BUFFER.
4	SHIFT THE SURFACE AND SECTOR REGISTER TO THE LOWER DATA BUFFER.
5	SHIFT AC 10 DATA TO THE UPPER DATA BUFFER. THE UPPER BUFFER SHOULD SINK IN THE SILO WHEN FULL.
6	ONE SINGLE CYCLE BREAK REQUEST. DIRECTION IS REGULATED BY FUNCTION IN THE COMMAND REGISTER.
7	CLEAR AC THEN READ THE LOWER DATA BUFFER TO THE AC.
8	NOT USED.
9	NOT USED.
10	USED AS DATA WITH OTHER BITS IN THE MAINTENANCE MODE.
11	NOT USED

8. PROGRAM DESCRIPTION (ACCEPT MODE)

THE FOLLOWING IS BRIEF DESCRIPTION OF THE STEPS TAKEN BY THE PROGRAM WHEN RUNNING THE ACCEPT MODE.

- A. A RANDOM FIELD IS GENERATED. IF FIELD GENERATED IS A NON-EXISTING FIELD, THE MAXIMUM FIELD AVAILABLE WILL BE USED.

- B. A RANDOM BLOCK LENGTH IS GENERATED (128 OR 256 WORD SECTORS).
- C. A RANDOM AMOUNT OF SEQUENTIAL SECTORS TO TRANSFER IS GENERATED. IF THE FIELD PREVIOUSLY SELECTED WAS AN EXTENDED FIELD OR IF HALF BLOCK TRANSFERS WERE SELECTED (128 WORD SECTORS), THE AMOUNT OF SECTORS WILL BE LIMITED TO 17(8). IF THE FIELD SELECTED WAS FIELD 0 AND IF FULL BLOCK TRANSFERS WERE SELECTED (256 WORD SECTORS), THE AMOUNT OF SECTORS WILL BE LIMITED TO 7(8).
- D. A RANDOM STARTING SECTOR WILL BE GENERATED. THE RANDOM AMOUNT OF EXTRA SECTORS PREVIOUSLY GENERATED WILL BE ADDED TO THIS STARTING SECTOR, DETERMINING THE ACTUAL LENGTH OF THE DATA TRANSFER. IF THE STARTING SECTOR WAS 14 AND THE AMOUNT OF EXTRA SECTORS WAS 6, SECTORS 14, 15, 16, 17, 00, 01, AND 02 WILL BE USED FOR TRANSFERRING DATA.
- E. AN INITIAL SOFTWARE WORD COUNT WILL BE CALCULATED.
- F. AN INITIAL RANDOM CURRENT ADDRESS WILL BE GENERATED. IF THE FIELD PREVIOUSLY GENERATED WAS FIELD 0, THE CURRENT ADDRESS WILL BE LIMITED WITHIN THE END OF THE PROGRAM +4000 LOCATIONS.
- G. THE BUFFER SELECTED WILL BE FILLED WITH RANDOM DATA, CHECKSUMMED, AND THE CHECKSUM SAVED. (NOTE: BUFFER IS DEPENDENT ON FIELD, WORD COUNT, BLOCK LENGTH, AND CURRENT ADDRESS PREVIOUSLY SELECTED.)
- H. THE PROGRAM WILL THEN POLE THE DISK DRIVES. DRIVE SELECTION IS SEQUENTIAL, THAT IS DISK0, DISK1, DISK2, ETC.
- I. DATA WILL BE WRITTEN ON THE SELECTED DISK DRIVE TO COMPLETE THE SEEK OPERATION USING THE RANDOM PARAMETERS GENERATED ABOVE. AS DATA IS WRITTEN, A BACK GROUND PROGRAM WILL CLEAR THE BUFFER AREA ALREADY WRITTEN ON THE DISK.
- J. WHEN THE WRITE AND CLEAR IS COMPLETE, DATA WILL BE READ OFF THE CURRENT DRIVE INTO THE BUFFER AREA. AS DATA IS READ, A BACK GROUND PROGRAM WILL CHECKSUM THE BUFFER INFORMATION ALREADY READ OFF THE DISK.
- K. WHEN THE READ AND CHECKSUM IS COMPLETE, THE CHECKSUM FOUND WILL BE COMPARED TO THE CHECKSUM SAVED PREVIOUS TO THE WRITE OPERATION. IF CHECKSUMS DO NOT COMPARE OR IF A CRC ERROR HAS OCCURRED, A WORD BY WORD COMPARE WILL BE MADE TO DETERMINE AND TYPE OUT THE BAD DATA FOUND.
- L. STEPS A-H WILL BE REPEATED AND THE DRIVE POLE WILL BE STARTED AT THE CURRENT DRIVE +1.
- M. FOR ALL POSSIBLE ERRORS, SEE SECTION 5 IN THIS DOCUMENT.

9.

CONSOLE PACKAGE ADDENDUM

9.1. DESCRIPTION

THE CONSOLE PACKAGE HAS BEEN ADDED TO THIS DIAGNOSTIC TO ALLOW THE PROGRAM TO RUN WITH NO HARDWARE SWITCH REGISTER AND TO HAVE COMMUNICATIONS WITH THE DIAGNOSTIC VIA A TERMINAL. THE DIAGNOSTIC CAN BE RUN IN TWO MODES WITH THE CONSOLE PACKAGE . 1) RUNNING WITH THE CONSOLE PACKAGE ACTIVE . THIS ALLOWS THE OPERATOR CONTROL OF THE DIAGNOSTIC THROUGH THE TERMINAL. THE DIAGNOSTIC WILL ASK FOR THE VALUE OF THE PSEUDO SWITCH REGISTER, BEFORE CONTINUING WITH EXECUTION OF THE DIAGNOSTIC. ALL ERRORS AND PASS COMPLETES WILL BE PRINTED AT THE TERMINAL. NO HALTS WILL BE EXECUTED. 2) CONSOLE PACKAGE NOT ACTIVE-THIS WILL RESULT IN THE NORMAL STANDALONE OPERATION OF THE PROGRAM AS DISCRIBED IN SECTIONS 1 THROUGH 8 OF THIS DUCUMENT.

9.2 RESTRICTIONS

- 1) RUNNING THE CONSOLE PACKAGE REQUIRES THAT THE PSEUDO SWITCH REGISTER BE USED.
- 2) ONCE RUNNING THE CONSOLE PACKAGE NONACTIVE AND NOW DESIRE TO RUN IT ACTIVE. ONE MUST RELOAD THE DIAGNOSTIC AND INITILIZE FOR A ACTIVE CONSOLE PACKAGE.

9.3 INITIALIZATION

FOR A ACTIVE CONSOLE PACKAGE

- 1.) SET LOCATION 21 BIT0=0 TO INDICATE USE PSEUDO SWITCH REGISTER.
- 2.) SET LOCATION 22 BIT3=1 TO INDICATE CONSOLE PACKAGE ACTIVE.

FOR A NON ACTIVE CONSOLE PACKAGE

- 1.) SET LOCATION 21 BIT0=1 TO INDICATE NOT TO USE PSEUDO SWITCH REGISTER, BUT TO USE HARDWARE SWITCHES.
- 2.) SET LOCATION 22 BIT3=0 TO INDICATE CONSOLE PACKAGE NOT ACTIVE.

9.4 CONTROL CHARACTERS

CONTROL CHARACTERS ARE USED TO GIVE THE OPERATOR THE

ABILITY TO PERFORM THE FOLLOWING FUNCTIONS.
NOTE: THE PROGRAM WILL RESPOND TO THE CONTROL
CHARACTER IN FIVE (5) SECONDS OR LESS.

CONTROL C

THIS WILL START THE LOADER THAT IS
IN LOCATION 7600.

CONTROL R

THIS WILL RESTART THE PROGRAM AND
REASK THE SWITCH REGISTER QUESTION AS
DESCRIBED IN SECTION 9.6.

CONTROL E

THIS WILL CONTINUE THE PROGRAM FROM
AN ERROR IF ALLOWED BY THE DIAGNOSTIC
OR FROM A WAITING STATEMENT.

CONTROL L

THIS WILL SWITCH THE TERMINAL MESSAGES
FROM THE DISPLAY TO A LINE PRINTER.
TO RESTORE THE MESSAGES ON THE TERMINAL
CONTROL L MUST BE TYPED AGAIN. IF
NO PRINTER IS AVAILABLE AND CONTROL L
IS TYPED THE RESULT WILL BE THAT THE
CONSOLE PACKAGE WILL WAIT FOR CONTROL C OR R.
THE CONTROL L WILL OUTPUT TO THE LINE
PRINTER AND THE PROGRAM WILL
ATTEMPT TO CONTINUE AS IF A CONTROL E
WAS TYPED IN.

CONTROL D

THIS WILL ALLOW THE ABILITY TO CHANGE
THE SWITCH REGISTER DURING PROGRAM
OPERATION. TYPING THIS CHARACTER WILL RESULT
IN AN INTERIGATION OF THE SWITCH REGISTER
QUESTION AS DESCRIBED IN SECTION 9.6.

CONTROL S

THIS WILL STOP PROGRAM EXECUTION AND WAIT IN A
LOOP FOR A CONTINUE. THE ONLY WAY TO CONTINUE
WILL BE TO TYPE A CONTROL Q, R OR C .
THIS IS A NONPRINTING CHARACTER.

CONTROL Q

THIS IS TO CONTINUE A PROGRAM AFTER A CONTROL
S IS TYPED. THIS IS A NONPRINTING CHARACTER.

9.5

WAITING MESSAGE

THE WAITING MESSAGE IS USED TO ALLOW THE OPERATOR TIME
TO MAKE A DECISION AS TO WHAT CONTROL CHARACTER
TO TYPE. THIS MESSAGE MAY APPEAR AT THE END
OF PASS MESSAGE IF THE HALT ON PASS BIT IS SET. THE CONTROL
CHARACTERS MAY NOW BE USED TO PERFORM THE NEEDED FUNCTION.

THE WAITING MESSAGE MAY BE PRINTED AFTER A ERROR MESSAGE
IF THE HALT ON ERROR BIT IS SET, HERE AGAIN THE CONTROL
CHARACTERS MAY BE USED.
THE WAITING MESSAGE MAY BE PRINTED IF OPERATOR INTERVENTION
IS REQUIRED.

9.6 SWITCH REGISTER MESSAGE

THIS MESSAGE IS USED TO SETUP THE PSEUDO SWITCH REGISTER
BEFORE PROGRAM EXECUTION TAKES PLACE, THE SWITCH REGISTER
IS SETUP WHEN THE FOURTH CHARACTER IS ENTERED OR A CARRIAGE
RETURN IS TYPED

SR=0000 4000

UNDER SCORING INDICATES OPERATOR RESPONSE

9.7 END OF PASS

THE NORMAL PASS COMPLETE TYPEOUT AS DESCRIBED IN SECTION
5.5 IS USED.

9.8 ERRORS

THE STANDARD ERROR REPORTS AS DESCRIBED IN SECTION 5
OF THIS DOCUMENT WILL BE USED.

9.9 SWITCH REGISTER SETTINGS

THE STANDARD SWITCH SETTINGS AS DESCRIBED IN SECTION 3
OF THIS DOCUMENT WILL BE USED.

9.10 PARAMETER CONTROL WORDS

THE CONSOLE PACKAGE USES THE LOCATIONS 20 21 22 FOR THE
FOLLOWING PURPOSES.

LOCATION 20
PSEUDO SWITCH REGISTER

LOCATION 21
HARDWARE IDENTIFIER 1

LOCATION 22

HARDWARE IDENTIFIER 2

LOCATION 0021

BIT ---	OCTAL VALUE -----	FUNCTION WHEN 0 -----	FUNCTION WHEN 1 -----
0	4000	USE PSEUDO SWITCHES	USE HARDWARE SWITCHES
1	2000	NO OPTION 1	HAS OPTION 1
2	1000	NO OPTION 2	HAS OPTION 2
3	400	NO 8A SIMULATOR	HAS 8A SIMULATOR
4	200	NO OPTION SIMULATOR	HAS OPTION SIMULATOR
5	100	NOT ON 8A XOR	ON 8A XOR
6	40	NOT PDP8-E TYPE CPU	PDP8-E TYPE CPU

7-11

8A MEMORY SIZE EX. 1K=00
2K=01
7K=06

32K=31

LOCATION 0022

BIT ---	OCTAL VALUE -----	FUNCTION WHEN 0 -----	FUNCTION WHEN 1 -----
0	4000	NOT ON ACT8A LINE	ON ACT 8A LINE
1	2000	NOT ON ACT 8E LINE	ON ACT 8E LINE
2	1000	NOT YET DEFINED	
3	400	DEACTIVE CONSOLE PACKAGE	ACTIVE CONSOLE PACKAGE

9.11

LOCATION CHANGES -----

THE FOLLOWING FIELD 1 LOCATIONS CAN BE CHANGED TO MEET THE SPECIFIC NEED FOR MODIFICATION OF THE DIAGNOSTIC.

0246

IS THE LOCATION FOR THE VALUE OF THE NUMBER OF PROGRAM PASSES NEED TO PRINT THE END OF PASS MESSAGE.

1037

IS THE LOCATION SET FOR THE NUMBER OF FILLER CHARACTERS AFTER A CRLF SET TO FOUR (4)

10.

APT-8 HOOKS -----

10.1 DESCRIPTION

TWO INTERFACES HAVE BEEN PROVIDED WHICH ALLOW THIS DIAGNOSTIC TO RUN UNDER THE STANDARD APT-8 SYSTEM. THESE INTERFACES ARE:

1. TIMING INTERFACE

2. ERROR INTERFACE

EACH WILL BE EXPLAINED IN DETAIL.

10.2 SETUP

THE FOLLOWING INFORMATION MUST BE INDICATED DURING THE INITIAL PROGRAM START UP.

1. SINGLE OR MULTIPLE DRIVE TESTING.
2. DRIVE OR DRIVES TO BE TESTED.
3. DIAGNOSTIC RUNNING UNDER APT-8.
4. THE AMOUNT OF MEMORY IN 1K INCREMENTS.

IF SINGLE DRIVE TESTING BIT 5 OF ADDRESS 22 MUST BE SET TO A ONE (1) WITH BITS 6-11 CONTAINING THE DRIVE TO BE TESTED. IF MULTIPLE DRIVES ARE TO BE DONE BIT MUST BE SET TO A ZERO (0) AND BIT 6-11 CONTAINING THE HIGHEST NUMBER DRIVE TO BE TESTED. WHEN MULTIPLE DRIVE TESTING ONLY A SPECIFIC NUMBER OF DRIVES CAN BE INDICATED. THE PROGRAM ASSUMES THE DRIVES ARE TO BE DONE BEGINNING WITH DRIVE ZERO (0) AND FINISHING WITH THE HIGHEST DRIVE INDICATED. IF MULTIPLE DRIVES OTHER THAN CONSECUTIVELY NUMBERED DRIVES BEGINNING WITH DRIVE ZERO (0) ARE TO BE DONE, THEY MUST BE DONE AS SINGLE DRIVES AND TESTED INDEPENDANTLY.

THE PROGRAM ALLOWS DRIVES 0-7 TO BE SELECTED. USER SHOULD NOT EXCEED 0-3 DRIVES FOR THE MK8E CONTROL.

BIT ZERO OF ADDRESS 22 MUST BE SET TO A ONE TO INDICATE THAT THE PROGRAM WILL RUN UNDER APT-8.

NOTE: IT SHOULD BE NOTED AT THIS TIME THAT WHILE RUNNING UNDER APT-8 THE HARDWARE SWITCH REGISTER IS INOPERATIVE. ONLY THE HALT AND SINGLE STEP SWITCH WILL EFFECT THE PROGRAM RUN.

AMOUNT OF MEMORY IN 1K INCREMENTS SHOULD BE STORED IN BITS 7-11 OF LOCATION 21. AN ADDITION OF 1 TO THE NUMBER OF BITS IN 7-11 INCREASES MEMORY SIZE BY 1K. EX. 4K=3/8K=7. REMEMBER TO RETAIN STATUS OF BITS WHEN MODIFYING LOCATION 21.

APT-8 INTERFACES:

10.3.1. TIMING

APT-8 IS NOTIFIED OF PROGRAM RUN BETWEEN .2 SEC AND 2.0 SEC ON A 1.2 MICROSECOND MEMORY CYCLE. THIS WILL ALLOW THE DIAGNOSTIC TO RUN WITHOUT CAUSING AN APT-8 TIMEOUT ERROR IF THE DIAGNOSTIC IS TO BE RUN ON THE SLOWER MOS MEMORY.

10.3.2. ERRORS

ONLY THE DRIVE IN ERROR IS REPORTED TO APT-8 SYSTEM.
SYSTEM. ERRORS WHICH CAUSE A PROGRAMMED HALT CAUSE A TIMEOUT
ERROR. IF A PROGRAMMED HALT SHOULD OCCUR, THE ERROR PC WILL
APPEAR IN THE AC ON THE DEVICE UNDER TEST. PROGRAMMED HALTS
ARE EXPLAINED EARLIER IN THIS DOCUMENT.

11. PROGRAM LISTING

```

/RK8E/RK8L DATA RELIABILITY PROGRAM: MU=88-DHRKC=M
/
/MAINDEC=88-DHRKC=M-L
/COPYRIGHT 1972,1975,1976,1977 DIGITAL EQUIP. CORP.
/
/MAYNARD, MASS. 01754
/
0001      FIELD 1
/
/CONSOL SRC=V2=RB= CONSOLE PACKAGE
/
/THE PROGRAM SHOULD CHECK FOR A CONTROL CHARACTER FROM THE TERMINAL
/EVERY FIVE(5) SECONDS OR SOONER.

/LOCATIONS THAT NEED TO BE SET UP FOR USING THE CONSOLE PACKAGE.

/CNTVAL IN XC8PASS THIS LOCATION DETERMINDS THE NUMBER OF
/PROGRAM COMPLETIONS THAT ARE NEEDED BEFORE THE PASS MESSAGE IS TYPED
/THE VALUE SHOULD PUT THE PASS MESSAGE OUT IN THE RANGE OF 1 TO 5 MINUTES.
/THIS SHOULD BE A POSITIVE NUMBER.

/C88TRY THIS IS FOUND IN CNTRL ROUTINE CONTROL R PART
/IT IS THE RETURN WHEN CONTROL R IS ENTERED (RESTART PROGRAM)
/THE RETURN JUMPS TO X008H WHICH CONTAINS C88TRY SO PUT THE LABEL C88TRY
/WHERE YOU WANT TO RESTART THE PROGRAM.

/SETUP1 IN XC8ERR THIS IS THE MASK BIT FOR HALT ON ERROR
/PLACE THE CORRECT BIT IN THIS LOCATION FOR HALTING ON ERRORS.

/SETUP2 IN XC8PASS THIS IS THE MASK FOR HALT A END OF PASS.

/THE CALL TABLE IS A CONDITIONAL ASSEMBLY.
/TO ASSEMBLE THE CALL REMOVE THE / BEFORE CONSOL=0.
/IN COMBINING THE CONSOL PACKAGE TO A DIAGNOSTIC.
/THE CALL TABLE IS TO BE AT THE BEGINNING OF A PROGRAM.

```

```

0000      CONSOL=0
0001      PSKF= 0001
0002      PCLF= 0002
0003      PSKE= 0003
0004      PSTB= 0004
0005      PSIE= 0005
0006      GTF= 0006
0007      ACL= 0007
0008      CAF= 0008
0009      MQL= 0009
0010      MQA= 0010
/
0020      *20
/
0020      0000      F15WR, 0
0021      4000      F10P1, 4000

```

```

0022      0000      F10P2, 0
/
/IFDEF CONSOL <

0024      *24

4424      C8PASS= JMS I .
0024      0200      XC8PAS . /C8 PASS COMPLETION ROUTINE
4425      C8CKSW= JMS I .
0025      0262      XC8SW . /CHECK SW REG SETTING
4426      C8TTYI= JMS I .
0026      0272      XC8TTY . /FETCH CONSOL CHAR
4427      C8CNTR= JMS I .
0027      0400      XC8CNT . /CHECK FOR CONTROL CHAR
4430      C8PRNT= JMS I .
0030      0303      XC8PNT . /C8 PRINT A BUFFER
4431      C8SWIT= JMS I .
0031      0656      XC8PSW . /SET UP PSEUDO SW, REG
4432      C8OCTA= JMS I .
0032      1000      XC8OCT . /CONVERT TO ASCII AND PRINT
4433      C8CRLF= JMS I .
0033      1023      XC8CRL . /DO A CARRIAGE RETURN+ LINE FEED
4434      C8ECHO= JMS I .
0034      1063      XC8ECH . /CHECK INPUT CHAR
4435      C8TYPE= JMS I .
0035      1077      XC8TYP . /C8 PRINT ONE CHAR
4436      C8ERR= JMS I .
0036      1207      XC8ERR . /C8 ERROR HANDLER
4437      C8INQU= JMS I .
0037      0635      XC8ING . /LOOK FOR OPERATOR INTERVENTION
4440      C8CKPA= JMS I .
0040      1041      XC8CKP . /CHECK IF CONTROL CHAR
4441      C8PAUS= JMS I .
0041      0337      XC8PAU . /IF CONSOL PACKAGE RETURN CALL PLUS ONE
/IF NOT USING CONSOL REPLACE CALL WITH
/ A HLT AND THEN GO TO THE HALT

/*****
/ *20 /PSEUDO SWITCH REGISTER
/ *21 /HARDWARE INDICATORS
/ 4000=USE FRONT PANEL SWITCH REGISTER
/ 0000=USE THE PSEUDO SWITCH REGISTER LOC.20
/ *22 /SYSTEM CONFIGURATION
/ 4000=CONSOL PACKAGE SET ACTIVE
/ 0000=CONSOL PACKAGE SET DEACTIVE
/ *23 /RESERVED FOR FUTURE USE
/
0200      *200
/
/*****
/C8PASS

```

```

/THIS IS CALLED AT THE END OF EACH PROGRAM COMPLETION
/THE VALUE OF** CNTVAL** WILL BE DETERMINED BY THE TIME IT TAKES
/THE PROGRAM TO COMPLETE THIS MANY CPASS TO BE IN THE 1 TO 4 MINUTE
/RANGE
/      CPASS**JMS      XC8PAS
/EX. OF CALL      CPASS
/      /      HLT      /HALT IF NON CONSOL PACKAGE
/      JMP      START1 /CONTINUE RUNNING THIS PROGRAM

```

```

/RETURN TO LOCATION CALL PLUS ONE WITH THE AC=0 IF NON CONSOL PACKAGE AND HLT
/IF CONTINUE TO RUN THEN RETURN TO CALL PLUS2 AC=0
/THE LOCATION SETUP2 IS THE MASK BIT FOR THE HALT AT END OF PASS
/CHECK THAT IT IS CORRECT FOR THE CURRENT PROGRAM

```

```

/CALLS USED BY XC8PAS ARE  CHKCLA-XC8CNLF-XC8OCTA-XC8SW-XC8PNT-XC8ING-

```

```

0200 0000      XC8PAS, 0
0201 7200      CLA
0202 4777'      JMS      CHKCLA      /IS WORD 22 BIT 3 ACTIVE CONSOLE?
0203 5212      JMP      DOPACK      /IS CLASSIC
0204 4776'      JMS      C8GET      /GET REGISTERS.
0205 4262      JMS      XC8SW      /DEACTIVE CONSOL CHECK SR SETTING
0206 0375      AND      (400      /FOR HALT ON END OF CPASS
0207 7640      SZA CLA      /1= HALT 0 CONTINUE
0210 5600      JMP I      XC8PAS      /GO TO HALT
0211 5230      JMP      C8BY1      /CONTINUE ON RUNNING PROGRAM
0212 4232      DOPACK, JMS      CKCOUT /CLASS CHECK CPASS COUNT
0213 5230      JMP      C8BY1      /CPASS COUNT NOT DONE REDO PROGRAM
0214 2250      ISZ      PASCNT      /CPASS COUNT DONE SET CPASS COUNT
0215 4774'      JMS      XC8CNLF
0216 4303      JMS      XC8PNT      /CPKNT BUFFER
0217 0253      MESPAS
0220 1250      TAD      PASCNT      /GET NUMBER
0221 4773'      JMS      XC8OCTA      /CONVERT IT TO ASCII
0222 4774'      JMS      XC8CNLF      /DO A CARRIAGE RETURN
0223 4776'      JMS      C8GET      /GET REGISTERS.
0224 4262      JMS      XC8SW      /CHECK A HALT AT END OF CPASS
0225 0375      SETUP2, AND      (400 /MASK BIT
0226 7640      SZA CLA      /HALT 01 NO SKIP CONTINUE =0
0227 4772'      JMS      XC8ING      /STOP PROGRAM EXECUTION-LOOK FOR INPUT
0230 2200      C8BY1, ISZ      XC8PAS /BUMP RETURN
0231 5600      JMP I      XC8PAS
0232 0000      CKCOUT, 0
0233 1251      TAD      DOSET      /CHECK IF SET UP NEEDED
0234 7640      SZA CLA      /0=SET UP CPASS COUNT VALUE
                                /1=CPASS COUNT VALUE OK
                                /CPASS COUNT VALUE ON
0235 5242      JMP      NOSET      /GET COUNT VALUE FOR THIS PROG
0236 1252      TAD      CNTVAL      /SET TO NEGATIVE
0237 7040      CMA
0240 3247      DCA      DOCNT      /STORE IN HERE
0241 2251      ISZ      DOSET      /INDICATE VALUE SET UP
0242 2247      NOSET, ISZ      DOCNT /COUNT THE NUMBER OF PASSES
0243 5230      JMP      C8BY1      /EXIT FOR ANOTHER PASS

```

```

0244 3251      DCA      DOSET      /SET TO CPKNT CPASS
0245 2232      ISZ      CKCOUT      /BUMP RETURN FOR
0246 5632      JMP I      CKCOUT      /CPASS C8TYPE OUT
0247 0000      DOCNT, 0
0250 0000      PASCNT, 0
0251 0000      DOSET, 0
0252 0000      CNTVAL, 0
0253 0410      MESPAS, TEXT      "DHRKC PASS "
0254 2213
0255 0310
0256 4040
0257 2001
0260 2323
0261 4000

```

```

/*****

```

```

/C8CKSW

```

```

/THIS ROUTINE CAN BE USED INPLACE OF A READ THE SWITCHES LAS.
/ROUTINE THAT WILL CHECK WHERE TO READ THE
/C8 SWITCHES FROM IE. FROM PANEL OR PSEUDO SWITCH REGISTER
/THE SELECTION IS DETERMINED BY THE STATE OF BIT 0 IN LOCATION 21.

```

```

/C8CKSW=      JMS XC8SW
/EX.      JMS      XC8SW      /READ THE C8SWIT REGISTER
                                /RETURN WITH THE CONTENTS OF SWITCH REGISTER

```

```

/RETURN TO NEXT LOCATION FOLLOWING CALL WITH THE AC= TO VALUE OF C8SWIT SETTING

```

```

/CALLS USED ARE-XC8CKPA-

```

```

0262 0000      XC8SW, 0
0263 4771'      JMS      XC8CKPA      /GO CHECK THE IF ANY CONTRL
0264 7000      NOP
0265 1021      TAD      21      /GET MU FUN INDICATOR
0266 7710      SPA CLA      /CHECK IF FROM PANEL 4000
0267 7614      7614      /DO LAS AND SKIP GET FROM PANEL WITH LAS
0270 1020      TAD      20      /PSEUDO SWITCH
0271 5662      JMP I      XC8SW      /EXIT WITH STATUS BIT IN AC.

```

```

/*****

```

```

/C8TTY1
/THIS ROUTINE WILL LOOK FOR A INPUT FROM THE TERMINAL
/AND REMOVE ANY PARITY BITS, THEN MAKE IT 8 BIT ASCII.
/      C8TTY1= JMS XC8TTY
/EX.      JMS      XC8TTY1      /READ CHAR FROM THE CONSOL DEVICE
                                /RETURN TO CALL PLUS ONE AC CONTAINS THE CHAR

```

/CALLS USED-NONE- BUT C8CHAR IS OFF PAGE AND IN ROUTINE CALLED XC8ECHO

```

/
/
0272 0000 XC8TTY, 0
0273 0031 KSF /LOOK FOR KEYBOARD FLAG
0274 5273 JMP -1
0275 0036 KRB /GET CHAR
0276 0370 AND (177 /MASK FOR 7 BITS
0277 1367 TAD (200 /ADD THE EIGHTH BIT
0300 3766 DCA C8CHAR /STORE IT
0301 1766 TAD C8CHAR
0302 5672 JMP I XC8TTY /EXIT

```

/*****

/C8PRNT

/THIS ROUTINE WILL TYPE THE CONTENTS OF THE C8 PRINT BUFFER, THE LOCATION
/OF THE BUFFER WILL BE IN THE ADDRS FOLLOWING THE CALL, PRINTING OF THE BUFFER
/WILL STOP WHEN A 00 CHAR IS DETECTED, C8CHARACTERS ARE PACKED 2 PER WORD.

/ C8PRNT= JMS XC8PNT

```

/EX. JMS XC8PNT /C8PRNT THE CONTENTS OF THE FOLLOWING BUFFER
/ MESS77 /LOCATION OF C8PRNT BUFFER

```

/C8PRNT WILL USE THE LOCATION FOLLOWING THE CALL AS THE POINTER FOR THE
/C8PRNT ROUTINE, RETURN TO CALL PLUS TWO WITH AC= 0

/CALLS USED ARE-XC8TYPE-XC8PNT

```

0303 0000 XC8PNT, 0
0304 7300 CLA CLL
0305 1703 TAD I XC8PNT /GET C8PRNT BUFFERS STARTING LOCATION
0306 3336 DCA PTSTOR /STORE IN PTSTOR
0307 2303 ISZ XC8PNT /BUMP RETURN
0310 1736 C8DO1, TAD I PTSTOR /GET DATA WORD
0311 0365 AND (7700 /MASK FOR LEFT BYTE
0312 7450 SNA /CHECK IF 00 TERMINATE
0313 5703 JMP I XC8PNT /EXIT
0314 7500 SMA /IS AC MINUS
0315 7020 CHL /MAKE CHAR A 300 AFTER ROTATE
0316 7001 IAC /MAKE CHAR A 200 AFTER ROTATE
0317 7012 RTR
0320 7012 RTR
0321 7012 RTR
0322 4764 JMS XC8TYPE /PUT CHAR IN BITS 4-11 MAKE IT 8 BIT ASCII
0323 1736 TAD I PTSTOR /C8PRNT IT ON CONSOLE
/GET DATA WORD

```

```

0324 0363 AND (0077 /MASK FOR RIGHT BYTE
0325 7450 SNA /CHECK IF 00 TERMINATOR
0326 5703 JMP I XC8PNT //EXIT
0327 1362 TAD (3740 /ADD FUDGE FACTOR TO DETERMINE IF 200
0330 7500 SMA /OR 300 IS TO BE ADD TO CHAR
0331 1361 TAD (100 /ADD 100
0332 1360 TAD (240 /ADD 200
0333 4764 JMS XC8TYPE /C8TYPE ONLY BITS 4-11
0334 2336 ISZ PTSTOR /BUMP POINTER FOR NEXT WORD
0335 5310 JMP C8DO1 /DO AGAIN
0336 0000 PTSTOR, 0 /STOP FOR C8PRNT BUFFER

```

/*****

/C8PAUS

/THIS ROUTINE WILL CHECK IF THE CONSOL PACKAGE IS ACTIVE, IF ACTIVE
/IT WILL RETURN TO CALL PLUS ONE AC= 0, AND DO THAT INSTRUCTION.
/IF THE CONSOL PACKAGE IS NOT ACTIVE THE CALL WILL BE REPLACED
/WITH A 7402 HALT AND THEN RETURN TO THE HALT.

/ C8PAUS= JMS XC8PAU

```

/EX. JMS XC8PAUS /CHECK IF ON ACTIVE CONSOL IF NOT HALT HERE
/ ANYTHING /RETURN HERE IF ON ACTIVE CONSOL
/
/

```

/CALLS USED ARE-CHKCLA=

```

0337 0000 XC8PAU, 0
0340 7300 CLA CLL
0341 4777 JMS CHKCLA /CHECK LOC 22 BIT 3 CONSOL BIT
0342 5350 JMP C8DO3 /GO TO CONSOL PART RETURN CALL+1
0343 7040 CMA /DEACTIVE CONSOL PACKAGE PUT HLT IN CALL
0344 1337 TAD XC8PAU /GET CURRENT RETURN ADDR
0345 3337 DCA XC8PAU /SET UP RETURN
0346 1357 TAD (7402 /GET CODE FOR HLT
0347 3737 DCA I XC8PAU /PUT HLT IN CALL LOCATION
0350 5737 C8DO3, JMP I XC8PAU /GO TO HALT OR RETURN TO NEXT LOCATION

```

```

0357 7402
0360 0240
0361 0100
0362 3740
0363 0077
0364 1077
0365 7700
0366 1075
0367 0200
0370 0177
0371 1041

```

0372 0635
0373 1000
0374 1023
0375 0400
0376 0624
0377 1200
0400

PAGE

/*****

/C0CNTN

/THIS ROUTINE WILL CHECK FOR THE PRESENCE OF CONTROL CHARACTERS

/IT WILL CHECK FOR THE FOLLOWING CHAR C-H-Q-L-S

/ C0CNTN= JMS XC0CNT

/EX.

JMS XC0CNTN

/CHECK FOR CONTROL CHARACTER

/

JMP ANYTHING

/LOC FOLLOWING CALL IS FOR CONTINUING THE PROGRAM

/

JMP ANYTHING

/LOC. IS FOR RETURN IF INMODE SET AND NOT CNTRL CHAR

/RETURN IS TO CALL PLUS ONE IF CONTINUE

/RETURN IS TO CALL PLUS TWO IF INMODE SET AND NOT CONTROL CHAR

/RETURN IS TO CALL PLUS TWO IF INMODE IS NOT SET AND NO

/CONTROL CHAR .,THIS WILL PRINT THE CHARACTER AND A ?

/CLEAR THE AC AND RETURN CALL+2.

/CALLS USED ARE=CMKCLA=XC0TYPE=XC0CRLF=C0GET=UPAROW=XC0TYI=XC0PSH=

/

/

/

0400 0000

XC0CNT, 0

0401 3777'

DCA

ACSAVE

/SAVE THE AC

0402 4776'

JMS

CMKCLA

/CHECK LOC.22 BITS FOR CONSOLE BIT

0403 5206

JMP

+3

/ON ACTIVE CONSOLE

0404 1777'

TAD

ACSAVE

/DEACTIVE CONSOLEGET AC FOR RETURN

0405 5600

JMP I

XC0CNT

/EXIT NOT ON ACTIVE CONSOLE

0406 6004

GTF

0407 3775'

DCA

FLSAVE

0410 7501

MCA

0411 3774'

DCA

MQSAVE

/SAVE THE MQ

0412 3255

DCA

INDEXA

/SET DISPLACEMENT INTO TABLE B

0413 1257

TAD

XTABLA

/GET ADDRS OF TABLE A

0414 3256

DCA

GETDAT

/CONTAINS POINTER TO CONTROL CHAR

0415 1656

REDDA,

TAD I

GETDAT

/SET CNTRL CHAR FROM TABLE

0416 7450

SNA

END OF TABLE

/CHECK FOR A 0 END OF TABLE

0417 5226

JMP

DONEA

/END OF TABLE NO CONTROL CHAR

0420 1773'

TAD

C0CHAR

/COMPARE CHAR TO CONTROL CHAR

0421 7650

SNA

CLA

/0 IF MATCH

0422 5243

JMP

GOITA

/MATCH

0423 2255

ISZ

INDEXA

/NO MATCH NOT END OF TABLE REDD

0424 2256

ISZ

GETDAT

/BUMP INDEX FOR EXIT WHEN CONTROL FOUND

0425 5215

JMP

REDDA

/BUMP GETDAT FOR COMPARE OF NEXT CNTRL CHAR.

0426 1772'

DONEA,

TAD

INMODE

/CHECK IF PROGRAM EXPECTS CHAR

0427 7640

SZA

CLA

/10CHAR EXPECTED 0= NO CHAR EXPECTED

0430 5240

JMP

EXITA

/CHAR EXPECTED

0431 1773'

TAD

C0CHAR

/GET CHAR= NOT CONTROL+ NOT EXPECTED

0432 4771'

JMS

XC0TYPE

/C0PKNT CHAR

0433 1370

TAD

C277

/GET CODE FOR "7"

0434 4771'

JMS

XC0TYPE

0435 4767'

JMS

XC0CRLF

0436 2200

ISZ

XC0CNT

/BUMP RETURN

0437 5600

JMP I

XC0CNT

/EXIT CALL+2

0440 2200

EXITA,

ISZ

XC0CNT

/BUMP RETURN FOR MAIN PROGRAM CHECK OF CHAR

0441 1773'

TAD

C0CHAR

/PUT CHAR IN AC.

0442 5600

JMP I

XC0CNT

/EXIT

0443 1773'

GOITA,

TAD

C0CHAR

/GET THE CONTENTS OF CHAR

0444 1366

TAD

C100

/ADD 100 TO FORM A GOOD ASCII CHARACTER

0445 3773'

DCA

C0CHAR

/RESTORE CORRECT CHAR

0446 1260

TAD

XTABLB

/GET START OF TABLE B

0447 1255

TAD

INDEXA

/GET NOW P \ INTO TABLE

0450 3254

DCA

G0TOA

/STORE IT

0451 1654

TAD I

G0TOA

/GET THE ROUTINE STARTING ADDRESS

0452 3254

DCA

G0TOA

/STORE IT IN HERE

0453 5654

JMP I

G0TOA

/GOTO CONTROL CHAR ROUTINE

0454 0000

G0TOA,

0000

0000

/ADD OF CNTRL ROUTINE TO EXECUTE

0455 0000

INDEXA,

0000

0000

/DISPLACEMENT INTO CNTRL TABLE

0456 0000

GETDAT,

0000

0000

/LOCATION OF ADDRS OF CONTROL CHAR.

0457 0461

XTABLA,

TABLA

TABLA

/ADDMS OF TABLEA

0460 0471

XTABLB,

TABLB

TABLB

/ADDMS OF TABLEB

0461 7575

TABLA,

7575

7575

/CNTRL C BACK TO MONITOR 203

0462 7564

7564

7564

/CNTRL L SWITCH ERROR PRINTING DEVICE 214

0463 7557

7557

7557

/CNTRL Q START DISPLAYING CHAR, AGAIN 221

0464 7556

7556

7556

/CNTRL R BACK TO BEGINNING OF PROGRAM 222

0465 7555

7555

7555

/CNTRL S STOP SENDING CHAR TO DISPLAY WAIT FOR CNTRL Q 223

0466 7573

7573

7573

/CNTRL E CONTINUE WITH PROGRAM 205

0467 7574

7574

7574

/CNTRL D CHANGE SWITCH REGISTER ON FLY

0470 0000

0000

0000

0471 0551

TABLB,

CNTRL

CNTRL

0472 0537

CNTRL

CNTRL

0473 0500

CNTRL

CNTRL

0474 0511

CNTRL

CNTRL

0475 0521

CNTRL

CNTRL

0476 0545

CNTRL

CNTRL

0477 0000

CNTRL

CNTRL

/

/CNTRL Q

/START SENDING CHAR. TO THE DISPLAY

/THIS WILL RETURN CONTROL TO CALL THAT WAS SET BY

/THE CALL FOR CONTROL S.

/

0500 3772'

CNTRLQ,

DCA

INMODE

/SET SUFT FLAG FOR UNEXPECTED CHAR

0501 1335

TAD

C0SETS

/CHECK IF CONTROL S TYPED IN

0502 7640

SZA

CLA

0503 5306

JMP

BYNETH

/CONTROL S TYPED IN

0504 4765'

JMS

C0GET

/NO CONTROL S TYPED PREVIOUSLY

0505 5600

JMP I

XC0CNTN

/LEAVE VIA CNTH ENTRY ADDRESS

0506 3335

BYNETH,

DCA

C0SETS

/CLEAR THE SUFT FLAG

0507 4765'

JMS

C0GET

/RESTORE REGISTERS

0510 5736

JMP I

C0NETH

/EXIT TO ADDRESS SET BY CONTROL S

```

/
/CONTROL R
/GO TO THE QUESTION C8SWIT
0511 3764' CNTRLR, DCA TTYLPT /CLEAR THE TYPE FLAG SET TO TTY
0512 3335 DCA C8SETS /CLEAR SOFT FLAG FOR CNTRL S
0513 3772' DCA INMODE
0514 4763' JMS UPAROW /PRINT THE " AND C8CHAR
0515 3762' C8BY4, DCA C8SWST /CLEAR FLAG FOR CNTRL D OR R
0516 6203 CIP CIP 0
0517 5720 JMP I XDUSW /GO TO ADDRS OF C8SWIT
0520 0200 XDUSW, BGN /DUSW IS LABEL FOR C8SWIT QUESTION
/
/CONTROL S
/STOP SENDING CHAR, TO DISPLAY UNTIL A "Q IS RECEIVED
/
0521 1335 CNTRL8, TAD C8SETS /IF1 DO NOT STORE IN C8RETR
0522 7640 BZA CLA
0523 5327 JMP C8D07 /DONT SET UP C8RETR
0524 7001 IAC /MAKE RETURN CALL PLUS 2
0525 1200 TAD XC8CNT /GET RETURN FOR THIS CALL
0526 3336 DCA C8RETR /STORE IT HERE FOR USE BE CNTRL Q
0527 2335 C8D07, I8Z C8SETS /SET FLAG TO SAVE CALL
0530 4761' JMS XC8TTYI /LOOK FOR THE INPUT
0531 4765' JMS C8GET /GET REGISTERS
0532 4200 JMS XC8CNTR /CHECK FOR THE CONTROL CHAR
0533 7200 CLA
0534 5321 JMP CNTRL8 /IF NOT A CNTRL Q R C REASK
0535 0000 C8SETS, 0
0536 0000 C8RETR, 0
/
/ SWITCH OUTPUT FROM ONE OUTPUT DEVICE TO ANOTHER- THE TWO OUTPUTS ARE THE
/ CONSOLE AND THE PRINTER WITH DEVICE CODE 06.
/
0537 1764' CNTRL8, TAD TTYLPT /GET PRESENT C8SWIT INDICATOR
0540 7000 CMA /COMPLEMENT IT
0541 3764' DCA TTYLPT /STORE NEW C8SWIT
0542 4763' JMS UPAROW /C8PRNT " AND CHAR ON NEW DEVICE
0543 4765' JMS C8GET /RESTORE THE REGISTERS
0544 5600 JMP I XC8CNT /EXIT
/
/CONTROL E
/CONTINUE RUNNING FROM A INQUIRE OR ERROR
/
0545 4763' CNTRL8, JMS UPAROW /PRINT THE CONTROL CHAR
0546 3762' DCA C8SWST /CLEAR ENTRY FLAG.
0547 4765' JMS C8GET /GET THE REGISTERS
0550 5600 JMP I XC8CNT /RETURN TO CALL PLUS ONE
/
/CONTROL C

```

```

/RETURN TO MONITOR CONTROL C
0551 3764' CNTRL8, DCA TTYLPT /CLEAR THE LPT FLAG TO PRINT ON DISPLAY
0552 3762' DCA C8SWST /CLEAR ENTRY FLAG.
0553 4763' JMS UPAROW /C8PRNT " AND LETTER IN CHAR
0554 6203 CIP CIP 0 /GO TO 0 FLD
0555 6007 CAF /CLEAR THE WORLD
0556 5760 JMP I (7600 /GO TO DIAGNOSTIC MONITOR
/*****
/
/
0560 7600
0561 0272
0562 0745
0563 0615
0564 1121
0565 0624
0566 0100
0567 1023
0570 0277
0571 1077
0572 1076
0573 1075
0574 1346
0575 1347
0576 1200
0577 1345
0600
/
/CONTROL D
/CHANGE THE SWITCH REGISTER ANYTIME CNTRL D AND RETURN TO
/ THE PROGRAM RUNNING.
/
0600 4215 CNTRL8, JMS UPAROW
0601 1213 TAD C8SETD /CHECK IF THE RETURN ADDRS IS SAFE
0602 7640 BZA CLA
0603 5207 JMP C8D011 /DO NOT CHANGE THE RETURN ADDRS
0604 1777' TAD XC8CNT /GET THE RETURN ADDRS AND SAVE IT
0605 3214 DCA C8RETD /SAVE THE RETURN HERE
0606 2213 I8Z C8SETD /INDICATE RETURN SAVED DONT DESTROY
0607 4256 C8D011, JMS XC8PSW /GO CHANGE THE SWITCH REGISTER
0610 3213 DCA C8SETD /CLEAR THE FLAG
0611 4224 JMS C8GET /RESTORE THE AC MQ LINK ETC
0612 5614 JMP I C8RETD /RETURN TO THE PROGRAM
/
0613 0000 C8SETD, 0
0614 0000 C8RETD, 0
/
/THIS WILL TYPE A UP ARROW AND THE CHAR IN C8CHAR.
0615 0000 UPAROW, 0 /C8PRNT THE " AND THE CHAR C8TYPED IN

```

```

0016 1376      TAD      (356      /CODE FOR -
0017 4775*     JMS      XC8TYPE
0020 1774*     TAD      C8CHAR    /C8TYPE THE CHAR
0021 4775*     JMS      XC8TYPE
0022 4773*     JMS      XC8CRLF
0023 5015      JMP I     UPAROW    /EXIT

```

```

0024 0000      C8GET, 0
0025 7200      CLA
0026 1772*     TAD      MQSAVE
0027 7421      MQL          /RESTORE MQ
0030 1771*     TAD      FLSAVE
0031 7004      RAL          /RESTORE THE LINK
0032 7200      CLA
0033 1770*     TAD      ACSAVE
0034 5024      JMP I     C8GET    /RESTORE THE AC
                                /GET THE REGISTERS

```

```

/C8INQU
/C8INQU ROUTINE WILL PRINT A WAITING
/AND THE PROGRAM IS EXPECTING A CONTROL CHAR INPUT
/IF CONTINUE FROM CONTROL CHAR RETURN IS CALL PLUS ONE
/IF NO CONTROL CHAR ENTERED THEN WAITING IS REPRINTED
/AND PROGRAM WAITS FOR A CONTROL CHAR AGAIN.

```

```

/      C8INQU =      JMS XC8INQ

```

```

/EX.      JMS      XC8INQ      /C8 WILL PRINT A WAITING AND WAIT FOR INPUT
/          DD ANYTHING        /RETURN IS CALL PLUS ONE AC #0 CONTINUE

```

```

/CALLS USED ARE=CHKCLA=XC8PNT=XC8TYI=C8GET=XC8CNTR=

```

```

0035 0000      XC8INQ, 0
0036 7300      CLA CLL
0037 4767*     JMS      CHKCLA    /CHECK LOC 22 BIT 3 CONSOLE BIT
0040 7410      SKP          /ACTIVE CONSOLE PACKAGE
0041 5035      JMP I     XC8INQ    /NOT CONSOLE LEAVE
0042 4766*     JMS      XC8PNT
0043 0051      WATME8
0044 4765*     JMS      XC8TYI    /INQUIRY WAITING
0045 4224      JMS      C8GET     /GET CHARACTER
0046 4777*     JMS      XC8CNTR
0047 5035      JMP I     XC8INQ    /CHECK IF CONTROL CHARACTER
0050 5236      JMP      XC8INQ+1  /EXIT AND CONTINUE
0051 2701      WATME8, TEXT "WAITING " /REABK
0052 1124
0053 1116

```

```

0054 0740
0055 0000

```

```

/C88WIT

```

```

/ROUTINE WILL CHECK IF CONSOLE IS ACTIVE IF IT IS ACTIVE DISPLAY
/SW QUESTION. IN NOT ACTIVE IT WILL NOT PRINT THE SW QUESTION BUT
/RETURN TO CALL PLUS ONE AC#0.
/C88WIT WILL SET UP THE PSEUDO SWITCH
/REGISTER WITH THE NEW DATA ENTERED
/
/      C88WIT =      JMS XC8PSW

```

```

/EX.      JMS      XC8PSW

```

```

/SET UP PSEUDO C88WIT REGISTER IF
/ON THE CONSOLE PACKAGE. RETURN IS CALL PLUS ONE AC = 0

```

```

/CALLS USED ARE=CHKCLA=XC8PSW=XC8PNT=XC8OCTA=XC8TYPE=

```

```

0056 0000      XC8PSW, 0
0057 4767*     JMS      CHKCLA    /CHECK LOC 22 BIT 3 CONSOLE BIT
0060 7410      SKP          /ACTIVE CONSOLE
0061 5036      JMP I     XC8PSW    /DEACTIVE CONSOLE PACKAGE
0062 1345      TAD      C88WST
0063 7640      SZA CLA
0064 5764*     JMP      C8BY4
0065 2345      ISZ      C88WST
0066 4766*     C88OPS, JMS      XC8PNT /FIRST ENTRY SET FLAG
0067 0747      MESA
0068 1020      TAD      20
0069 4763*     JMS      XC8OCTA    /GET CONTENTS OF SW
0070 1362      TAD      (40
0071 4775*     JMS      XC8TYPE    /CONVERT IT TO ASCII
0072 2761*     ISZ      INMODE     /GET SPACE
0073 4760*     JMS      XC8ECHO
0074 4315      JMS      TSTCHA
0075 1774*     TAD      C8CHAR
0076 3020      DCA      20        /SET FLAG FOR CHAR EXPECTED
0077 1774*     TAD      C8CHAR    /LOOK FOR INPUT
0078 3020      DCA      20        /NOT CONTROL TEST IT IS LEGAL
0079 3020      DCA      20        /STORE NEW CHAR IN SW REG
0080 3020      DCA      20
0081 1357      TAD      (=3
0082 3346      DCA      TMPCNT    /GET A MINUS 3
0083 4760*     JMS      XC8ECHO
0084 4315      JMS      TSTCHA
0085 1020      TAD      20
0086 7106      RTL CLL
0087 7004      RAL          /GET NEXT CHAR
0088 1774*     TAD      C8CHAR
0089 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0090 2346      ISZ      TMPCNT    /GET C88WIT REGISTER
0091 5303      JMP      GETCH1    /ROTATE IT LEFT 3 PLACES
0092 5342      JMP      ENDIT
0093 4760*     JMS      XC8ECHO
0094 4315      JMS      TSTCHA
0095 1020      TAD      20
0096 7106      RTL CLL
0097 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0098 1774*     TAD      C8CHAR
0099 3020      DCA      20        /SAVE NEW CONTENTS
0100 2346      ISZ      TMPCNT
0101 5303      JMP      GETCH1
0102 5342      JMP      ENDIT
0103 4760*     JMS      XC8ECHO
0104 4315      JMS      TSTCHA
0105 1020      TAD      20
0106 7106      RTL CLL
0107 7004      RAL          /GET NEXT CHAR
0108 1774*     TAD      C8CHAR
0109 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0110 2346      ISZ      TMPCNT
0111 5303      JMP      GETCH1
0112 5342      JMP      ENDIT
0113 4760*     JMS      XC8ECHO
0114 4315      JMS      TSTCHA
0115 1020      TAD      20
0116 7106      RTL CLL
0117 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0118 1774*     TAD      C8CHAR
0119 3020      DCA      20        /SAVE NEW CONTENTS
0120 2346      ISZ      TMPCNT
0121 5303      JMP      GETCH1
0122 5342      JMP      ENDIT
0123 4760*     JMS      XC8ECHO
0124 4315      JMS      TSTCHA
0125 1020      TAD      20
0126 7106      RTL CLL
0127 7004      RAL          /GET NEXT CHAR
0128 1774*     TAD      C8CHAR
0129 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0130 2346      ISZ      TMPCNT
0131 5303      JMP      GETCH1
0132 5342      JMP      ENDIT
0133 4760*     JMS      XC8ECHO
0134 4315      JMS      TSTCHA
0135 1020      TAD      20
0136 7106      RTL CLL
0137 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0138 1774*     TAD      C8CHAR
0139 3020      DCA      20        /SAVE NEW CONTENTS
0140 2346      ISZ      TMPCNT
0141 5303      JMP      GETCH1
0142 5342      JMP      ENDIT
0143 4760*     JMS      XC8ECHO
0144 4315      JMS      TSTCHA
0145 1020      TAD      20
0146 7106      RTL CLL
0147 7004      RAL          /GET NEXT CHAR
0148 1774*     TAD      C8CHAR
0149 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0150 2346      ISZ      TMPCNT
0151 5303      JMP      GETCH1
0152 5342      JMP      ENDIT
0153 4760*     JMS      XC8ECHO
0154 4315      JMS      TSTCHA
0155 1020      TAD      20
0156 7106      RTL CLL
0157 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0158 1774*     TAD      C8CHAR
0159 3020      DCA      20        /SAVE NEW CONTENTS
0160 2346      ISZ      TMPCNT
0161 5303      JMP      GETCH1
0162 5342      JMP      ENDIT
0163 4760*     JMS      XC8ECHO
0164 4315      JMS      TSTCHA
0165 1020      TAD      20
0166 7106      RTL CLL
0167 7004      RAL          /GET NEXT CHAR
0168 1774*     TAD      C8CHAR
0169 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0170 2346      ISZ      TMPCNT
0171 5303      JMP      GETCH1
0172 5342      JMP      ENDIT
0173 4760*     JMS      XC8ECHO
0174 4315      JMS      TSTCHA
0175 1020      TAD      20
0176 7106      RTL CLL
0177 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0178 1774*     TAD      C8CHAR
0179 3020      DCA      20        /SAVE NEW CONTENTS
0180 2346      ISZ      TMPCNT
0181 5303      JMP      GETCH1
0182 5342      JMP      ENDIT
0183 4760*     JMS      XC8ECHO
0184 4315      JMS      TSTCHA
0185 1020      TAD      20
0186 7106      RTL CLL
0187 7004      RAL          /GET NEXT CHAR
0188 1774*     TAD      C8CHAR
0189 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0190 2346      ISZ      TMPCNT
0191 5303      JMP      GETCH1
0192 5342      JMP      ENDIT
0193 4760*     JMS      XC8ECHO
0194 4315      JMS      TSTCHA
0195 1020      TAD      20
0196 7106      RTL CLL
0197 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0198 1774*     TAD      C8CHAR
0199 3020      DCA      20        /SAVE NEW CONTENTS
0200 2346      ISZ      TMPCNT
0201 5303      JMP      GETCH1
0202 5342      JMP      ENDIT
0203 4760*     JMS      XC8ECHO
0204 4315      JMS      TSTCHA
0205 1020      TAD      20
0206 7106      RTL CLL
0207 7004      RAL          /GET NEXT CHAR
0208 1774*     TAD      C8CHAR
0209 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0210 2346      ISZ      TMPCNT
0211 5303      JMP      GETCH1
0212 5342      JMP      ENDIT
0213 4760*     JMS      XC8ECHO
0214 4315      JMS      TSTCHA
0215 1020      TAD      20
0216 7106      RTL CLL
0217 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0218 1774*     TAD      C8CHAR
0219 3020      DCA      20        /SAVE NEW CONTENTS
0220 2346      ISZ      TMPCNT
0221 5303      JMP      GETCH1
0222 5342      JMP      ENDIT
0223 4760*     JMS      XC8ECHO
0224 4315      JMS      TSTCHA
0225 1020      TAD      20
0226 7106      RTL CLL
0227 7004      RAL          /GET NEXT CHAR
0228 1774*     TAD      C8CHAR
0229 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0230 2346      ISZ      TMPCNT
0231 5303      JMP      GETCH1
0232 5342      JMP      ENDIT
0233 4760*     JMS      XC8ECHO
0234 4315      JMS      TSTCHA
0235 1020      TAD      20
0236 7106      RTL CLL
0237 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0238 1774*     TAD      C8CHAR
0239 3020      DCA      20        /SAVE NEW CONTENTS
0240 2346      ISZ      TMPCNT
0241 5303      JMP      GETCH1
0242 5342      JMP      ENDIT
0243 4760*     JMS      XC8ECHO
0244 4315      JMS      TSTCHA
0245 1020      TAD      20
0246 7106      RTL CLL
0247 7004      RAL          /GET NEXT CHAR
0248 1774*     TAD      C8CHAR
0249 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0250 2346      ISZ      TMPCNT
0251 5303      JMP      GETCH1
0252 5342      JMP      ENDIT
0253 4760*     JMS      XC8ECHO
0254 4315      JMS      TSTCHA
0255 1020      TAD      20
0256 7106      RTL CLL
0257 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0258 1774*     TAD      C8CHAR
0259 3020      DCA      20        /SAVE NEW CONTENTS
0260 2346      ISZ      TMPCNT
0261 5303      JMP      GETCH1
0262 5342      JMP      ENDIT
0263 4760*     JMS      XC8ECHO
0264 4315      JMS      TSTCHA
0265 1020      TAD      20
0266 7106      RTL CLL
0267 7004      RAL          /GET NEXT CHAR
0268 1774*     TAD      C8CHAR
0269 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0270 2346      ISZ      TMPCNT
0271 5303      JMP      GETCH1
0272 5342      JMP      ENDIT
0273 4760*     JMS      XC8ECHO
0274 4315      JMS      TSTCHA
0275 1020      TAD      20
0276 7106      RTL CLL
0277 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0278 1774*     TAD      C8CHAR
0279 3020      DCA      20        /SAVE NEW CONTENTS
0280 2346      ISZ      TMPCNT
0281 5303      JMP      GETCH1
0282 5342      JMP      ENDIT
0283 4760*     JMS      XC8ECHO
0284 4315      JMS      TSTCHA
0285 1020      TAD      20
0286 7106      RTL CLL
0287 7004      RAL          /GET NEXT CHAR
0288 1774*     TAD      C8CHAR
0289 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0290 2346      ISZ      TMPCNT
0291 5303      JMP      GETCH1
0292 5342      JMP      ENDIT
0293 4760*     JMS      XC8ECHO
0294 4315      JMS      TSTCHA
0295 1020      TAD      20
0296 7106      RTL CLL
0297 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0298 1774*     TAD      C8CHAR
0299 3020      DCA      20        /SAVE NEW CONTENTS
0300 2346      ISZ      TMPCNT
0301 5303      JMP      GETCH1
0302 5342      JMP      ENDIT
0303 4760*     JMS      XC8ECHO
0304 4315      JMS      TSTCHA
0305 1020      TAD      20
0306 7106      RTL CLL
0307 7004      RAL          /GET NEXT CHAR
0308 1774*     TAD      C8CHAR
0309 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0310 2346      ISZ      TMPCNT
0311 5303      JMP      GETCH1
0312 5342      JMP      ENDIT
0313 4760*     JMS      XC8ECHO
0314 4315      JMS      TSTCHA
0315 1020      TAD      20
0316 7106      RTL CLL
0317 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0318 1774*     TAD      C8CHAR
0319 3020      DCA      20        /SAVE NEW CONTENTS
0320 2346      ISZ      TMPCNT
0321 5303      JMP      GETCH1
0322 5342      JMP      ENDIT
0323 4760*     JMS      XC8ECHO
0324 4315      JMS      TSTCHA
0325 1020      TAD      20
0326 7106      RTL CLL
0327 7004      RAL          /GET NEXT CHAR
0328 1774*     TAD      C8CHAR
0329 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0330 2346      ISZ      TMPCNT
0331 5303      JMP      GETCH1
0332 5342      JMP      ENDIT
0333 4760*     JMS      XC8ECHO
0334 4315      JMS      TSTCHA
0335 1020      TAD      20
0336 7106      RTL CLL
0337 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0338 1774*     TAD      C8CHAR
0339 3020      DCA      20        /SAVE NEW CONTENTS
0340 2346      ISZ      TMPCNT
0341 5303      JMP      GETCH1
0342 5342      JMP      ENDIT
0343 4760*     JMS      XC8ECHO
0344 4315      JMS      TSTCHA
0345 1020      TAD      20
0346 7106      RTL CLL
0347 7004      RAL          /GET NEXT CHAR
0348 1774*     TAD      C8CHAR
0349 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0350 2346      ISZ      TMPCNT
0351 5303      JMP      GETCH1
0352 5342      JMP      ENDIT
0353 4760*     JMS      XC8ECHO
0354 4315      JMS      TSTCHA
0355 1020      TAD      20
0356 7106      RTL CLL
0357 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0358 1774*     TAD      C8CHAR
0359 3020      DCA      20        /SAVE NEW CONTENTS
0360 2346      ISZ      TMPCNT
0361 5303      JMP      GETCH1
0362 5342      JMP      ENDIT
0363 4760*     JMS      XC8ECHO
0364 4315      JMS      TSTCHA
0365 1020      TAD      20
0366 7106      RTL CLL
0367 7004      RAL          /GET NEXT CHAR
0368 1774*     TAD      C8CHAR
0369 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0370 2346      ISZ      TMPCNT
0371 5303      JMP      GETCH1
0372 5342      JMP      ENDIT
0373 4760*     JMS      XC8ECHO
0374 4315      JMS      TSTCHA
0375 1020      TAD      20
0376 7106      RTL CLL
0377 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0378 1774*     TAD      C8CHAR
0379 3020      DCA      20        /SAVE NEW CONTENTS
0380 2346      ISZ      TMPCNT
0381 5303      JMP      GETCH1
0382 5342      JMP      ENDIT
0383 4760*     JMS      XC8ECHO
0384 4315      JMS      TSTCHA
0385 1020      TAD      20
0386 7106      RTL CLL
0387 7004      RAL          /GET NEXT CHAR
0388 1774*     TAD      C8CHAR
0389 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0390 2346      ISZ      TMPCNT
0391 5303      JMP      GETCH1
0392 5342      JMP      ENDIT
0393 4760*     JMS      XC8ECHO
0394 4315      JMS      TSTCHA
0395 1020      TAD      20
0396 7106      RTL CLL
0397 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0398 1774*     TAD      C8CHAR
0399 3020      DCA      20        /SAVE NEW CONTENTS
0400 2346      ISZ      TMPCNT
0401 5303      JMP      GETCH1
0402 5342      JMP      ENDIT
0403 4760*     JMS      XC8ECHO
0404 4315      JMS      TSTCHA
0405 1020      TAD      20
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0408 1774*     TAD      C8CHAR
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0416 7106      RTL CLL
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0418 1774*     TAD      C8CHAR
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0422 5342      JMP      ENDIT
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0424 4315      JMS      TSTCHA
0425 1020      TAD      20
0426 7106      RTL CLL
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0428 1774*     TAD      C8CHAR
0429 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0430 2346      ISZ      TMPCNT
0431 5303      JMP      GETCH1
0432 5342      JMP      ENDIT
0433 4760*     JMS      XC8ECHO
0434 4315      JMS      TSTCHA
0435 1020      TAD      20
0436 7106      RTL CLL
0437 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0438 1774*     TAD      C8CHAR
0439 3020      DCA      20        /SAVE NEW CONTENTS
0440 2346      ISZ      TMPCNT
0441 5303      JMP      GETCH1
0442 5342      JMP      ENDIT
0443 4760*     JMS      XC8ECHO
0444 4315      JMS      TSTCHA
0445 1020      TAD      20
0446 7106      RTL CLL
0447 7004      RAL          /GET NEXT CHAR
0448 1774*     TAD      C8CHAR
0449 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0450 2346      ISZ      TMPCNT
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0452 5342      JMP      ENDIT
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0454 4315      JMS      TSTCHA
0455 1020      TAD      20
0456 7106      RTL CLL
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0459 3020      DCA      20        /SAVE NEW CONTENTS
0460 2346      ISZ      TMPCNT
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0462 5342      JMP      ENDIT
0463 4760*     JMS      XC8ECHO
0464 4315      JMS      TSTCHA
0465 1020      TAD      20
0466 7106      RTL CLL
0467 7004      RAL          /GET NEXT CHAR
0468 1774*     TAD      C8CHAR
0469 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0470 2346      ISZ      TMPCNT
0471 5303      JMP      GETCH1
0472 5342      JMP      ENDIT
0473 4760*     JMS      XC8ECHO
0474 4315      JMS      TSTCHA
0475 1020      TAD      20
0476 7106      RTL CLL
0477 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0478 1774*     TAD      C8CHAR
0479 3020      DCA      20        /SAVE NEW CONTENTS
0480 2346      ISZ      TMPCNT
0481 5303      JMP      GETCH1
0482 5342      JMP      ENDIT
0483 4760*     JMS      XC8ECHO
0484 4315      JMS      TSTCHA
0485 1020      TAD      20
0486 7106      RTL CLL
0487 7004      RAL          /GET NEXT CHAR
0488 1774*     TAD      C8CHAR
0489 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0490 2346      ISZ      TMPCNT
0491 5303      JMP      GETCH1
0492 5342      JMP      ENDIT
0493 4760*     JMS      XC8ECHO
0494 4315      JMS      TSTCHA
0495 1020      TAD      20
0496 7106      RTL CLL
0497 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0498 1774*     TAD      C8CHAR
0499 3020      DCA      20        /SAVE NEW CONTENTS
0500 2346      ISZ      TMPCNT
0501 5303      JMP      GETCH1
0502 5342      JMP      ENDIT
0503 4760*     JMS      XC8ECHO
0504 4315      JMS      TSTCHA
0505 1020      TAD      20
0506 7106      RTL CLL
0507 7004      RAL          /GET NEXT CHAR
0508 1774*     TAD      C8CHAR
0509 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0510 2346      ISZ      TMPCNT
0511 5303      JMP      GETCH1
0512 5342      JMP      ENDIT
0513 4760*     JMS      XC8ECHO
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0542 5342      JMP      ENDIT
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0544 4315      JMS      TSTCHA
0545 1020      TAD      20
0546 7106      RTL CLL
0547 7004      RAL          /GET NEXT CHAR
0548 1774*     TAD      C8CHAR
0549 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0550 2346      ISZ      TMPCNT
0551 5303      JMP      GETCH1
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0553 4760*     JMS      XC8ECHO
0554 4315      JMS      TSTCHA
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0557 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
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0560 2346      ISZ      TMPCNT
0561 5303      JMP      GETCH1
0562 5342      JMP      ENDIT
0563 4760*     JMS      XC8ECHO
0564 4315      JMS      TSTCHA
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0566 7106      RTL CLL
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0568 1774*     TAD      C8CHAR
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0583 4760*     JMS      XC8ECHO
0584 4315      JMS      TSTCHA
0585 1020      TAD      20
0586 7106      RTL CLL
0587 7004      RAL          /GET NEXT CHAR
0588 1774*     TAD      C8CHAR
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0594 4315      JMS      TSTCHA
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0600 2346      ISZ      TMPCNT
0601 5303      JMP      GETCH1
0602 5342      JMP      ENDIT
0603 4760*     JMS      XC8ECHO
0604 4315      JMS      TSTCHA
0605 1020      TAD      20
0606 7106      RTL CLL
0607 7004      RAL          /GET NEXT CHAR
0608 1774*     TAD      C8CHAR
0609 3020      DCA      20        /CHECK IF CR+ GOOD CHAR
0610 2346      ISZ      TMPCNT
0611 5303      JMP      GETCH1
0612 5342      JMP      ENDIT
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0614 4315      JMS      TSTCHA
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0617 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
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0619 3020      DCA      20        /SAVE NEW CONTENTS
0620 2346      ISZ      TMPCNT
0621 5303      JMP      GETCH1
0622 5342      JMP      ENDIT
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0624 4315      JMS      TSTCHA
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0626 7106      RTL CLL
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0634 4315      JMS      TSTCHA
0635 1020      TAD      20
0636 7106      RTL CLL
0637 7004      RAL          /GET CHAR+ ADD IT TO PREVIOUS CONTENTS
0638 1774*     TAD      C8CHAR
0639 3020      DCA      20        /SAVE NEW CONTENTS
0640 2346      ISZ      TMPCNT
0641 5303      JMP      GETCH1
0642 5342      JMP      ENDIT
0643 4760*     JMS      XC8ECHO
0644 4315      JMS      TSTCHA
0645 1020      TAD      20
0
```

```

0715 0000 TSTCHA, 0
0716 7041 CIA
0717 1356 TAD (215
0720 7650 SNA CLA
0721 5342 JMP ENDIT
0722 1774 TAD C8CHAR
0723 1355 TAD (-260
0724 7710 SPA CLA
0725 5336 JMP ERR1
0726 1774 TAD C8CHAR
0727 1354 TAD (-270
0730 7700 SNA CLA
0731 5336 JMP ERR1
0732 1774 TAD C8CHAR
0733 0353 AND (7
0734 3774 DCA C8CHAR

0735 5715 JMP I TSTCHA
0736 1352 ERR1, TAD (277
0737 4775 JMS XC8TYPE
0740 4773 JMS XC8CRLF
0741 5266 JMP C8RDP8
0742 4773 ENDIT, JMS XC8CRLF
0743 3345 DCA C8HST
0744 5656 JMP I XC8PSW
0745 0000 C8HST, 0

0746 0000 TMPCNT, 0
0747 2322 MESA, TEXT "SR= "
0750 7540
0751 0000

```

```

0752 0277
0753 0007
0754 7510
0755 7520
0756 0215
0757 7775
0760 1063
0761 1076
0762 0040
0763 1000
0764 0515
0765 0272
0766 0303
0767 1200
0770 1345
0771 1347
0772 1346
0773 1023
0774 1075
0775 1077
0776 0336
0777 0400

```

```

1000 PAGE
      /C8OCTA
      /OCTAL TO ASCII CONVERSION
      /THIS ROUTINE WILL TAKE THE OCTAL NUMBER IN THE AC AND CONVERT IT TO ASCII
      /THE RESULT WILL BE PRINTED ON THE CONSOLE TERMINAL
      / C8OCTA= JMS XC8OCT
      /
      /EX. JMS XC8OCTA /AC CONTAINS NUMBER TO BE CHANGE
      / RETURN IS TO CALL PLUS ONE AC=0
      /
      /CALLS USED ARE=XC8TYPE=

1000 0000 XC8OCT, 0
1001 7106 CLL RTL
1002 7006 RTL
1003 3221 DCA C8TMP1
1004 1377 TAD (-4
1005 3222 DCA C8CKP
1006 1221 XC8O4, TAD C8TMP1
1007 0376 AND (0007
1010 1375 TAD (260
1011 4277 JMS XC8TYPE
1012 1221 TAD C8TMP1
1013 7006 RTL
1014 7004 RAL
1015 3221 DCA C8TMP1
1016 2222 ISZ C8CKP
1017 5206 JMP C8O4
1020 5600 JMP I XC8OCT
1021 0000 C8TMP1, 0
1022 0000 C8CKP, 0

      /*****
      /C8CRLF
      /C8TYPE CR AND LF WITH FILLERS FOLLOWING EACH LF AND CR
      /
      / C8CRLF= JMS XC8CRL
      /
      /EX. JMS XC8CRLF /C8PHNT A CR AND LF WITH FILL
      / /RETURN TO CALL PLUS ONE AC =0
      /CALLS USED ARE=XC8TYPE=

1023 0000 XC8CRLF, 0
1024 7300 CLA CLL
1025 1374 TAD (215
1026 4277 JMS XC8TYPE
1027 1237 TAD FILLER
1030 7040 CMA
1031 3240 DCA FILCNT

```

```

1032 1373      TAD      (212      /GET CODE FOR LF
1033 4277      JMS      XC8TYPE
1034 2240      ISZ      FILCNT      /CHECK ON FILLER CHAR
1035 5233      JMP      C8D02      /TYPE A NON PRINTING CHAR
1036 5623      JMP I      XC8CRL      /EXIT
1037 8004      FILLER, 8004      /FILLER SET FOR 4 CHAR
1040 8000      FILCNT, 0      /COUNTER FOR FILL

```

```

//*****
/C8CKPA
/THIS ROUTINE WILL CHECK IF A CHARACTER WAS ENTERED FROM THE
/TERMINAL. IF THE FLAG IS SET AND THE CONSOLE PACKAGE IS
/ACTIVE A CHECK IS MADE TO DETERMINE IF IT IS A CONTROL CHAR.
/IF IT WAS A CONTROL CHAR THEN ITS CONTROL FUNCTION IS PERFORMED.
/IF NOT A CONTROL CHARACTER OR A CONTROL E-D-L-O- IT WILL DO
/THE CONTROL FUNCTION AND RETURN TO CALL PLUS 2.
/A NON CONTROL CHARACTER WILL BE PRINTED AND A "7" IT WILL RETURN TO
/CALL PLUS 2.
/IF NO FLAG IS SET OR THE CONSOLE IS NOT ACTIVE THE RETURN IS TO
/CALL PLUS 1.

```

```

/      C8CKPA= JMS      XC8CKP

```

```

/EX.      JMS      XC8CKPA      /CALL TO CHECK IF CONTROL CHAR SET
/      ANYTHING(SKIP)      /RETURN IF NOT FLAG OR NOT CONSOLE ACTIVE
/      ANYTHING(JMP EXIT SKIP CHAIN)      /RETURN IF NOT CONTROL OR CONTINUE CONTROL

```

```

/CALLS USED ARE-XC8TTYI-XC8CNTR-C8GET-

```

```

1041 8000      XC8CKP, 0
1042 3772      DCA      AC8AVE      /SAVE THE AC
1043 6004      GTF      /SAVE THE FLAG
1044 3771      DCA      FL8AVE      /SAVE THE FLAG
1045 7501      MOA      /PUT MO IN AC
1046 3770      DCA      MO8AVE      /SAVE THE MO
1047 6031      KSF      /CHECK THE KEYBOARD FLAG
1050 5201      JMP      C8BY3      /EXIT TO CALL PLUS 1
1051 4767      JMS      CHKCLA      /CHECK LOC 28 BIT 3 CONSOLE BIT
1052 7410      SKP      /ACTIVE CONSOLE PACKAGE
1053 5201      JMP      C8BY3      /EXIT TO CALL PLUS 1
1054 4766      JMS      XC8TTYI      /GET THE CHAR
1055 4765      JMS      C8GET      /GET THE FLAG
1056 4764      JMS      XC8CNTR      /CHECK IF CONTROL CHAR.
1057 7000      NOP      /RETURN IF A CONTINUE CHAR.
1060 2241      ISZ      XC8CKP      /BUMP RETURN FOR CALL PLUS 2
1061 4765      C8BY3, JMS      C8GET      /GET REGISTERS
1062 5641      JMP I      XC8CKP      /END GOOD BY

```

```

//*****

```

```

/C8ECHO
/THIS ROUTINE WILL LOOK FOR A CHAR FROM THE KEYBOARD. STORE IT IN LOCATION CHAR
/CHECK IF IT WAS A CONTROL CHARACTER- SET INMODE- PRINT CHARACTER

```

```

/      C8ECHO =      JMS XC8ECHO
/EX.      JMS      XC8ECHO      /LOOK FOR CONSOLE CHAR C8PRINT IT
/      /RETURN CALL PLUS ONE AC = CHAR C8TYPED IN

```

```

/CALLS USED ARE-XC8TTYI-XC8CNTR-C8GET-XC8ECHO-XC8TYPE

```

```

/
XC8ECHO, 0
1063 8000      JMS      XC8TTYI      /WAIT FOR CHAR FROM KEYBOARD
1064 4766      JMS      C8GET      /RESTORE THE REGISTERS
1065 4765      ISZ      INMODE      /SET INMODE IDENTIFYING THIS AS A EXPECTED CHAR
1066 2276      JMS      XC8CNTR      /GO CHECK IF IT IS A CONTROL CHAR
1067 4764      JMP I      XC8ECHO      /WAS A CONTROL CHAR- CONTINUE RUNNING
1070 5663      JMS      XC8TYPE      /NOT A CONTROL CHAR C8PRINT IT
1071 4277      DCA      INMODE      /CLEAR FLAG THAT CHAR EXPECTED
1072 3276      TAD      C8CHAR      /GET CHAR IN AC
1073 1275      JMP I      XC8ECHO      /EXIT
1074 5663      C8CHAR, 0
1075 8000      INMODE, 0
1076 8000

```

```

//*****

```

```

/C8TYPE
/THIS ROUTINE WILL C8PRINT ON THE CONSOLE OR THE LPT WITH DEVICE CODE 66.
/
/      C8TYPE= JMS XC8TYP

```

```

/EX.      JMS      XC8TYPE      /C8PRINT THE CHAR IN THE AC.
/      /RETURN CALL PLUS ONE AC = 8000
/      /DO NOT CLEAR THE LINK IN THIS ROUTINE NEEDED BY C8OCT

```

```

/CALLS USED ARE-C8HANG-XC8CNTR-XC8PNT-XC8CHLP-XC8INGU-

```

```

1077 8000      XC8TYP, 0
1080 3320      DCA      PNTBUF      /STORE CHAN
1081 1321      TAD      TTYLPT      /CHECK 0=TTY 7777=LPT
1082 7640      SZA      CLA
1083 5312      JMP      XDULPT      /DO OUT PUT ON LPT
1084 1320      TAD      PNTBUF
1085 6046      TFS
1086 6041      TSF
1087 5306      JMP      =-1
1088 6042      TCF
1089 5316      JMP      C8BY5
1090 1320      XDULPT, TAD      PNTBUF      /GET CHAR
1091 6046      PSTB      PCLF      /C8PRINT IT
1092 4322      JMS      C8HANG      /CHECK KEYBOARD IF HUNG
1093 6042      PCLF      /CLEAR THE FLAG
1094 7600      C8BY5, 7600      /CLEAR THE AC

```

```

1117 5677      JMP I   XC8TYP      /EXIT
1120 0000      PNTBUF, 0
1121 0000      TTYLPT, 0

1122 0000      C0MANG, 0
1123 7200      CLA
1124 1316      TAD
1125 3320      DCA      C0BYS      /GET CONSTANT 7600
1126 6661      DCA      PNTBUF     /PNTBUF IS NOW A COUNTER
1127 7410      PSKF      /SKIP ON PRINTER DONE
1130 5722      SKP      /NOT DONE YET
1131 2345      JMP I   C0MANG     /SAM FLAG DONE
1132 5326      ISZ      C0CONT     /FIRST COUNTER FAST ONE
1133 2320      JMP      ,=4       /CHECK IF FLAG SET YET
1134 5331      ISZ      PNTBUF     /MADE 4096 COUNTS ON FAST COUNTER
1135 1764      JMP      ,=5       /KEEP IT UP FOR 5 SEC
1136 3322      TAD      XC0CNTR    /GET THE RETURN ADDRESS IN CONTROL
1137 3321      DCA      C0MANG     /SAVE IT IN MANG
1140 4763      DCA      TTYLPT     /ALLOW PRINTING ON TTY
1141 1146      JMS      XC0PNT
1142 4223      MESHANG
1143 4762      JMS      XC0CRLF    /LPT ERROR
1144 5722      JMS      XC0INQU    /PRINT WAITING
1145 0000      JMP I   C0MANG     /CONTINUE TO SAVE ADDRESS
1146 1420      C0CONT, 0          /COUNTER FOR TIMER
1147 2400      MESHANG,TEXT      "LPT ERROR"

1150 0522
1151 2217
1152 2200

1162 0635
1163 0303
1164 0400
1165 0624
1166 0272
1167 1200
1170 1346
1171 1347
1172 1345
1173 0212
1174 0215
1175 0200
1176 0007
1177 7774
1200

```

PAGE

```

/*****
/*****
/*****

```

```

/THIS ROUTINE WILL CHECK LOCATION 22 THE HARD WARE CONFIG WORD,
/TO SEE IF THE CONSOLE BIT 3 (400) IS SET IF SET THEN RETURN
/TO CALL PLUS TWO FOR A ACTIVE CONSOLE PACKAGE AC=0
/IF NOT SET THEN TO CALL PLUS ONE FOR A DEACTIVE CONSOLE PACKAGE.

```

```

1200 0000      CHKCLA, 0
1201 7200      CLA
1202 1022      TAD      22
1203 0377      AND      (400)      /GET THE CONTENTA OF LOCATION 22
1204 7650      SNA CLA      /MASK FOR BIT 3 (400)
1205 2200      ISZ      CHKCLA      /
1206 5600      JMP I   CHKCLA      /ACTIVE CONSOLE PACKAGE RETURN
                                   /CALL PLUS ONE (1) FOR ACTIVE
                                   /DEACTIVE CONSOLE PACKAGE RETURN
                                   /CALL PLUS TWO (2)

```

/C0ERR

```

/THIS ROUTINE WILL DETERMINE WHAT TO DO WHEN A C0ERR IS ENCOUNTERED
/Will check if classic system, will check C0SWIT registers.
/
/   C0ERR= JMS XC0ERR
/EX.      JMS      XC0ERR      /GO TO C0ERR CALL IF NOT CONSOLE
                                   /RETURN IS CALL PLUS ONE AC =0000
/

```

```

/CALLS USED ARE-CHKCLA-XC0CRLF-XC0SWM-XC0INQU-XC0PNT-XC0OCTA-

```

```

1207 0000      XC0ERR, 0
1210 6002      IOF
1211 3345      DCA      ACSAVE      /SAVE AC
1212 6004      GTF
1213 3347      DCA      PLSAVE      /SAVE THE FLAGS
1214 7501      MGA
1215 3346      DCA      MQSAVE      /SAVE THE MQ
1216 7340      CLA CLL CMA
1217 1207      TAD      XC0ERR      /SUBTRACT A 1 FOR TRUE LOCATION
1220 3344      DCA      PCSAVE      /GET RETURN LOCATION
1221 6201      CDF
1222 7340      CLA CLL CMA
1223 1776      TAD I   (CLASIK)    /GET REAL PC.
1224 3316      DCA      REALPC      /SAVE IT.
1225 6211      CDF      10
1226 4200      JMS      CHKCLA      /CHECK LOC.22 BIT 3 CONSOLE BIT
1227 7410      SKP      /ACTIVE CONSOLE PACKAGE
1230 5270      JMP      NTCLAS     /NOT CLASSIC SYSTEM
1231 4775      JMS      C0GET      /GET REGISTERS.
1232 4774      JMS      XC0SWM     /CHECK SWITCH REG FOR BIT THAT INDICATES
                                   /NO ERROR MESSAGE
1233 0373      SETUP1, AND      10000 /MASK FOR BIT FOR NO ERROR PRINTING
                                   /IF THIS ERROR MESSAGE IS TO ALWAYS
                                   /BE PRINTED LEAVE AND VALUE AT 0000
                                   /SKIP IF BIT IS 0 PRINT ERROR MESSAGE
                                   /DO NOT PRINT

1234 7640      SZA CLA
1235 5262      JMP      C0U010
1236 4772      JMS      XC0CRLF
1237 4771      JMS      XC0PNT
1240 1320      ERRMES
1241 4771      JMS      XC0PNT      /PRINT THE ERROR MESSAGE
1242 1330      MESPC
1243 1316      TAD      REALPC
1244 4770      JMS      XC0OCTA    /PRINT THE PC STATEMENT
1245 4771      JMS      XC0PNT     /GET PC
                                   /CONVERT 4 DIGIT PC TO ASCII
1246 1333      MESAC              /PRINT THE AC MESS

```

```

1247 1345      TAD      AC8AVE
1250 4770'     JMS      XC8OCTA
1251 4771'     JMS      XC8PNT
1252 1336      MESMQ
1253 1346      TAD      MQ8AVE
1254 4770'     JMS      XC8OCTA
1255 4771'     JMS      XC8PNT
1256 1341      MESFL
1257 1347      TAD      FL8AVE
1260 4770'     JMS      XC8OCTA
1261 4772'     JMS      XC8CRLF
1262 4775'     C80010, JMS      C8GET
1263 4774'     JMS      XC8SW
1264 7610      SKP CL A
1265 5300      JMP      C8BY2
1266 4767'     JMS      XC8INQ
1267 5300      JMP      C8BY2
1270 4775'     NTCLAS, JMS      C8GET
1271 4774'     JMS      XC8SW

1272 7610      SKP CL A
1273 5607      JMP I      XC8ERR
1274 1366      TAD      (7402
1275 3744      DCA I      PC8AVE
1276 4775'     JMS      C8GET
1277 5744      JMP I      PC8AVE
1300 4775'     C8BY2, JMS      C8GET
1301 5607      JMP I      XC8ERR

/
/
1302 7402      ROUINS, HLT
1303 7000      NOP
1304 3317      DCA      MYAC
1305 0201      CDF      B
1306 1020      TAD      SWR
1307 3765      DCA I      (SWR)
1310 1776      TAD I      (CLASIK)
1311 3315      DCA      CLRTRN
1312 1317      TAD      MYAC
1313 0202      CIF      B
1314 5715      JMP I      CLRTRN

1315 0000      CLRTRN, 0
1316 0000      REALPC, 0
1317 0000      MYAC, 0

/
1320 0410      ERKMS, TEXT "DHRKC=H FAILED "
1321 2213
1322 0310
1323 4040
1324 0601
1325 1114
1326 0504
1327 4000
1330 4040      MESPC, TEXT " PC:"

```

```

1331 2003
1332 7200
1333 4040      MESAC, TEXT " AC:"
1334 0103
1335 7200
1336 4040      MESMQ, TEXT " MQ:"
1337 1521
1340 7200
1341 4040      MESFL, TEXT " FL:"
1342 0614
1343 7200
1344 7777      PC8AVE, 7777
1345 7777      AC8AVE, 7777
1346 7777      MQ8AVE, 7777
1347 7777      FL8AVE, 7777
/
1365 0020
1366 7402
1367 0635
1370 1000
1371 0303
1372 1023
1373 0000
1374 0262
1375 0624
1376 1514
1377 0400
0000

```

FIELD 0

0000	00000000	00000000	11101111	11111111	11000000	00000000	00000000	00000000
0100	00000000	00000000	00000000	00000000	00000000	00000000	00000000	00000000
0200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0500	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
0700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1100	11111111	11111111	11111111	11111111	11111111	11100000	00111111	11111111
1200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111
1300	11111111	11111111	11111111	11111111	11111111	00000000	00000111	11111111
1400								
1500								
1600								
1700								
2000								
2100								
2200								
2300								
2400								
2500								
2600								
2700								
3000								
3100								
3200								
3300								
3400								
3500								
3600								
3700								

4000
4100
4200
4300
4400
4500
4600
4700
5000
5100
5200
5300
5400
5500
5600
5700
6000
6100
6200
6300
6400
6500
6600
6700
7000
7100
7200
7300
7400
7500
7600
7700

```
/
/NOTE: LOCATION 0 CONTAINS THE REVISION
/LEVEL (IN ASCII) ON PROGRAM LOAD.
/
/ALL KNOWN HALTS
/
1400 0556 ERHLT0 /SKIP TRAP DL3C
1401 0563 ERHLT2 /SKIP TRAP DCLR
1402 2561 ERHLT3 /SKIP TRAP DLAG
1403 2544 ERHLT5 /SKIP TRAP DRST
1404 0547 ERHLT6 /SKIP TRAP DLDC
1405 3130 INTER1 /NO DISK INTERRUPT
1406 2362 INTER2 /UNDEFINED INTERRUPT
1407 0206 FLDHLT /PROGRAM WILL ONLY RUN IN FIELD 0
1410 2702 NODSKS /NO DISKS AVAILABLE TO RUN
1411 0603 STPHLT /PROGRAM STOP FROM SWR4=1
1412 2755 CHNHLT /IOT CHANGE MALT
1413 1707 BADHLT /COMPUTER MUST BE DOWN, CHECKSUM FAILED
/
1414 3136 BIGSTP /BUT WORD-BY-WORD COMPARE WORKED.
/STOP FOR ALL ERROR HALTS.
/
6740 DL3C=6740 /LOAD SECTOR COUNTER
6741 DSKP=6741 /SKIP ON TRANSFER DONE OR ERROR
6742 DCLR=6742 /CLEAR DISK CONTROL LOGIC
6743 DLAG=6743 /LOAD ADDRESS AND GO
6744 DLCA=6744 /LOAD CURRENT ADDRESS
6745 DRST=6745 /READ STATUS REGISTER
6746 DLDC=6746 /LOAD COMMAND REGISTER
/
4406 LAS=JMS I XLAS
4407 CLASIC=JMS I XCLAS
4427 RANDAT=JMS I XRNWRD
4430 DISCON=JMS I XDUMP
4431 SPAC=JMS I XSPAC
4432 ONEIN=JMS I XOCT1
4433 FORIN=JMS I XOCT4
4434 SETGEN=JMS I XSTGEN
/
4435 SETFLD=JMS I XSTFLD
4437 YESNO=JMS I XCHKYN
/
4436 SELCHK=JMS I XCKPOT
4440 RANGEN=JMS I XRNDOO
4442 RESRAN=JMS I XRSRAN
4441 DISKGO=JMS I XDSKGO
4443 RECAL=JMS I XRESTR
4444 RECEIV=JMS I XWAIT
4446 ERROR=JMS I XERRO
4447 RDSTAT=JMS I XRDST
4453 LDADD=JMS I XLDDAD
4450 DSKSKP=JMS I XSKKP
```

```
4451 LDCMD=JMS I XLDCM
4452 LOCUR=JMS I XLDCA
4454 CLRALL=JMS I XCLDR
4455 PRNTER=JMS I XPRN
4456 OCTEL=JMS I XPROCT
4445 TYPE=JMS I XPRINT
4457 CRLF=JMS I XCNLF
4426 GENDAT=JMS I XGN DAT
4424 CHK22=JMS I XCHK22
4425 KTICK=JMS I XKTCK
/
0000 /
/
0000 0310 310 /REVISION "H" INTERRUPT SERVICE RETURN
0001 5001 5001 /DCA SAVAC SAVE AC AT INT.
0002 0002 0002 /RAL SHIFT LINK AT TIME OF INT.
0003 0003 0003 /DCA SYLNK SAVE LINK AT TIME OF INT.
0004 0004 0004 /JMP I 5 RETURN TO INT. SERVICE
0005 0005 0005 /RETURN POINTER
/
0006 1546 XLAS, MYLAS
0007 1514 XCLAS, CLASIK
/
0010 *10
/
0010 0000 AUTO10, 0
/
0011 0000 AUTO11, 0
/
0012 0000 AUTO12, 0
/
0013 0004 K0004, 0004
0014 0070 K0070, 0070
0015 0100 K0100, 0100
0016 0200 K0200, 0200
0017 0400 K0400, 0400
/
0020 *20
/
0020 0000 SWR, 0
0021 4000 OP1, 4000
0022 0000 OP2, 0
/
0023 2156 KAERRO, AERRO
0024 0523 XCHK22, CHK22
0025 1154 XKTCK, XKTICK
0026 1737 XGN DAT, GNDAT
0027 2600 XRNWRD, RNWRD
0030 2637 XDUMP, DUMP
0031 1506 XSPAC, SPAC
0032 2400 XOCT1, OCT1
0033 2430 XOCT4, OCT4
0034 1753 XSTGEN, STGEN
0035 2703 XSTFLD, STFLD
```

```

0036 2060 XCKPOT, CHKPOT
0037 2035 XCHKYN, CHKYN
0040 1715 XRANDOM, RANDOM
0041 2200 XD8KGO, DSKGO
0042 1761 XRSRAN, RSRAN
0043 3052 XRESTR, RESTOR
0044 2000 XWAIT, WAIT
0045 2620 XPRINT, PRINT
0046 1200 XEMRO, ERRO
0047 2541 XROST, ROST
0050 0751 XSKKP, SKKP
0051 0542 XLUCH, LUCH
0052 2530 XLCCA, LCCA
0053 2554 XLDAO, LDAO
0054 0560 XCLDR, CLOR
0055 1450 XPRN, PRN
0056 1426 XFROCT, FROCT
0057 1414 XCNLF, UPONE
0060 0000 AMOUNT, 0
0061 0001 K0001, 0001
0062 0003 K0003, 0003
0063 0006 K0006, 0006
0064 0007 K0007, 0007
0065 0010 K0010, 0010
0066 0017 K0017, 0017
0067 0260 K0260, 0260
0070 0277 K0277, 0277
0071 0770 A0770, 0770
0072 7007 A7007, 7007
0073 4000 K4000, 4000
0074 4100 K4100, 4100
0075 1000 K1000, 1000
0076 1777 K1777, 1777
0077 7700 K7700, 7700
0100 7760 K7760, 7760
0101 7777 K7777, 7777
0102 0077 K0077, 0077
0103 6201 KCDF, CDF
0104 7400 K7400, 7400
/
/
/
0105 7764 M12, -12
/
/
/
0106 7774 M4, -4
0107 7770 M10, -10
0110 7775 K7775, 7775
/
/
/
0111 0000 TRASH1, 0
0112 0000 TRASH2, 0
0113 0000 TRASH3, 0
0114 0000 UPDATE, 0
0115 0000 POLDSK, 0

```

```

0116 0000 OPNTAL, 0
0117 0000 BUFTAL, 0
0120 0000 PCNEG, 0
0121 0000 STNEG, 0
0122 0000 EXNEG, 0
0123 0000 CMNEG, 0
0124 0000 INTDA, 0
0125 0000 DAREG, 0
0126 0000 CANEG, 0
0127 0000 WCNEG, 0
0130 0000 FWNNEG, 0
0131 0000 ASNEG, 0
0132 0000 WAREG, 0
0133 0000 ADREG, 0
0134 0000 DGREG, 0
0135 0000 DBREG, 0
0136 0000 INTCH, 0
0137 0000 STATRY, 0
0140 0000 DATTRY, 0
0141 0000 CMKSAV, 0
0142 0000 FNDUSUM, 0
0143 0000 MAXFLD, 0
0144 7607 MAXTIM, 7607
0145 3240 MAXTRK, 3240
0146 3600 BGNBUF, STRBUF
0147 0000 CONSEC, 0
0150 7777 CLKCNT, -1
/
/
/
0151 0756 DATPOT, DAT1
0152 3522 TIMPOT, 00TM1
0153 3537 STAPOT, D0HRD-3
0154 3512 RUNPOT, D8K0B
/
/
/
0155 0000 CRCENT, 0
0156 0000 CRCFLG, 0
0157 0000 DATFLG, 0
0160 0000 SPFLD, 0
0161 0000 SPTRK1, 0
0162 0000 SPTRK2, 0
0163 0000 SPSEC, 0
0164 0000 SPBLK, 0
0165 0000 ERFLG, 0
0166 0000 SAVAC, 0
0167 0000 SVLNK, 0
0170 0000 FINTIM, 0
0171 0000 TRYCNT, 0
0172 3213 XTEXT, TEXPC
0173 3142 PRNDAT, TYPDAT
0174 0000 SAVCM, 0
0175 0000 CLNBAK, 0
/
/
/
0176 3131 BGHLT, BIGHLT
0200
/
/
/

```

```

/START OF PROGRAM BY OPERATOR;
/AT 0200, TTY INTERIGATION;
/AT 0201, CHANGE IOT DEVICE CODES;
/AT 0202, RESTART AT SEEK ROUTINE;
/
0200 4777' BGN, JMS APT8 /TO REGULAR TEST
0201 5776' JMP CHANG /CHANGE IOT ROUTINE
0202 5775' JMP RUN
0203 3156 DCA CRCFLG /CLEAR CRC FLAG
0204 6224 RIF
0205 7440 SZA /FIELD 0777?
0206 4576 FLUMLT, JMS I BGMHT /WILL ONLY RUN IN FIELD 0777?
0207 1103 TAD KCOF
0210 3211 DCA ,+1
0211 7402 MLT /MAKE DF=IF
/
/SETUP INTERRUPT SERVICE;
/
0212 1362 TAD ACUCA
0213 3001 DCA 1 /SETUP AC DCA
0214 1250 TAD KRUT
0215 3002 DCA 2 /SETUP ROTATE LINK
0216 1361 TAD LNKDCA
0217 3003 DCA 3 /SETUP SAVE LINK
0220 1360 TAD K5405
0221 3004 DCA 4 /SETUP JMP RETURN
0222 1363 TAD BRKRET
0223 3005 DCA 5 /RETURN POINTER
/
/CLEAR DATA INFORMATION TABLE
/AT END OF PROGRAM;
/
0224 1077 STRTEX, TAD K7700
0225 3111 DCA TRASH1 /CLEAR COUNTER
0226 1774' TAD RANJMS
0227 3773' DCA SWDAT /SET INSTRUCTION SWITCH
0230 7340 CLA CLL CMA
0231 1152 TAD TIMPOT
0232 3010 DCA AUTO10 /LOCATION POINTER
0233 3410 DCA I AUTO10 /CLEAR
0234 2111 ISZ TRASH1
0235 5233 JMP ,+2 /MORE TO CLEAR
0236 3157 DCA DATFLG
0237 5775' SKPNOP, JMP RUN
/
/
/PRINT PROGRAM NAME AND
/ASK OPERATOR ABOUT AMOUNT
/OF MEMORY;
/
0240 4457 CRLF
0241 4455 PRINTER
0242 3307 MES1 /PRINT "RK8E/RK8L DATA RELIABILITY"
0243 4455 PRINTER
0244 3346 MES5 /PRINT "AMOUNT OF MEMORY"

```

```

0245 4432 ONEIN
0246 0070 0070 /RECEIVE ONE OCTAL
0247 5243 JMP ,+4 /LIMITS
0250 7004 KRUT, RAL ,+4 /INPUT ERROR
0251 7006 RTL
0252 7040 CMA
0253 3143 DCA MAXFLD /COMPLEMENT
0254 4772' JMS CLAFLD /MAXIMUM FIELD POINTER
0255 3111 ALLAGN, DCA TRASH1 /CHECK FOR CLASSIC.
0256 1107 TAD M10
0257 3112 DCA TRASH2
0260 3060 DCA AMOUNT /A FEW POINTERS
/
/ASK OPERATOR ABOUT DISK(S) TO TEST;
/
0261 1111 NEXT, TAD TRASH1
0262 1154 TAD RUNPOT
0263 3113 DCA TRASH3
0264 4455 PRINTER /SAVE RUN POINTER
0265 3325 MES2 /PRINT "EXERCISE"
0266 7340 CLA CLL CMA
0267 4455 PRINTER
0270 3332 MES3 /PRINT "DISK"
0271 1067 TAD K0260
0272 1111 TAD TRASH1
0273 4445 TYPE /ADD IN DISK NUMBER
0274 1070 TAD K0277 /TYPE DISK NUMBER
0275 4445 TYPE ?
0276 4444 RECEIV /RECEIVE KEY INPUT
0277 4437 YESNO /WAS IT YES OR NO
0300 5255 JMP ALLAGN /NEITHER
0301 5304 JMP ,+3 /WAS A NO
0302 2060 ISZ AMOUNT /AMOUNT OF DISK FOUND
0303 7340 CLA CLL CMA /AC TO 7777 FOR EXISTING DISK
0304 3513 DCA I TRASH3 /SETUP RUN POINTER
0305 2111 ISZ TRASH1
0306 2112 ISZ TRASH2
0307 5261 JMP NEXT /ASK ABOUT NEXT DISK
/
/
/ASK IF ACCEPT MODE;
/
0310 1060 TAD AMOUNT /GET AMOUNT FOUND
0311 7650 SNA CLA /WERE ANY FOUND
0312 5224 JMP STRTEX /OPERATOR ERROR NO DISK INPUT
0313 4455 PRINTER /PRINT "ACCEPT MODE?"
0314 3363 MES6
0315 4444 RECEIV /RECEIVE INPUT
0316 4437 YESNO /YES OR NOT???
0317 5313 JMP ,+4 /NEITHER ALL AGAIN
0320 7610 SKP CLA /MANUAL TEST
0321 5771' JMP ASKSUR /ASK "ARE YOU SURE"
/
/
/IF ACCEPT MODE, INTERAGATE

```

```

/ABOUT CONSTANT FIELD1
/
0322 4455 MANUAL, PRINTER /PRINT "FIELD?"
0323 3404 MESSAGE
0324 4444 RECEIV /RECEIVE Y OR N
0325 4437 YESNO /CHECK FOR Y OR N
0326 5322 JMP MANUAL /NEITHER Y OR N
0327 5345 JMP ASKNX1 /WAS A N, ASK ABOUT NEXT
0330 4431 SPACE /SPACE OUT ONE
0331 4432 ONEIN /GET 1 OCTAL
0332 0070 0070 /LIMITS
0333 5322 JMP MANUAL /INPUT ERROR ASK AGAIN
0334 7104 CLL RAL
0335 7006 RTL
0336 3160 DCA SPFLD /SAVE INPUT
0337 1160 TAD SPFLD
0340 1143 TAD MAXFLD /COMPARE TO MAXIMUM
0341 7700 SMA CLA /U.K.?
0342 5322 JMP MANUAL /INPUT ERROR
0343 7340 CLA CLL CMA /SETUP FIELD FLAG
0344 3770 DCA FLOPLG

/
/INTERIGATE ABOUT CONSTANT TRACK1
/
0345 4455 ASKNX1, PRINTER /PRINT "TRACK?"
0346 3410 MESSAGE
0347 4444 RECEIV /RECEIVE Y OR N
0350 4437 YESNO /CHECK FOR Y OR N
0351 5345 JMP ASKNX1 /ERROR, ASK AGAIN
0352 5767 JMP ASKNX2 /N, ASK ABOUT NEXT
0353 4431 SPACE
0354 4432 ONEIN /RECEIVE 1 IN OCTAL
0355 0010 0010 /LIMITS
0356 5345 JMP ASKNX1 /ERROR, ASK AGAIN
0357 5766 JMP SAVE1 /TU SAVE SOME ROOM.

/
0360 5405 K5405, 5405
0361 3167 LNKDCA, DCA SVLNK
0362 3166 ACUCA, DCA SAVAC
0363 2304 BRKRET, RETURN

/
0366 0400
0367 0406
0370 3572
0371 0513
0372 1404
0373 2601
0374 0522
0375 0600
0376 2730
0377 2070
0400 PAGE
/
/

```

```

/INTERIGATE ABOUT CONSTANT
/BLOCK LENGTH1
/
0400 3161 SAVE1, DCA SPTRK1 /SAVE EXTENDED TRACK BIT
0401 4433 FORIN /GET FOUR IN OCTAL.
0402 5777 JMP ASKNX1 /ERROR, ASK AGAIN
0403 3162 DCA SPTRK2 /SAVE CYL., SURFACE, AND SECTOR
0404 7340 CLA CLL CMA
0405 3776 DCA TRKFLG /SETUP TRACK FLAG

/
0406 4455 ASKNX2, PRINTER /PRINT "BLOCK LENGTH?"
0407 3424 MESSAGE
0410 4444 RECEIV /RECEIVE INPUT
0411 4437 YESNO /CHECK FOR Y OR N
0412 5206 JMP ASKNX2 /ERROR, ASK AGAIN
0413 5225 JMP ASKNX3 /N, ASK ABOUT NEXT
0414 4431 SPACE /Y, SPACE OUT 1
0415 4432 ONEIN /RECEIVE 1 IN OCTAL
0416 0010 0010 /LIMITS
0417 5206 JMP ASKNX2 /ERROR, ASK AGAIN
0420 7640 SZA CLA /SET HALF BLOCK?
0421 7340 CLA CLL CMA /YES
0422 3164 DCA SPBLK /SETUP BLOCK NUMBER
0423 7340 CLA CLL CMA
0424 3775 DCA MLFFLG /SETUP BLOCK FLAG

/
/INTERIGATE ABOUT CONSTANT
/SECTORS1
/
0425 4455 ASKNX3, PRINTER /PRINT "EXTRA SECTORS?"
0426 3414 MESSAGE
0427 4444 RECEIV /RECEIVE INPUT
0430 4437 YESNO /CHECK FOR Y OR N
0431 5225 JMP ASKNX3 /INPUT ERROR
0432 5264 JMP ASKNX5 /N, ASK ABOUT NEXT
0433 4431 SPACE /SPACE OUT 1
0434 4432 ONEIN /RECEIVE 1 IN OCTAL
0435 0010 0010 /LIMITS
0436 5225 JMP ASKNX3 /ERROR, ASK AGAIN
0437 7104 CLL RAL
0440 7006 RTL
0441 3163 DCA SPSEC /SAVE IT
0442 4432 ONEIN /RECEIVE 1 IN OCTAL
0443 0070 0070 /LIMITS
0444 5225 JMP ASKNX3 /INPUT ERROR, ASK AGAIN
0445 1163 TAD SPSEC /ADD IN LAST
0446 3163 DCA SPSEC /SAVE ALL
0447 1164 TAD SPBLK
0450 7640 SZA CLA /BLOCK LENGTH 0????
0451 5254 JMP .+5 /NO LIMIT IS 17.
0452 1160 TAD SPFLD
0453 7640 SZA CLA /FIELD 0?????
0454 1065 TAD K0010 /LIMIT IS 17.
0455 1064 TAD K0007

```

```

0456 7140      CLL CMA
0457 1163      TAD      SPSEC
0460 7630      S2L CLA
0461 5225      JMP      ASKNX3
0462 7340      CLA CLL CMA
0463 3774      DCA      SECFLG
/
/
/
/INTERIGATE ABOUT "OPERATOR
/SELECT DATA";
/
0464 4455      ASKNX5, PRINTER
0465 3433      MES13
0466 1322      TAD      RANJMS
0467 3773      DCA      SWDAT
0470 4444      RECEIV
0471 4437      YESNO
0472 5264      JMP      ASKNX5
0473 5313      JMP      ASKSUR
0474 1346      TAD      KSKP
0475 3773      DCA      SWDAT
0476 1105      TAD      M12
0477 3111      DCA      TRASH1
0500 7340      CLA CLL CMA
0501 1151      TAD      DATPOT
0502 3010      DCA      AUTO10
0503 4457      CNLF
0504 4433      FORIN
0505 5264      JMP      ASKNX5
0506 3410      DCA I      AUTO10
0507 2111      ISZ      TRASH1
0510 5303      JMP      .+5
0511 7340      CLA CLL CMA
0512 3157      DCA      DATFLG
/
/ASK IF HE'S SURE;
/
0513 4455      ASKSUR, PRINTER
0514 3436      MES14
0515 4444      RECEIV
0516 4437      YESNO
0517 5313      JMP      ASKSUR
0520 5772      JMP      STRTEX
0521 5771      JMP      RUN
0522 4426      RANJMS, GENDAT
/
/THIS ROUTINE TESTS FOR BEING ON APT,
/IF ON APT RETURN IS PLUS ONE, IF NOT RETURN IS PLUS TWO,
/
0523 0000      CHEK22, 0
0524 1022      TAD      22
0525 7700      SNA CLA
0526 2323      ISZ      CHEK22
0527 5723      JMP I      CHEK22
/ON APT?
/NO, UPDATE RETURN,
/AND RETURN,

```

```

/ROUTINE TO NOTIFY APT.
/
0530 0000      KTIME, 0
0531 4424      CHK22
0532 7410      SKP
0533 5730      JMP I      KTIME
0534 6002      IOF
0535 6201      CDF      00
/ON APT,
/NOT ON APT, GO ABOUT NORMAL RUN,
/TURN INTERRUPT SYSTEM OFF
/DATA FIELD SHOULD ALWAYS
/BE ZERO IN PROGRAM RUN,
/CHANGED TO CURRENT DATA FIELD,
/
0536 6272      CIF      70
0537 4741      JMS I      K6500
0540 5730      JMP I      KTIME
/RETURN,
0541 6500      K6500, 6500
/
/
/SUBROUTINE TO LOAD COMMAND REGISTER
/
0542 0000      LDCM, 0
0543 3123      DCA      CMNEG
0544 1123      TAD      CMNEG
0545 6746      IOT6, 0LDC
0546 7610      KSKP, SKP CLA
0547 4576      ERHLT6, JMS I      BGHLT
0550 1122      TAD      EXREG
0551 7110      CLL RAM
0552 7030      S2L CLA
0553 1016      TAD      K0200
0554 6740      IOT0, 0LSC
0555 7610      SKP CLA
0556 4576      ERHLT0, JMS I      BGHLT
0557 5742      JMP I      LDCM
/LOAD COMMAND REGISTER
/LOAD EXT. DRIVE
/SKIP TRAP IOT0
/EXIT
/
/SUBROUTINE TO ISSUE "DCLR" CLEAR IOT
CLDR, 0
0560 0000      IOT2, DCLR
0561 6742      JMP I      CLDR
0562 5760      ERHLT2, JMS I      BGHLT
0563 4576
/DCLR "CLEAR IOT"
/EXIT
/ERRROR SKIP TRAP
/
0571 0600
0572 0224
0573 2601
0574 3574
0575 3575
0576 3573
0577 0345
0600
PAGE
/
/SETUP ADDRESSING, COMMAND,
/AND DATA PARAMETERS;

```

```

/MAKE FIELD1
/
0600 4406 RUN, LAS
0601 0016 AND K0200
0602 7640 SZA CLA
0603 7402 STPHLT, HLT
0604 1777 TAD FLDFLG
0605 7650 SNA CLA
0606 5211 JMP ,+3
0607 1160 TAD SPFLD
0610 5230 JMP HNPLD
0611 7301 CLA CLL IAC
0612 1143 TAD MAXFLD
0613 7650 SNA CLA
0614 5230 JMP HNPLD
0615 4440 RANGEN
0616 0014 AND K0070
0617 7450 SNA
0620 5230 JMP HNPLD
0621 3136 DCA INTCM
0622 1136 TAD INTCM
0623 1143 TAD MAXFLD
0624 7710 SPA CLA
0625 5231 JMP HNPLD+1
0626 1143 TAD MAXFLD
0627 7040 CMA
0630 3136 RNPLD, DCA INTCM

/MAKE BLOCK LENGTH1
/
0631 1776 TAD MLFFLG
0632 7650 SNA CLA
0633 4440 RANGEN
0634 1164 TAD SPBLK
0635 0015 AND K0100
0636 1136 TAD INTCM
0637 3136 DCA INTCM
0640 1136 TAD INTCM
0641 0015 AND K0100
0642 7640 SZA CLA
0643 1016 TAD K0200
0644 1104 TAD K7400
0645 3112 DCA TRASH2
0646 1112 TAD TRASH2
0647 7041 CIA
0650 3114 DCA UPDATE
0651 1136 TAD INTCM
0652 0350 AND A0170
0653 7640 SZA CLA
0654 1005 TAD K0010
0655 1004 TAD K0007
0656 3111 DCA TRASH1

/MAKE AMOUNT OF SECTORS
/TO TRANSFER1

```

```

/GET THE SWITCHES.
/MASK HALT SW.
/TIME TO HALT?
/HAUT FROM SHR4=1.
/GET FIELD FLAG
/WAS IT SET?
/NO, USE RANDOM FIELD
/YES, GET OPERATOR FIELD
/GO

/GET MAXIMUM FIELD POINTER
/ANY FIELDS THERE
/NO EXTENDED FIELDS TO USE
/YES, GET A RANDOM FIELD
/MASK
/COULD BE 0
/WAS DON'T HAVE TO CHECK LIMITS
/SAVE FIELD FOUND

/ADD IN MAXIMUM FIELD POINTER
/IN LIMITS????
/YES, USE IT
/NO, USE MAXIMUM IN THE MACHINE

```

```

0657 1775 TAD SECFLG
0660 7650 SNA CLA
0661 4440 RANGEN
0662 1163 TAD SPSEC
0663 0111 AND TRASH1
0664 3147 DCA CONSEC
0665 1147 TAD CONSEC
0666 7040 CMA
0667 3111 DCA TRASH1

/MAKE WORD COUNT1
/
0670 1112 TAD TRASH2
0671 2111 ISZ TRASH1
0672 5270 JMP ,+2
0673 3127 DCA WCREG

/MAKE CURRENT ADDRESS1
/
0674 4440 RANGEN
0675 3126 DCA CAREG
0676 1136 TAD INTCM
0677 0014 AND K0070
0700 7640 SZA CLA
0701 5317 JMP FILLUP
0702 1146 TAD 06NBUF
0703 7140 CMA CLL
0704 1126 TAD CAREG
0705 7620 SNL CLA
0706 5315 JMP CONCUR
0707 1127 TAD WCREG
0710 7041 CIA
0711 1126 TAD CAREG
0712 1016 TAD K0200
0713 7630 SZL CLA
0714 5317 JMP FILLUP
0715 1146 CONCUR, TAD 06NBUF
0716 3126 DCA CAREG

/ROUTINE TO FILL AND CHECK SUM BUFFER
/
0717 4425 FILLUP, KTICK
0720 4434 SETGEN
0721 1106 TAD M4
0722 3137 DCA STATRY
0723 4435 REPILL, SETFLO
0724 3325 DCA ,+1
0725 7402 HLT
0726 3141 DCA CHKSAV
0727 4427 NEWRD, RANDAT
0730 3111 DCA TRASH1
0731 1111 TAD TRASH1
0732 3411 DCA I AUTO11
0733 7100 CLL

/GENERATE RANDOM CA
/SAVE IT

/MASK FIELD BITS
/EXTENDED FIELD????
/INITIAL CA O,K,***

/GENERATE DATA
/SAVE OUTPUT WORK
/GET BACK WORK
/STORE IN BUFFER

```

```

0734      1111      TAD      TRASH1      /GET BACK WORD
0735      1141      TAD      CMKSAV      /AND IN LAST
0736      7430      BZL              /LINK SET??
0737      7001      IAC              /ADD IT IN
0740      3141      DCA      CMKSAV      /SAVE FOR NEXT
0741      2117      ISZ      BUPTAL      /UPDATE BUFFER TALLY
0742      5327      JMP      NEWRD      /MORE WORDS TO GO
0743      0201      CDF      0
0744      1165      TAD      ERPLG
0745      7650      SNA      CLA      /ENRDR FLAG SET???
0746      5774      JMP      POLNEX      /POLE DRIVES
0747      5773      JMP      REWRT      /YES, MUST BE A WRITE ERROR

0750      0170      AS170, 0170

/
/SUBROUTINE TO ISSUE "DSKP" DISK SKIP IOT
/
0751      0000      DSKP, 0
0752      6741      IOT1, DSKP      /DISK SKIP IOT
0753      7410      SKP              /DID NOT SKIP
0754      2351      ISZ      DSKP
0755      5751      JMP I DSKP      /EXIT

/
/PLACE FOR DATA IN MANUAL MODE
/
0756      0000      DAT1, 0000
0757      0000      DAT2, 0000
0760      0000      DAT3, 0000
0761      0000      DAT4, 0000
0762      0000      DAT5, 0000
0763      0000      DAT6, 0000
0764      0000      DAT7, 0000
0765      0000      DAT8, 0000
0766      0000      DAT9, 0000
0767      0000      DAT10, 0000
0770      0000      DAT11, 0000
0771      0000      DAT12, 0000

/
0773      1047
0774      1000
0775      3574
0776      3575
0777      3572
1000

PAGE
/
/ROUTINE TO SELECT DRIVE NO.
/SEQUENTIAL SELECTION 0,1,2,3,0,1,ETC.
/
1000      2115      POLNEX, ISZ      POLDISK      /UPDATE POLE POINTER
1001      1115      TAD      POLDISK      /GET POINTER
1002      4436      SELCHK      /CHECK IF DISK ON SYSTEM,
1003      5200      JMP      POLNEX      /NO, TRY NEXT DRIVE
1004      1115      TAD      POLOSK
1005      7112      CLL      RTM
1006      0061      AND      K0001

```

```

1007 3122 DCA EXREG /SET EXT, DRIVE BIT
/
/DRIVE COMPLETED, START
/WRITE SEQUENCE!
/SELECT DISK ADDRESS.
/
1010 1115 GOTIT, TAD POLDISK /GET DRIVE NO.
1011 0062 AND K0003 /MASK
1012 7104 CLL RAL /MOVE TO 9-10.
1013 1136 TAD INTCH /ADD IN OTHER.
1014 3136 DCA INTCH /SAVE INITIAL COMMAND.
1015 1777* TAD TRKFLG /GET TRACK FLAG
1016 7650 SNA CLA /HAS IT SET???
1017 4440 RANGEN /GET RANDOM DA.
1020 1161 TAD SPTRK1 /GET OPERATOR CONSTANT INPUT.
1021 0061 AND K0001 /MASK EXT, BIT.
1022 1136 TAD INTCH /ADD IN OTHER.
1023 3136 DCA INTCH /SAVE COMPLETE INITIAL COMMAND.
1024 1777* TAD TRKFLG /GET TRACK FLAG
1025 7650 SNA CLA /HAS IT SET???
1026 4440 RANGEN /USE RANDOM
1027 1162 TAD SPTRK2 /GET INPUT
1030 3124 DCA INTDA /SAVE INITIAL DA.
1031 1777* TAD TRKFLG
1032 7640 SZA CLA /INPUT BY OPERATOR?
1033 5247 JMP REWRT /LET HIM FAIL??
1034 1136 TAD INTCH
1035 7010 RAR
1036 7620 SNL CLA /EXT BIT SET?
1037 5247 JMP REWRT /NO. DON'T LIMIT DA.
1040 1145 TAD MAXTRK
1041 1124 TAD INTDA
1042 7630 SZL CLA /BEYOND MAXIMUM LIMIT?
1043 5247 JMP REWRT /NO. DONT LIMIT.
1044 1124 TAD INTDA
1045 7040 CMA
1046 3124 DCA INTDA /YES, SET TO LEGAL LIMIT.
/
/WRITE INFORMATION!
/CLEAR BUFFER ON THE FLY!
/
1047 4441 REWRT, DISKGO /GO WRITE
1050 4400 4400 /WRITE DATA POINTER
1051 5263 JMP REREAD /WRITE O.K.
1052 7340 CLA CLL CMA
1053 3165 DCA ERFLG /SET WRITE ERROR FLAG
1054 4442 RESRAN /RESET GENERATOR
1055 2137 ISZ STATRY /UPDATE WRITE ME=TRY
1056 5776* JMP REFILL /TRY AGAIN
/
/CHECK FOR LOOP ON WRITE!
/
1057 4406 LAS /GET SWITCH 0
1060 7710 TRYTIM, SPA CLA /LOOP ON WRITE???
1061 5775* JMP REFILL-2 /YES, TRY WRITE AGAIN

```

```

1062 5351      JMP      STRREL      /RESTORE ALL UNIVES ON ERROR
1063 1260      REREAD, TAD      TRYTIM
1064 3171      DCA      TRYCNT      /SETUP FOR SOFT ERROR RETNY
1065 3165      DCA      ERFLG      /CLEAR ERROR FLAG
1066 1106      TAD      M4
1067 3137      DCA      STATRY      /SETUP TRY COUNTER
1070 1106      TAD      M4
1071 3140      DCA      DATTRY      /SETUP TRY COUNTER
1072 3155      DCA      CMCNT      /CLEAR CRC COUNTER!!!!

```

```

/
/READ INFORMATION!
/CHECK BUFFER ON THE FLY!
/

```

```

1073 4441      RDTRY, DISKGO      /READ DATA
1074 0400      0400      /READ DATA POINTER
1075 7610      SKP CLA      /DATA READ O.K.
1076 5305      JMP      RDSTA      /STATUS ERROR
1077 3155      DCA      CMCNT      /CLEAR CRC COUNTER!

```

```

/
/CHECK DATA ON NO STATUS ERRORS!
/

```

```

1100 4774*     JMS      DTCHK      /CHECK DATA
1101 5324      JMP      RERUN      /DATA O.K.
1102 2140      ISZ      DATTRY      /UPDATE READ RE-TRY
1103 5273      JMP      RDTRY      /TRY AGAIN
1104 5323      JMP      RERUN-1     /TRY TO SEEK IT
1105 1121      RDSTA, TAD      STNEG  /GET STATUS READ
1106 0065      AND      K0010      /MASK CRC
1107 7450      SNA      /CRC ERROR????
1110 5320      JMP      UPTRY      /NO, TRY READ AGAIN
1111 3156      DCA      CRCFLG      /YES, SET FLAG
1112 2155      ISZ      CMCNT      /UPDATE CRC POINTER

```

```

/
/CHECK DATA AFTER CRC ERROR!
/

```

```

1113 4774*     JMS      DTCHK      /CHECK DATA
1114 7610      SKP CLA      /IS A HARD ERROR!
1115 7340      CLA CLL CMA      /SET RETRY COUNTER!
1116 3165      DCA      ERFLG      /SETUP FOR 64 RETRYS IF AC=7777
1117 7410      SKP      /CHECK ON RETRY!!!!
1120 3155      UPTRY, DCA      CMCNT
1121 2137      ISZ      STATRY      /UPDATE TRY POINTER
1122 5273      JMP      RDTRY      /TRY AGAIN
1123 3165      DCA      ERFLG      /IS A HARD ERROR
1124 3155      RERUN, DCA      CMCNT  /CLEAR CRC COUNT
1125 3156      DCA      CRCFLG      /CLEAR CRC FLAG
1126 4773*     JMS      CKTIM      /CHECK TIME POINTERS
1127 1165      TAD      ERFLG
1130 7650      SNA CLA      /IS IT 64 RETRYS FOR SOFT ERROR?
1131 5334      JMP      ,+3      /NO DON'T BOTHER
1132 2171      ISZ      TRYCNT      /YES, UPDATE RETRY COUNTER
1133 5266      JMP      REREAD+3   /TRY AGAIN

```

```

/
/CHECK FOR LOOP ON READ!
/

```

```

1134 4406      LAS      /GET SWITCH 1
1135 7104      CLL RAL
1136 7710      SPA CLA      /LOOP????
1137 5263      JMP      REREAD      /YES, LOOP
1140 1137      TAD      STATRY      /TEST FOR HARD ERROR
1141 7650      SNA CLA
1142 5351      JMP      STRREL      /YES
1143 3165      DCA      ERFLG      /CLEAR ERROR FLAG

```

```

/
/CHECK FOR TYPE STATUS
/REPORT!
/

```

```

1144 4406      LAS      /MASK
1145 0017      AND      K0400      /TYPE STATUS REPORT????
1146 7640      SZA CLA      /YES
1147 4772*     JMS      TPSTA      /DU NEXT DRIVE
1150 5771*     JMP      RUN

```

```

/
/RESTORE DRIVE AFTER ERROR
/

```

```

1151 1123      STRREL, TAD      CMNEG  /GET DRIVE NO.
1152 4443      RECAL      /RESTORE
1153 5771*     JMP      RUN      /START NEXT DRIVE

```

```

/
/ROUTINE TO DETERMINE IF TIMING NEEDS TO BE FOR APT SYSTEM.
/

```

```

1154 0000      XKTICK, 0
1155 4424      CHK22      /TEST FOR APT
1156 7410      SKP      /NO, RETURN TO NORMAL RUN
1157 5754      JMP I      XKTICK
1160 6201      CDF      0
1161 2150      ISZ      CLKCNT      /LONG COUNTER FOR APT
1162 5366      JMP      EXTICK      /NORMAL RETURN
1163 1100      TAD      K7760      /INIT COUNTER
1164 3150      DCA      CLKCNT
1165 4770*     JMS      KTIME      /NOTIFY APT
1166 6201      EXTICK, CDF 0
1167 5754      JMP I      XKTICK
1170 0530
1171 0600
1172 3000
1173 2450
1174 1600
1175 0721
1176 0723
1177 3573

```

```

1200 PAGE

```

```

/
/ERROR HANDLER!
/UPDATE "SOFT" OR "HARD" TALLYS!
/PRINT ERROR TEXT AND DATA!
/CHECK INHIBIT ERROR SW!
/

```

```

1200 0000
1201 7001

```

```

ERNO, 0
IAC

```

```

/UPDATE AC FLAG

```

```

1202 3374      DCA      PCNTR2      /SAVE NON-RECOVERABLE POINTER;
/
/COMPUTE WAY TO "HARD"/"SOFT" TALLY;
/
1203 1377      TAD      K7773
1204 3375      DCA      PCNTR3
1205 1123      TAD      CMREG      /LINE COUNTER
1206 0863      AND      K0006      /GET LAST COMMAND
1207 7170      CLL CML  CMA RAR      /MASK DRIVE NUMBER
1210 3373      DCA      PCNTR1
1211 1062      TAD      K0003      /SETUP COUNTER
1212 2373      ISZ      PCNTR1
1213 5211      JMP      .+2
1214 1153      TAD      STAPOT      /COMPUTE WAY TO BUFFER
1215 3373      DCA      PCNTR1      /POINTER TO BUFFER
/
/DETERMINE IF ERROR IS "HARD" OR "SOFT";
/
1216 1156      TAD      CRCFLG      /GET CRC FLAG
1217 7650      SNA CLA      /CRC ERROR???
1220 5251      JMP      NTSOFT      /NO, WAS DEFINITELY A HARD ERROR;
1221 1600      TAD I      ERRO      /GET ERROR POINTER;
1222 7650      SNA CLA      /WAS IT FIRST TIME?
1223 5255      JMP      NTERR      /NO ERROR, ADDITIONAL CRC DATA;
1224 1125      TAD      DAREG      /COMPARE FAILING SECTOR TO
1225 0866      AND      K0017      /SECTOR WHERE DATA ERROR
1226 7041      CIA
1227 1131      TAD      ASNEG
1230 7640      SZA CLA      /SAME SECTOR?
1231 5251      JMP      NTSOFT      /NO, "HARD" ERROR
1232 7340      CLA CLL  CMA
1233 1155      TAD      CRCCNT
1234 7450      SNA
1235 5245      JMP      SOFT
1236 1110      TAD      K7775
1237 7650      SNA CLA
1240 2373      ISZ      PCNTR1
1241 1773      TAD I      PCNTR1
1242 7440      SZA
1243 1101      TAD      K7777
1244 5254      JMP      NTERR=1
1245 1101      SOFT, TAD      K7777
1246 1773      TAD I      PCNTR1
1247 3773      DCA I      PCNTR1
1250 2373      ISZ      PCNTR1
1251 1101      NTSOFT, TAD      K7777
1252 2773      ISZ I      PCNTR1
1253 7610      SKP CLA
1254 3773      DCA I      PCNTR1
/
/CHECK INHIBIT SW;
/
1255 4423      NTERR, JMS I  KAERRO
1256 4406      LAB
1257 7106      CLL RTL
/REPORT ERROR TO APT IF REQUIRED

```

```

1260 7710      SPA CLA
1261 5356      JMP      ERROEX+1      /INHIBIT ERRORS???
/
/CHECK FOR NO HEADER ON SECOND DATA ERROR;
/
1262 1600      DDMEAD, TAD I  ERRO
1263 7650      SNA CLA
1264 5355      JMP      ERROEX
/
/TYPE ERROR MESSAGE;
/
1265 4457      CRLF
1266 4457      CRLF
1267 1374      TAD      PCNTR2
1270 7640      SZA CLA
1271 5275      JMP      .+4
1272 7340      CLA CLL  CMA
1273 4455      PRNTER
1274 3335      MES4
1275 1600      TAD I      ERRO
1276 1376      TAD      MEUTAD
1277 3120      DCA      PCREG
1280 1520      TAD I      PCREG
1281 3304      DCA      .+3
1282 7340      CLA CLL  CMA
1283 4455      PRNTER
1284 7402      MLY
1285 7340      CLA CLL  CMA
1286 4455      PRNTER
1287 3303      MES0
1288 4457      CRLF
1289 1200      TAD      ERRO
1290 3120      DCA      PCREG
1291 2200      ISZ      ERRO
1292 1600      TAD I      ERRO
1293 3371      DCA      ESARE
1294 2200      ISZ      ERRO
1295 1172      TAD      XTXT
1296 3374      DCA      PCNTR2
1297 1372      TAD      XREG
1298 3010      DCA      AUTD10
1299 1105      TAD      M12
1300 3373      DCA      PCNTR1
1301 1371      STHAUT, TAD      ESARE
1302 7500      SMA
1303 5363      JMP      NOTEX
1304 7104      CLL RAL
1305 3371      DCA      ESARE
1306 2375      ISZ      PCNTR3
1307 7610      SKP CLA
1308 4457      CRLF
1309 1374      TAD      PCNTR2
1310 2374      ISZ      PCNTR2
1311 2374      ISZ      PCNTR2
1312 3343      DCA      .+5

```

```

1341 7340      CLA CLL CMA
1342 4455      PRNTER
1343 7402      HLT
1344 1410      TAD I AUTO10
1345 4456      UCTEL
1346 2373      AGAIN, ISZ PCNTR1
1347 5325      JMP STKAUT
1350 1520      TAD I PCNEG
1351 1106      TAD M4
1352 7650      SNA CLA
1353 4573      JMS I PRNDAT
1354 5360      JMP ,+4
1355 4573      ERROEX, JMS I PRNDAT
1356 2200      ISZ ERNO
1357 2200      ISZ ERNO
1360 7301      CLA CLL IAC
1361 4454      CLRALL
1362 5600      JMP I ERNO
1363 7104      NOTEX, CLL RAL
1364 3371      DCA ESAVE
1365 2374      ISZ PCNTR2
1366 2374      ISZ PCNTR2
1367 2010      ISZ AUTO10
1370 5346      JMP AGAIN

/
1371 0000      ESAVE, 0
1372 0117      XREG, PCREG-1
1373 0000      PCNTR1, 0
1374 0000      PCNTR2, 0
1375 0000      PCNTR3, 0
1376 1377      MEDTAD, BUFPNT-1
1377 7773      K7773, 7773

/
1400 PAGE
/
/POINTERS FOR TEXT INFORMATION:
/
1400 3247      BUFPNT, ERTX1
1401 3255      ERTX2
1402 3264      ERTX3
1403 3276      ERTX4

/
/ROUTINE TO CHECK FOR CLASSIC AND LIMIT
/TRANSFERS TO FIELD 0 IF AVAILABLE.
/
1404 0000      CLAFLO, 0
1405 1022      TAD 22
1406 0017      AND K0400
1407 7650      SNA CLA
1410 5604      JMP I CLAFLO
1411 7340      CLA CLL CMA
1412 3143      DCA MAXFLO
1413 5604      JMP I CLAFLO

/
/ROUTINE TO DO CRLF

```

/ON CLASSIC?

/NO,

/LIMIT TO FIELD 0.

```

1414 0000      /
1415 7300      UPONE, 0
1416 1224      CLA CLL
1417 4445      TAD K0215
1420 1225      TYPE K0212
1421 4445      TAD K0212
1422 4445      TYPE K0212
1423 5614      JMP I UPONE

/
1424 0215      K0215, 0215
1425 0212      K0212, 0212

/
/ROUTINE TO PRINT FOUR OCTAL
/
1426 0000      PROCT, 0
1427 7006      RTL
1430 7006      RTL
1431 3214      DCA UPONE
1432 1106      TAD M4
1433 3250      DCA PRN
1434 1214      TAD UPONE
1435 0064      AND K0007
1436 1067      TAD K0260
1437 4445      TYPE
1440 1214      TAD UPONE
1441 7006      RTL
1442 7006      RAL
1443 3214      DCA UPONE
1444 2250      ISZ PRN
1445 5234      JMP ,+11
1446 4431      SPACE
1447 5626      JMP I PROCT

/
/SUBROUTINE TO PRINT TEXT
/
1450 0000      PRN, 0
1451 7650      SNA CLA
1452 4457      CRLF
1453 1650      TAD I PRN
1454 2250      ISZ PRN
1455 3226      DCA PROCT

/TYPE CRLF
/YES!!!!
/GET POINTER

1456 7300      MKPRN, CLA CLL
1457 1626      TAD I PROCT
1460 0077      AND K7700
1461 7450      SNA
1462 5304      JMP EXIT
1463 7500      SNA
1464 7020      CML
1465 7001      IAC
1466 7012      RTR
1467 7012      RTR

```

```
1470 7012      RTR
1471 4445      TYPE
1472 1626      TAD I   FRUCT
1473 0102      AND     K0077
1474 7450      SNA
1475 5304      JMP     EXIT
1476 1313      TAD     K3740
1477 7500      SNA
1480 1074      TAD     K4100
1501 4431      SPACE
1502 2226      ISZ     FRUCT      /SPACE OUT 1
1503 5256      JMP     MRPRN      /MORE TO PRINT
1504 7300      EXIT,   CLA CLL
1505 5650      JMP I   PRN

/
/ROUTINE TO SPACE OUT 1
/
1506 0000      SPAC,   0
1507 1312      TAD     K0240
1510 4445      TYPE
1511 5706      JMP I   SPAC

/
1512 0240      K0240, 240
1513 3740      K3740, 3740

/
/THIS ROUTINE WILL BE A SKIP INSTRUCTION FOR SYSTEMS WITHOUT CLASSIC
/OTHERWISE IT WILL EXECUTE THE NEXT INSTRUCTION IN FIELD 0 AND THEN
/SKIP THE INSTRUCTION AFTER THAT ONE.
/
1514 0000      CLASIK, 0
1515 3345      DCA     SAVEAC      /SAVE CURRENT AC.
1516 1714      TAD I   CLASIK      /GET INSTRUCTION TO EXECUTE,
1517 3344      DCA     ROUTMP      /SAVE IT.
1520 2314      ISZ     CLASIK
1521 1022      TAD     UP2
1522 0017      AND     K0400
1523 7640      SZA CLA
1524 5327      JMP     .+3          /ARE WE ON CLASSIC?
1525 1345      TAD     SAVEAC      /YES.
1526 5714      JMP I   CLASIK      /NO, THEN
1527 2314      ISZ     CLASIK      /EXIT.
1530 6211      COP     10
1531 1020      TAD     SWR
1532 3777      DCA I   (SWR)        /SAVE SWITCH REGISTER.
1533 1021      TAD     UP1
1534 3776      DCA I   (OP1)        /SAVE CONTROL 1.
1535 1022      TAD     OP2
1536 3775      DCA I   (OP2)
1537 1344      TAD     ROUTMP
1540 3774      DCA I   (ROUTMP)     /SAVE ROUTINE IN FIELD 1.
1541 1345      TAD     SAVEAC      /GET BACK AC.
1542 6212      CIF     10
1543 5774      JMP I   (ROUTMP)     /GO AND EXECUTE INSTRUCTION.

/
1544 0000      ROUTMP, 0
```

```
1545 0000      SAVEAC, 0
/
/ROUTINE TO GET THE SWITCHES.
/
1546 0000      MYLAS, 0
1547 4407      CLASIK      /CHECK FOR CLASSIC,
1550 4425      C0CKSW      /GET SWITCHES.
1551 7604      7604
1552 5746      JMP I   MYLAS      /EXIT.

/
/ROUTINE TO RESET REGISTERS FOR ERROR PRINTER
/
1553 0000      SETREG, 0
1554 1073      TAD     K4000      /GET STATUS
1555 3121      DCA     STREG      /SAVE FOR ERROR PRINTER
1556 7340      CLA CLL CMA      /DECREASE BY 1
1557 1111      TAD     TRASH1     /GET SECTOR POINTER
1560 0066      AND     K0017
1561 1112      TAD     TRASH2
1562 3125      DCA     DAREG      /ADD IN ADDRESS
1563 1170      TAD     FINTIM     /SAVE FOR ERROR PRINTER
1564 7640      SZA CLA          /CHECK IF FIRST SECTOR?
1565 5753      JMP I   SETREG      /IF SO, DON'T UPDATE COMMAND!
1566 1174      TAD     SAVCM      /NO, DON'T!
1567 3123      DCA     CMREG      /GET COMMAND REG.
1570 5753      JMP I   SETREG      /SAVE FOR ERROR PRINTER
1574 1302      /RETURN

1575 0022
1576 0021
1577 0020
1600 1600      PAGE

/
/ROUTINE TO CHECK DATA READ
/
1600 0000      DTCHK, 0
1601 1156      TAD     CRCFLG      /GET CRC FLAG
1602 7640      SZA CLA          /WAS IT SET?
1603 5212      JMP     WRDCHK      /YES, THEN WORD BY WORD CHECK!!!
1604 1142      TAD     FNDSUM      /GET CHECK SUM FOUND
1605 7041      CIA
1606 1141      TAD     CMKSAV      /COMPARE TO GOOD VALUE SAVED
1607 7650      SNA CLA          /WERE THEY THE SAME
1610 5600      JMP I   DTCHK      /YES, DATA O.K.
1611 7340      CLA CLL CMA
1612 3446      WRDCHK, DCA I   XERR0 /SETUP CHECKSUM ERROR FLAG
1613 1123      TAD     CMREG

1614 0015      AND     K0100
1615 7640      SZA CLA          /HALF BLOCK SET??
1616 1016      TAD     K0200      /YES!
1617 1104      TAD     K7400
1620 3112      DCA     TRASH2
1621 1112      TAD     TRASH2
1622 7040      CMA
```

```

1623 3314      DCA  MSKER
1624 7340      CLA CLL CMA
1625 3142      DCA  FNDSUM
1626 4442      RESRAN
1627 1130      TAD  FWNEG
1630 4435      SETFLD
1631 3246      DCA  GOCDF
1632 1112      TAD  TRASH2
1633 3361      DCA  RSRAN
1634 1124      TAD  INTDA
1635 3353      DCA  STGEN
1636 1361      DTR1, TAD  RSRAN
1637 0314      AND  MSKER
1640 3132      DCA  WAREG
1641 1353      TAD  STGEN
1642 0066      AND  K0017
1643 3131      DCA  ASREG
1644 4427      RANDAT
1645 3134      DCA  DGNEG
1646 7402      GOCDF, HLT/CDP
1647 1411      TAD I  AUTO11
1650 6201      CDF  0
1651 3135      DCA  DBNEG
1652 1011      TAD  AUTO11
1653 3133      DCA  ADREG
1654 1135      TAD  DBREG
1655 7041      CIA
1656 1134      TAD  DGNEG
1657 7650      SNA CLA
1660 5272      JMP  NOERR
1661 2142      ISZ  FNDSUM
1662 5310      JMP  NTWRKS
1663 1156      TAD  CRCFLG
1664 7650      SNA CLA
1665 1140      TAD  DATTRY
1666 2200      ISZ  DTCHK
1667 4446      ERROR
1670 0004      0004
1671 7760      7760
1672 2361      NOERR, ISZ  RSRAN
1673 5300      JMP  ,+5
1674 2353      ISZ  STGEN
1675 7000      NOP
1676 1112      TAD  TRASH2
1677 3361      DCA  RSRAN
1680 2117      ISZ  BUFTAL
1681 5236      JMP  UTM1
1682 1446      TAD I  XENRO
1683 7650      SNA CLA
1684 3155      DCA  CRCCNT
1685 2446      ISZ I  XENRO
1686 5600      JMP I  DTCHK
1687 4576      BADHLT, JMS I  BGHLT
1690 4446      NTWRKS, ERROR
1711 0000      0000

```

```

/SET FIRST TIME FLAG
/NO, SETUP RANDOM GENERATOR
/GET FINAL WC
/GET AUTO11+ BUFTAL+ FIELD
/SAVE FIELD CDF

/GENERATE DATA
/SAVE GOOD DATA POINTER
/CDP TO BUFFER FIELD
/GET BAD DATA WORD
/HOME OF
/SAVE BAD WORD
/GET ADDRESS
/SAVE FOR PRINTER
/GET DATA READ

/COMPARE TO GOOD VALUE
/WERE THEY THE SAME
/YES, NO ERROR
/FIRST TIME PRINT???
/NO, JUST ADDRESS AND DATA
/GET CRC FLAG
/IF SET NO NON-RECOVERABLE.
/NO, GET NON-RECOVERABLE FLAG.
/UPDATE FOR ERROR RETURN
/ERROR DATA
/POINTER
/POINTER

/UPDATE BUFFER TALLY
/MORE WORDS TO CHECK
/GET ERROR INDICATOR;
/WAS THERE AN ERROR?
/NO, CLEAR CRC COUNTER
/CHECK FOR COMPUTER ERROR?
/ALL O.K.
/COMPUTER MUST BE DOWN, CHECKSUM
/OTHER ERRORS IN BUFFER

```

```

1712 0000      0000
1713 5272      JMP  NDERR
1714 0000      /MSKER, 0
1715 0000      /ROUTINE TO GENERATE RANDOM NUMBERS
1716 7301      RANDOM, 0
1717 1373      CLA CLL IAC
1720 1374      TAD  RAD1
1721 1375      TAD  RAD2
1722 3373      DCA  RAD1
1723 7004      RAL
1724 1373      TAD  RAD1
1725 1374      TAD  RAD2
1726 1375      TAD  RAD3
1727 3374      DCA  RAD2
1730 7004      RAL
1731 1373      TAD  RAD1
1732 1374      TAD  RAD2
1733 1375      TAD  RAD3
1734 3375      DCA  RAD3
1735 1375      TAD  RAD3
1736 5715      JMP I  RANDOM
1737 0000      /EXIT, RANDOM NUMBER IN AC
1740 7301      /GENERATOR FOR RANDOM DATA
1741 1367      GNDAT, 0
1742 1370      CLA CLL IAC
1743 7106      TAD  RAN1
1744 3367      TAD  RAN2
1745 1370      CLL RTL
1746 7012      DCA  RAN1
1747 1367      TAD  RAN2
1750 3370      DCA  RAN2
1751 1370      TAD  RAN2
1752 5737      JMP I  GNDAT
1753 0000      /ROUTINE TO SAVE RANDOM GENERATOR
1754 1367      STGEN, 0
1755 3371      TAD  RAN1
1756 1370      DCA  SAV1
1757 3372      TAD  RAN2
1760 5753      DCA  SAV2
1761 0000      /ROUTINE TO RESET RANDOM GENERATOR
1762 1371      RSRAN, 0
1763 3367      TAD  SAV1
1763 3367      DCA  RAN1

```

```

1764 1372          TAD          SAV2
1765 3370          DCA          RAN2
1766 5761          JMP I        RSMAN

/
1767 1234          RAN1,       1234
1770 5670          RAN2,       5670

/
1771 0000          SAV1,       0
1772 0000          SAV2,       0
1773 1234          RAD1,       1234
1774 5670          RAD2,       5670
1775 4321          RAD3,       4321

/
/
2000          PAGE

/
/ROUTINE TO WAIT FOR KEY FROM OPERATOR.
/
2000 0000          WAIT,       0
2001 6032          KCC
2002 6031          KSF
2003 5202          JMP , -1
2004 6036          KRB
2005 0234          AND          K177
2006 1016          TAD          K0200
2007 3235          DCA          CHKYN
2010 1022          TAD          22
2011 0017          AND          K0400
2012 7650          SNA CLA
2013 5226          JMP          WAIT1
2014 1235          TAD          CHKYN
2015 6211          CDF          10
2016 3777*         DCA          C0CHAR
2017 2776*         ISZ          INMODE
2020 1777*         TAD          C0CHAR
2021 6201          CDF          0
2022 4407          CLASIC
2023 4427          C0CNTR
2024 7000          NOP
2025 7300          CLA CLL
2026 1235          WAIT1, TAD    CHKYN
2027 6006          TLO
2030 6001          TBF
2031 5230          JMP , -1
2032 6002          TCF
2033 5600          JMP I        WAIT

/
/EXIT
2034 0177          K177,       0177

/
/ROUTINE TO CHECK FOR YES OR NO
/
2035 0000          CHKYN,      0
2036 3200          DCA          WAIT
2037 1235          TAD          CHKYN
2040 3260          DCA          CKMPOT

```

```

/SAVE CHARACTER
/CHECK FOR CLASSIC
/MASK CLASSIC BIT
/CLASSIC=NON ZERO

/RESTORE CHAR, FOR CLASSIC

/SAVE CHARACTER,

/GET BACK AC,

/CHECK FOR CLASSIC,
/ROUTINE TO EXECUTE.

/CLEAR CLASSIC AC RETURN
/RESTORE CHARACTER

/EXIT

/SAVE POINTER
/GET PC STORED
/SAVE IT

```

```

2041 1200      TAD      WAIT
2042 2235      ISZ      CHMYN
2043 7041      CIA
2044 1257      TAD      K0316
2045 7650      SNA CLA
2046 5635      JMP I     CHMYN      /WAS IT A NO
2047 1200      TAD      WAIT      /YES
2050 2235      ISZ      CHMYN
2051 7041      CIA
2052 1256      TAD      K0331
2053 7650      SNA CLA
2054 5635      JMP I     CHMYN      /WAS IT A YES
2055 5660      JMP I     CHKPOT     /YES
                                   /WAS NEITHER
                                   /
                                   /
2056 0331      K0331, 0331
2057 0316      K0316, 0316
                                   /
                                   /ROUTINE TO CHECK DISK RUN POINTERS
                                   /
2060 0000      CHKPOT, 0
2061 0064      AND      K0007
2062 1154      TAD      MUNPOT
2063 3200      DCA      WAIT
2064 1600      TAD I     WAIT
2065 7640      SZA CLA
2066 2260      ISZ      CHKPOT     /GET RUN POINTER
2067 5660      JMP I     CHKPOT     /RUN THIS DRIVE
                                   /NO
                                   /EXIT
                                   /
                                   /ROUTINE TO TEST FOR APT AND SET UP APPROPRIATE
                                   /REGISTERS IN ON THE SYSTEM.
                                   /
2070 0000      APT0, 0
2071 4424      CHK22
2072 5301      JMP      .+7      /TEST FOR APT
2073 4407      CLASIC      /YES
2074 4431      C08WIT
2075 7000      NOP
2076 1355      TAD      K7000
2077 3775*     OCA      SKPNOP
2100 5351      JMP      EXAPT0
2101 1022      TAD      OP2
2102 0354      AND      K7377     /EXIT
2103 3022      DCA      UP2
2104 1355      TAD      K7000     /NOP CONSOLE PACKAGE
2105 3774*     DCA      MYLAS+3
                                   /NOP SWITCH REGISTER
                                   /NO OPERATOR INTERVENTION ALLOWED
2106 1022      TAD      OP2
2107 0064      AND      K0007
2110 3111      DCA      TRASH1    /GET # OF DRIVES
2111 1022      TAD      UP2
2112 0015      AND      K0100
2113 7650      SNA CLA
2114 5325      JMP      M0USKS    /SINGLE DRIVE * NON ZERO AC
                                   /NO.

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```

2115 7301      CLL CLA IAC
2116 3000      DCA AMOUNT
2117 1111      TAD TRASH1
2120 1154      TAD HUNPOT
2121 3111      DCA TRASH1
2122 7340      CLL CLA CMA
2123 3511      DCA I TRASH1
2124 5342      JMP MEMSET
2125 1111      MODSKS, TAD TRASH1
2126 7040      CMA
2127 3112      DCA TRASH2
2130 3111      DCA TRASH1
2131 1111      TAD TRASH1
2132 1154      TAD HUNPOT
2133 3113      DCA TRASH3
2134 7340      CLL CLA CMA
2135 3513      DCA I TRASH3
2136 2111      ISZ TRASH1
2137 2000      ISZ AMOUNT
2140 2112      ISZ TRASH2
2141 5331      JMP MODSKS+4
2142 1021      MEMSET, TAD 21
2143 7012      RTR
2144 0004      AND K0007
2145 7104      CLL RAL
2146 7006      RTL
2147 7040      CMA
2150 3143      DCA MAXFLD
2151 2270      EXAPTS, ISZ APT8
2152 2270      ISZ APT8
2153 5670      JMP I APT8
/
2154 7377      K7377, 7377
2155 7000      K7000, 7000
/
/THIS ROUTINE WILL NOTIFY APT OF AN ERROR.
/ONLY THE DRIVE IN ERROR IS ESTABLISHED.
/
2156 0000      AENROR, 0
2157 4424      CHK22
2160 7410      SKP
2161 5756      JMP I AENROR
2162 6002      IOF
2163 7200      CLA
2164 1115      TAD POLOSK
2165 0004      AND K0007
2166 0201      CDF 00
2167 0272      CIF 70
2170 5772      JMP I K6520
2171 7402      HLT
/NOTIFY APT
/SOMETHING WENT WRONG IF IT GETS HERE
/
2172 6520      K6520, 6520
2174 1551
2175 0237
2176 1076

```

```

2177 1075      PAGE
2200
/ROUTINE TO WRITE OR READ SECTORS SELECTED
/
2200 0000      DSKGO, 0
2201 7340      CLA CLL CMA
2202 3170      DCA FINTIM
2203 3156      DCA CRCFLG
2204 1126      TAD CAREG
2205 4452      LDCUR
2206 1127      TAD WCREG
2207 3130      DCA FWREG
2210 1124      TAD INTDA
2211 3111      DCA TRASH1
2212 1124      TAD INTDA
2213 0100      AND K7760
2214 3112      DCA TRASH2
2215 1136      TAD INTCM
2216 1600      TAD I DSKGO
2217 4451      LDCMD
2220 1123      TAD CMREG
2221 1075      TAD K1000
2222 3174      DCA SAVCM
2223 1111      TAD TRASH1
2224 0006      AND K0017
2225 1112      TAD TRASH2
2226 4453      LDADD
2227 6001      ION
/
/ROUTINE TO CLEAR OR CHECK SUM BUFFER ON THE FLY
/
2230 3777      GOBAK, DCA TIMER2
2231 3142      DCA FNUMSUM
2232 4435      SETFLD
2233 3254      DCA CMNCDF
2234 1170      TAD FINTIM
2235 7650      SNA CLA
2236 5241      JMP STNWRK
2237 4776      JMS TIME
2240 5234      JMP .+4
2241 1117      STNWRK, TAD BUPTAL
2242 7041      CIA
2243 1130      TAD FWREG
2244 7450      SNA
2245 5274      JMP WRRDON
2246 7041      CIA
2247 3175      DCA CLNBAK
2250 1175      TAD CLNBAK
2251 7041      CIA
2252 1117      TAD BUPTAL
2253 3117      DCA BUPTAL
2254 7402      CMNCDF, HLT
2255 1123      TAD CMNEG
2256 7700      SNA CLA
/
/SETUP FIRST TIME POINTER
/CLEAR CRC FLAG
/GET INITIAL CURRENT ADDRESS
/LOAD CURRENT ADDRESS
/SETUP FINAL WC
/GET INITIAL STARTING SECTOR
/SAVE
/GET DISK ADDRESS
/MASK
/SAVE
/GET INITIAL COMMAND
/GET READ OR WRITE
/LOAD COMMAND
/MAKE READ ALL OR WRITE ALL
/SAVE FOR SWITCH TO CONSECUTIVE MODE
/SECTOR TO DO
/MASK
/ADD TO TRACK
/LOAD AND GO
/TURN INTERRUPT ON
/
/CLEAR LONG TIMER
/CLEAR SUM CHECK
/GET FIELD TO BUFFER
/SAVE CDF
/TIME TO GO
/YES!!!!
/WAIT FOR FIRST INTERRUPT
/NOT HERE YET
/COMPARE TO SOFTWARE FINAL
/WAIT FOR DISK???
/YES!!!!
/SAVE DIFFERENCE
/UPDATE BUFFER TALLY
/CDF TO BUFFER FIELD
/READ OR WRITE

```

```

2257 5264      JMP      WASRD
2260 3411      GOCLR, DCA I AUTO11
2261 2175      ISZ      CLMBAK
2262 5260      JMP      GOCLR
2263 5274      JMP      WRKDON
2264 1142      WASRD, TAD FND00M
2265 7100      GOCHK, CLL
2266 1411      TAD I AUTO11
2267 7430      SZL
2270 7801      IAC
2271 2175      ISZ      CLMBAK
2272 5265      JMP      GOCHK
2273 3142      DCA      FND00M
2274 6201      WRKDON, COF 0
2275 1117      TAD      BUFTAL
2276 7650      SNA CLA
2277 5302      JMP      DSKEK
2300 4776      JMS      TIME
2301 5241      JMP      STRWRK
2302 2200      DSKEK, ISZ DSKGO
2303 5600      JMP I DSKGO

/
/INTERUPT SERVICE
/
2304 6741      RETURN, DSKP
2305 5353      JMP      NODSKP
2306 2111      ISZ      TRASH1
2307 7000      NOP
2310 1114      TAD      UPDATE
2311 1130      TAD      FWREG
2312 3130      DCA      FWREG
2313 6745      STATUS, DRST
2314 1073      TAD      K4000
2315 7440      SZA
2316 5337      JMP      STATER
2317 1130      TAD      FWREG
2320 7650      SNA CLA
2321 5365      JMP      TRDONE
2322 3170      DCA      FINTIM
2323 1174      TAD      SAVCM
2324 6746      RDLWRL, DLDC
2325 1111      TAD      TRASH1
2326 0066      AND      K0017
2327 1112      TAD      TRASH2
2330 6743      LOGGO, DLAG
2331 1167      RETRN, TAD SVLNK
2332 7110      CLL RAM
2333 1166      TAD      SAVAC
2334 6244      RMF
2335 6001      ION
2336 5400      JMP I 0
2337 4775      STATER, JMS SETREG
2340 1123      TAD      CMREG
2341 7710      SPA CLA
2342 7801      IAC

```

```

/WAS A READ!!
/WAS A WRITE, CLEAR BUFFER
/UPDATE TALLY
/MORE TO CLEAR
/DONE WITH SOME

/GET WORD

/UPDATE CLEAR POINTER
/MORE TO CHECKSUM
/SAVE IT

/LAST WORD DONE????
/EXIT
/TIME AND WAIT
/WAIT FOR INT, OR DONE!!!!
/EXIT

/DISK SKIP IOT
/NOT THE DISK
/UPDATE SECTOR NUMBER
/IT WON'T WORK WITHOUT IT!

/UPDATE WORD COUNT

/READ STATUS

/ONLY DONE FLAG?
/STATUS ERROR

/LAST TRANSFER?
/TRANSFER IS DONE
/CLEAR FIRST TIME POINTER!
/GET READ OR WRITE COMMAND
/LOAD COMMAND REGISTER
/GET SECTOR TO DO
/MASK OFF
/ADD IN TRACK
/LOAD DISK ANFD GO
/GET LINK

/GET AC
/RESTORE FIELDS
/TURN INTERRUPT ON
/EXIT
/SETUP REGISTERS!

/WRITE OR READ
/WRITE

```

```

2343 7801      IAC
2344 3347      DCA      .+3
2345 1137      TAD      STATRY
2346 4446      ERROR
2347 0000      0000
2350 7770      7770
2351 2200      ISZ      DSKGO
2352 5302      JMP      DSKEK
2353 3374      NODSKP, DCA TIMER3
2354 2374      ISZ      TIMER3
2355 5354      JMP      .-1
2356 4407      CLASSIC
2357 4440      CCKCPA
2360 7000      NOP
2361 6031      KSF
2362 4576      INTER2, JMS I 0GMLT
2363 6032      KCC
2364 5331      JMP      RETRN
2365 4775      TRDONE, JMS SETREG
2366 3170      DCA      FINTIM
2367 1167      TAD      SVLNK
2370 7110      CLL RAM
2371 1166      TAD      SAVAC
2372 6244      RMF
2373 5400      JMP I 0

/
/ TIMER3, 0
/
2375 1553
2376 3123
2377 3141
2400 0000      PAGE
/
/ROUTINE TO GET ONE IN OCTAL
/
2400 0000      OCT1, 0
2401 4444      RECEIV
2402 3354      DCA      LOAD
2403 1600      TAD I OCT1
2404 0064      AND      K0007
2405 1067      TAD      K0260
2406 7141      CLL CIA
2407 1354      TAD      LOAD
2410 7620      SNA CLA
2411 5226      JMP      INERR
2412 1600      TAD I OCT1
2413 0014      AND      K0070
2414 7110      CLL RAM
2415 7012      RTR
2416 1067      TAD      K0260
2417 7040      CMA
2420 1354      TAD      LOAD
2421 7630      SZL CLA
2422 5226      JMP      INERR
2423 1354      TAD      LOAD

```

```

/MODIFY HEADER POINTER
/GET TRY POINTER
/PRINT MESSAGE
/MODIFIED HEADER POINTER
/MESSAGE POINTER
/UPDATE FOR ERROR
/EXIT

/WAIT FOR DISK TO STOP.
/CHECK FOR CLASSIC.
/ROUTINE TO EXECUTE.

/KEYBOARD FLAG??
/ILLEGAL INTERRUPT

/EXIT BACK.
/SETUP REGISTERS!
/CLEAR FIRST TIME POINTER!

/REPLACE LINK
/REPLACE AC
/RESTORE MEMORY FIELDS+ FLAGS
/RETURN TO BACK GROUND

/RECEIVE
/SAVE IT
/GET LIMITS
/MASK

/GET INPUT
/IN LIMITS????
/NO, ERROR EXIT
/GET LIMITS
/MASK

/GET INPUT
/IN LIMITS????
/NO, ERROR
/GET INPUT

```

```

2424 0064      AND      K0007      /MASK
2425 2200      ISZ      OCT1
2426 2200      INERH, ISZ      OCT1
2427 5600      JMP I      OCT1      /GOOD EXIT

/ROUTINE TO RECEIVE FOUR OCTAL
/
2430 0000      OCT4, 0
2431 1106      TAD      M4
2432 3341      DCA      R0ST      /SETUP COUNTER
2433 3350      DCA      LDCA      /START WITH 0
2434 4432      ONEIN      /RECEIVE ONE OCTAL
2435 0070      0070      /LIMITS
2436 5630      JMP I      OCT4      /ENRUR EXIT
2437 1350      TAD      LDCA      /GET LAST
2440 2341      ISZ      R0ST      /UPDATE COUNTER
2441 7410      SKP
2442 5246      JMP      .+4      /EXIT
2443 7004      HAL
2444 7006      RTL
2445 5233      JMP      OCT4+3
2446 2230      ISZ      OCT4
2447 5630      JMP I      OCT4      /EXIT OCTAL IN AC

/ROUTINE TO UPDATE AND CHECK FOR PASS COMPLETE
/
2450 0000      CKTIM, 0
2451 1115      TAD      POLDISK
2452 0064      AND      K0007      /SETUP CURRENT DRIVE #
2453 3341      DCA      R0ST      /POINTER
2454 1341      TAD      R0ST
2455 1152      TAD      TIMPOT      /GET TIME POINTER
2456 3354      DCA      LDAD      /SAVE IT
2457 7301      CLA CLL IAC      /ONE FOR 0
2460 1147      TAD      CONSEC      /GET AMOUNT DONE
2461 1754      TAD I      LDAD      /ADD IN AMOUNT COMPLETED SO FAR
2462 3754      DCA I      LDAD      /SAVE IT
2463 7620      SNL CLA      /LINK UP???
2464 5650      JMP I      CKTIM      /NO, EXIT
2465 4440      RANGEN      /GET RANDOM NUMBER
2466 3777      DCA      RAN1      /RE-PRIME GENERATOR
2467 4440      RANGEN      /GET RANDOM NUMBER
2470 3776      DCA      RAN2      /RE-PRIME GENERATOR
2471 7100      CLL
2472 1354      TAD      LDAD
2473 1013      TAD      K0004
2474 3354      DCA      LDAD      /SECOND TIME POINTER
2475 2754      ISZ I      LDAD      /UPDATE IT
2476 1754      TAD I      LDAD      /GET COUNT
2477 1144      TAD      MAXTIM      /ADD IN FUDGE FACTOR
2500 7620      SNL CLA      /PASS COMPLETE???
2501 5650      JMP I      CKTIM      /NO, EXIT
2502 3754      DCA I      LDAD      /ZERO SECOND COUNTER
2503 1341      TAD      R0ST
2504 7040      CMA

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```

2505 3341      DCA      R0ST      /SETUP COUNTER
2506 1362      TAD      CMPPOT      /ADD IN POINTER
2507 1062      TAD      K0003
2510 2341      ISZ      R0ST      /COMPUTE BUFFER
2511 5307      JMP      .+2
2512 3341      DCA      R0ST      /SAVE ADDRESS POINTER
2513 7340      CLA CLL CMA
2514 2741      ISZ I      R0ST      /UPDATE PASS COMPLETE POINTER
2515 7610      SKP CLA
2516 3741      DCA I      R0ST      /HOLD AT 7777
2517 4457      CRLF
2520 4455      PRNTER      /PRINT "DISK"
2521 3477      MES17
2522 1115      TAD      POLDISK      /GET DISK POLE NUMBER
2523 0064      AND      K0007      /MASK
2524 1067      TAD      K0260
2525 4445      TYPE      /TYPE DISK NO,
2526 7340      CLA CLL CMA
2527 4455      PRNTER      /PRINT "PASS COMPLETE"
2530 3502      MES18
2531 4406      LAS
2532 0015      AND      K0100      /MASK
2533 7650      SNA CLA      /PASS COMPLETE DISCONNECT???
2534 5337      JMP      .+3      /NO WAY!!!!
2535 4430      DISCON      /DUMP DRIVE
2536 5775      JMP      RUN      /MORE TO TEST!!!!
2537 4774      JMS      TPSTA      /STATUS=COMPLETE TYPEOUT
2540 5650      JMP I      CKTIM      /EXIT

/SUBROUTINE TO READ STATUS REGISTER
/
2541 0000      R0ST, 0
2542 6745      IOT5, DRST      /READ STATUS IOT
2543 7410      SKP
2544 4576      ERHLT5, JMS I      HGHLT      /SKIP TRAP
2545 3121      DCA      STREG      /SAVE RESULTS
2546 1121      TAD      STNEG
2547 5741      JMP I      R0ST      /EXIT

/SUBROUTINE TO LOAD CURRENT ADDRESS REGISTER
/
2550 0000      LDCA, 0
2551 6744      IOT4, DLCA      /LOAD CURRENT ADDRESS IOT
2552 4425      KTICK      /NOTIFY APT
2553 5750      JMP I      LDCA      /EXIT

/SUBROUTINE TO LOAD TRACK ADDRESS REGISTER
/
2554 0000      LOAD, 0
2555 3125      DCA      OAMEG
2556 1125      TAD      OAMEG
2557 6743      IOT3, DLAG      /LOAD DISK ADDRESS REGISTER
2560 5754      JMP I      LOAD      /EXIT
2561 4576      ERHLT3, JMS I      HGHLT      /ENRUR SKIP TRAP

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2562 3541 / CMPPOT, D0CMP-3
2574 3000
2575 0600
2576 1770
2577 1767
2600 2600 PAGE
/
/ROUTINE TO GET RANDOM OR OPERATOR DATA
/
2600 0000 RNWRD, 0
2601 7402 SMDAT, HLT /MODIFIED SWITCH
2602 5600 JMP I RNWRD /EXIT
2603 6201 CDF 0 /HOME CDF
2604 1412 TAD I AUTO12 /GET DATA
2605 7402 RECDF, HLT /BUFFER CDF
2606 2116 ISZ OPMTAL /UPDATE TALLY
2607 5600 JMP I RNWRD /EXIT
2610 3220 DCA PRINT /SAVE WORD
2611 1105 TAD M12
2612 3116 DCA OPMTAL /REPLACE TALLY
2613 7340 CLA CLL CMA
2614 1151 TAD DATPOT
2615 3012 DCA AUTO12 /REPLACE AUTO INDEX
2616 1220 TAD PRINT /GET SAVED WORD
2617 5600 JMP I RNWRD /EXIT

/ROUTINE TO TYPE
/
2620 0000 PRINT, 0
2621 3237 DCA DUMP /STORE AC VALUE
2622 4424 CHK22 /SEE IF ON APT
2623 5235 JMP PREXIT /NO, EXIT
2624 1237 TAD DUMP /RETURN AC,
2625 4407 CLASIC /CHECK FOR CLASSIC,
2626 4435 CBTYP /ROUTINE TO EXECUTE,
2627 7410 SKP
2630 5620 JMP I PRINT /EXIT,
2631 6046 T1S
2632 6041 T3F
2633 5232 JMP ,=1
2634 6042 TCF
2635 7200 PREXIT, CLA
2636 5620 JMP I PRINT

/ROUTINE TO DUMP AND REPORT DISK STATUS
/
2637 0000 DUMP, 0
2640 4424 CHK22 /CHECK FOR APT
2641 5637 JMP I DUMP
2642 4455 PRINTER /PRINT "DISK "
2643 3477 MES17
2644 1115 TAD POLOSK
2645 0864 AND K0007 /SETUP CURRENT DRIVE #
2646 3200 DCA RNWRD /SAVE

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2647 1200 TAD RNWRD /GET DISK NUMBER
2650 1067 TAD K0260
2651 4445 TYPE /TYPE DISK NUMBER
2652 7340 CLA CLL CMA
2653 4455 PRINTER /PRINT "DISCONNECTED"
2654 3445 MES15
2655 4777 JMS TPSTA /TYPE STATUS REPORT
2656 1200 TAD RNWRD
2657 1154 TAD RUNPOT
2660 3200 DCA RNWRD /SAVE POINTER ADDRESS
2661 3600 DCA I RNWRD /CLEAR RUN POINTER
2662 3200 DCA RNWRD
2663 1106 TAD M4
2664 3220 DCA PRINT /CHECK FOR MORE POINTER
2665 1200 TAD RNWRD
2666 4436 SELCHK /CHECK SELECT POINTERS
2667 7610 SKP CLA /DISK NOT HERE
2670 5637 JMP I DUMP /MORE AVAILABLE
2671 2200 ISZ RNWRD
2672 2220 ISZ PRINT /UPDATE POINTERS
2673 5265 JMP ,=6
2674 4457 CRLF
2675 4455 PRINTER /PRINT "DISK"
2676 3477 MES17
2677 7340 CLA CLL CMA
2700 4455 PRINTER /PRINT "SYSTEM DOWN"
2701 3455 MES16
2702 4576 NOOSKB, JMS I BGHLT /ERROR, NO DISK AVAILABLE

/ROUTINE TO SETUP FIELD TO BUFFER+ AUTO11+ BUFFER TALLY
/
2703 0000 STFLD, 0
2704 7041 CIA
2705 1127 TAD WCMEG
2706 3117 DCA BUPTAL
2707 7340 CLA CLL CMA
2710 1126 TAD CAHEG /GET INITIAL CA
2711 3011 DCA AUTO11 /SAVE
2712 1157 TAD DATFLG /GET DATA FLAG
2713 7650 SNA CLA /WAS IT SET???
2714 5322 JMP ,+6 /NO, USE REGULAR
2715 1105 TAD M12
2716 3116 DCA OPMTAL /SETUP SPECIAL TALLY
2717 7340 CLA CLL CMA
2720 1151 TAD DATPOT
2721 3012 DCA AUTO12 /SETUP SPECIAL AUTO INDEX
2722 1136 TAD INTCH /GET LAST COMMAND
2723 0014 AND K0070 /MASK FIELD BITS
2724 1103 TAD KCDF /MAKE BUFFER CDF
2725 3205 DCA RECDF /SETUP SPECIAL CDF
2726 1205 TAD RECDF /GET BACK CDF
2727 5703 JMP I STFLD /EXIT, FIELD IN AC

/ROUTINE TO CHANGE DEVICE IOT CODES
/

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2730 4407  CHANG, CLASSIC /CHECK FOR CLASSIC.
2731 4431  C88WIT /ROUTINE TO EXECUTE.
2732 7000  NOP
2733 4406  LAS /GET SWITCHES
2734 0071  AND A0770 /MASK 3-8
2735 3776' DCA LDCM /SAVE DESIRED CODE
2736 1360  TAD CHNPOT /POINTER
2737 3111  DCA TRASH1 /ADDRESS POINTER
2740 1357  TAD CCNTR1 /AMOUNT TO DO
2741 3112  DCA TRASH2 /SETUP COUNTER
2742 1511  CHANGR, TAD I TRASH1 /GET ADDRESS POINTER
2743 3113  DCA TRASH3 /SAVE ADDRESS
2744 1513  TAD I TRASH3 /GET OLD CODE
2745 0072  AND A7007 /MASK OFF OLD CODE
2746 1776' TAD LDCM /AUD IN DESIRED CODE
2747 3513  DCA I TRASH3 /RESTORE
2750 2111  ISZ TRASH1 /UPDATE POINTER
2751 2112  ISZ TRASH2 /UPDATE CHANGE COUNTER
2752 5342  JMP CHANGR /MORE TO CHANGE
2753 4407  CLASSIC /CHECK FOR CLASSIC.
2754 4436  C88ERR /ROUTINE TO EXECUTE.
2755 7402  CHNHLT, HLT /IOTS CHANGED, HIT CONTINUE OR
2756 5775' JMP BGN /IF ON CONSOLE PACKAGE
/CONTROL Z TO START PROGRAM.

2757 7765  / CCNTR1, 7765
/ CHNPOT, CHNPOT+1
2760 2761  RETURN
2761 2304  STATUS
2762 2313  RDLWRL
2763 2324  LODGO
2764 2330  IOT0
2765 0554  IOT1
2766 0752  IOT2
2767 0561  IOT3
2770 2557  IOT4
2771 2551  IOT5
2772 2542  IOT6
2773 0545  /

2775 0200  /
2776 0542  /
2777 3000  /
3000 3000  PAGE
/ /ROUTINE TO TYPE STATUS REPORT
/ TPSTA, 0
3001 4424  CHK22
3002 5600  JMP I TPSTA
3003 4457  CRLF
3004 4455  PRNTER /PRINT "DISK HARD SOFT COMP"
3005 3372  MES7
3006 1107  TAD M10
3007 3245  DCA TSAVE1 /MAXIMUM TO DO

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3010 3246  DCA TSAVE2
3011 3247  DCA TSAVE3
3012 1246  CHKRES, TAD TSAVE2 /CLEAR SOME COUNTERS
3013 1062  TAD K0003
3014 3246  DCA TSAVE2
3015 1246  TAD TSAVE2
3016 1153  TAD STAPOT
3017 3251  DCA TSAVE3 /LOCATION OF DISK STATUS
3020 1247  TAD TSAVE3
3021 4436  SELCHK
3022 5241  JMP NOTSTA /CHECK RUN POINTER
3023 4457  CRLF /DISK NOT RUNNING
3024 4431  SPACE
3025 1247  TAD TSAVE3 /SPACE OUT ONE
3026 1067  TAD K0260 /GET DISK NO.
3027 4445  TYPE
3030 4431  SPACE /SPACE OUT ONE
3031 4431  SPACE /SPACE OUT ONE
3032 7346  CLA CLL CMA RTL
3033 3250  DCA TSAVE4 /COUNTER FOR FOUR WORDS
3034 1651  TAD I TSAVE5 /GET STATUS
3035 4456  OCTEL /TYPE IT
3036 2251  ISZ TSAVE5
3037 2250  ISZ TSAVE4
3040 5234  JMP ,=4
3041 2247  NOTSTA, ISZ TSAVE3 /UPDATE DRIVE NUMBER
3042 2245  ISZ TSAVE1 /MORE TO REPORT
3043 5212  JMP CHKRES /EXIT
3044 5600  JMP I TPSTA

3045 0000  TSAVE1, 0
3046 0000  TSAVE2, 0
3047 0000  TSAVE3, 0
3050 0000  TSAVE4, 0
3051 0000  TSAVE5, 0
/ /ROUTINE TO RECALIBRATE SELECTED DRIVE
/ /DISCONNECT DRIVE ON ERROR!
/
3052 0000  RESTOR, 0
3053 0063  AND K0006
3054 3200  DCA TPSTA /SAVE DRIVE NUMBER
3055 1077  TAD K7700
3056 3341  DCA TIMER2 /SETUP COUNTER
3057 2340  ISZ TIMER1
3060 5257  JMP ,=1
3061 2341  ISZ TIMER2 /WAIT FOR DISK TO COOL OFF!
3062 5257  JMP ,=3
3063 1200  TAD TPSTA
3064 4451  LDCMO /CURRENT DRIVE
3065 7326  CLA CLL CML RTL /LOAD COMMAND
3066 4454  CLRALL /ENABLE RECALIBRATE BIT
3067 4450  OSKSKP /"RECALIBRATE"
3070 5267  JMP ,=1 /DISK SKIP IOT
3071 4447  R0STAT /WAIT FOR FIRST DONE FLAG
/MEAU STATUS

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3072 7500      SMA
3073 5311      JMP      RESERR
3074 0076      AND      K1777
3075 7640      SZA CLA
3076 5311      JMP      RESERR
3077 4454      RESTA, CLRALL
3100 1016      TAD      K0200
3101 1200      TAD      TPSTA
3102 4451      LDCHD
3103 4450      DSKSKP
3104 5303      JMP      .-1
3105 4447      ROSTAT
3106 1073      TAD      K4000
3107 7650      SNA CLA
3110 5652      JMP I RESTOR
3111 7300      RESERR, CLA CLL
3112 4446      ERROR
3113 0003
3114 7500      7500
3115 4457      CRLF
3116 4457      CRLF
3117 4455      PRINTER
3120 3174      MES19
3121 4430      DISCON
3122 5652      JMP I RESTOR

/ROUTINE TO TIME AND WAIT
3123 0000      TIME, 0
3124 2340      ISZ      TIMER1
3125 5723      JMP I TIME
3126 2341      ISZ      TIMER2
3127 5723      JMP I TIME
3130 4576      INTER1, JMS I B0HLT

/ROUTINE TO COMBINE ERROR HALTS,
/WHEN THE COMPUTER HALTS THE AC
/WILL EQUAL THE PC ON THE FAILING
/HAULT INSTRUCTION.
3131 0000      B0HLT, 0
3132 7300      CLA CLL
3133 1331      TAD      B0HLT
3134 4407      CLASIC
3135 4436      CBERR
3136 7402      B0GSTP, HLT
3137 5332      JMP      .-3

3140 0000      TIMER1, 0
3141 0000      TIMER2, 0

/ROUTINE TO TYPE OUT DATA INFORMATION
3142 0000      TYPDAT, 0
3143 4455      PRINTER

```

/PRINT"RECALIBRATE ERROR DISCONNECT"

/DISCONNECT DISK
/MORE DISK AVAILABLE

/EXIT

/EXIT

/NO INTERRUPT OCCURRED, I GUESS!

/LOAD AC WITH PC.
/CHECK FOR CLASSIC.
/ROUTINE TO EXECUTE.
/AC=PC.
/NON-RECOVERABLE.

/PRINT "A8:"

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3144 3235      TEXAS
3145 1131      TAD      ASREG
3146 4456      OCTEL
3147 7340      CLA CLL CMA
3148 4455      PRINTER
3151 3237      TEXWA
3152 1132      TAD      WAREG
3153 4456      OCTEL
3154 7340      CLA CLL CMA
3155 4455      PRINTER
3156 3241      TEXAD
3157 1133      TAD      ADREG
3160 4456      OCTEL
3161 7340      CLA CLL CMA
3162 4455      PRINTER
3163 3243      TEXDG
3164 1134      TAD      DGREG
3165 4456      OCTEL
3166 7340      CLA CLL CMA
3167 4455      PRINTER
3170 3245      TEXDB
3171 1135      TAD      DBREG
3172 4456      OCTEL
3173 5742      JMP I TYPDAT

3174 2205      MES19, TEXT "RECALIBRATE ERROR DISCONNECT!"
3175 0301
3176 1411
3177 0222
3200 0124
3201 0540
3202 0522
3203 2217
3204 2240
3205 0411
3206 2303
3207 1716
3210 1605
3211 0324
3212 4100

3213 2003      TEXPC, TEXT "PC:"
3214 7200
3215 2324      TEXST, TEXT "ST:"
3216 7200
3217 0530      TEXEX, TEXT "EX:"
3220 7200
3221 0315      TEXCM, TEXT "CM:"
3222 7200
3223 1101      TEXIA, TEXT "IA:"
3224 7200
3225 0401      TEXDA, TEXT "DA:"
3226 7200
3227 0301      TEXCA, TEXT "CA:"
3230 7200

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3231	2703	TEXWC, TEXT	"WC1"
3232	7200		
3233	0627	TEFXW, TEXT	"FW1"
3234	7200		
3235	0123	TEXAS, TEXT	"AS1"
3236	7200		
3237	2701	TEXWA, TEXT	"WA1"
3240	7200		
3241	0104	TEXAD, TEXT	"AD1"
3242	7200		
3243	0407	TEXDG, TEXT	"DG1"
3244	7200		
3245	0402	TEXDB, TEXT	"DB1"
3246	7200		
/			
3247	2205	ERTX1, TEXT	"READ STATUS"
3250	0104		
3251	4023		
3252	2401		
3253	2425		
3254	2300		
3255	2722	ERTX2, TEXT	"WRITE STATUS"
3256	1124		
3257	0540		
3260	2324		
3261	0124		
3262	2523		
3263	0000		
3264	2205	ERTX3, TEXT	"RECALIBRATE STATUS"
3265	0301		
3266	1411		
3267	0222		
3270	0124		
3271	0540		
3272	2324		
3273	0124		
3274	2523		
3275	0000		
3276	0411	ERTX4, TEXT	"DISK DATA"
3277	2313		
3300	4004		
3301	0124		
3302	0100		
/			
3303	4005	MES0, TEXT	" ERROR"
3304	2222		
3305	1722		
3306	0000		
3307	2213	MES1, TEXT	"MK0E/RK0L DATA RELIABILITY"
3310	7005		
3311	5722		
3312	1370		
3313	1440		
3314	0401		
3315	2401		

3316	4022		
3317	0514		
3320	1101		
3321	0211		
3322	1411		
3323	2431		
3324	0000		
3325	0530	MES2, TEXT	"EXERCISE"
3326	0522		
3327	0311		
3330	2305		
3331	0000		
3332	4004	MES3, TEXT	" DISK"
3333	1123		
3334	1300		
3335	1617	MES4, TEXT	"NON-RECOVERABLE "
3336	1655		
3337	2205		
3340	0317		
3341	2605		
3342	2201		
3343	0214		
3344	0540		
3345	0000		
3346	0530	MES5, TEXT	"EXTENDED R/W MEMORY(0-7)?"
3347	2405		
3350	1604		
3351	0504		
3352	4022		
3353	5727		
3354	4015		
3355	0515		
3356	1722		
3357	3150		
3360	6055		
3361	6751		
3362	7700		
3363	0103	MES6, TEXT	"ACCEPT MODE?"
3364	0305		
3365	2024		
3366	4015		
3367	1704		
3370	0577		
3371	0000		
3372	0423	MES7, TEXT	"DSK HAND SOFT COMP"
3373	1340		
3374	1001		
3375	2204		
3376	4023		
3377	1706		
3400	2440		
3401	0317		
3402	1520		

3403	0000		
3404	0611	ME08, TEXT	"FIELD7"
3405	0514		
3406	0477		
3407	0000		
3410	2422	ME09, TEXT	"TRACK7"
3411	0103		
3412	1377		
3413	0000		
3414	0530	ME010, TEXT	"EXTRA SECTOR37"
3415	2422		
3416	0140		
3417	2305		
3420	0324		
3421	1722		
3422	2377		
3423	0000		
3424	0214	ME011, TEXT	"BLOCK LENGTH7"
3425	1703		
3426	1340		
3427	1405		
3430	1607		
3431	2410		
3432	7700		
3433	0401	ME013, TEXT	"DATA7"
3434	2401		
3435	7700		
3436	0122	ME014, TEXT	"ARE YOU SURE?"
3437	0540		
3440	3117		
3441	2540		
3442	2325		
3443	2205		
3444	7700		
3445	4004	ME015, TEXT	" DISCONNECTED!"
3446	1123		
3447	0317		
3450	1016		
3451	0503		
3452	2405		
3453	0441		
3454	0000		
3455	2331	ME016, TEXT	"SYSTEM SHUT DOWN, NO DISKS TO RUN!"
3456	2324		
3457	0515		
3460	4023		
3461	1025		
3462	2440		
3463	0417		
3464	2716		
3465	5440		
3466	1617		
3467	4004		
3470	1123		
3471	1323		

3472	4024		
3473	1740		
3474	2225		
3475	1641		
3476	0000		
3477	0411	ME017, TEXT	"DISK "
3500	2313		
3501	4000		
3502	4020	ME018, TEXT	" PASS COMPLETE!"
3503	0123		
3504	2340		
3505	0317		
3506	1520		
3507	1405		
3510	2405		
3511	4100		
/			
3512	0000	DSK00, 0	
3513	0000	DSK10, 0	
3514	0000	DSK20, 0	
3515	0000	DSK30, 0	
3516	0000	DSK40, 0	
3517	0000	DSK50, 0	
3520	0000	DSK60, 0	
3521	0000	DSK70, 0	
/			
3522	0000	D0TM1, 0	
3523	0000	D1TM1, 0	
3524	0000	D2TM1, 0	
3525	0000	D3TM1, 0	
3526	0000	D4TM1, 0	
3527	0000	D5TM1, 0	
3530	0000	D6TM1, 0	
3531	0000	D7TM1, 0	
3532	0000	D0TM2, 0	
3533	0000	D1TM2, 0	
3534	0000	D2TM2, 0	
3535	0000	D3TM2, 0	
3536	0000	D4TM2, 0	
3537	0000	D5TM2, 0	
3540	0000	D6TM2, 0	
3541	0000	D7TM2, 0	
/			
3542	0000	D0HRD, 0	
3543	0000	D0SOF, 0	
3544	0000	D0CMP, 0	
3545	0000	D1HRD, 0	
3546	0000	D1SOF, 0	
3547	0000	D1CMP, 0	
3550	0000	D2HRD, 0	
3551	0000	D2SOF, 0	
3552	0000	D2CMP, 0	

3553 0000 D3HRD, 0
3554 0000 D38OF, 0
3555 0000 D3CMP, 0
3556 0000 D4HRD, 0
3557 0000 D48OF, 0
3560 0000 D4CMP, 0
3561 0000 D5HRD, 0
3562 0000 D58OF, 0
3563 0000 D5CMP, 0
3564 0000 D6HRD, 0
3565 0000 D68OF, 0
3566 0000 D6CMP, 0
3567 0000 D7HRD, 0
3570 0000 D78OF, 0
3571 0000 D7CMP, 0
/
3572 0000 FLDPLG, 0
3573 0000 TRKFLG, 0
3574 0000 SECFLG, 0
3575 0000 MLFFLG, 0
/
/
/
3600 PAGE
/
3600 STRBUF=
/
8888

0000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111110
0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 11111111 11110011 11111111
0400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0500 11111111 11111111 11111111 11111111 11111111 11111111 11110000 11111111
0600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11011111
1000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 10001111
1600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111000
2000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11101111
2200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
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2500 11111111 11111111 11111111 11111111 11111111 11111111 11100000 00001111
2600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11110111
3000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111000
3600
3700

4000
4100

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

A0170	0750	C0GET	0624	CONCUR	0715	DAT3	0760
A0770	0071	C0HANG	1122	CONSEC	0147	DAT4	0761
A7007	0072	C0INQU	4437	CONSUL	0000	DAT5	0762
AC0CA	0362	C0OCTA	4432	CRCCNT	0155	DAT6	0763
ACL	7701	C0PASS	4424	CRCFLG	0156	DAT7	0764
ACSAVE	1345	C0PAUS	4441	CRLF	4437	DAT8	0765
ADREG	0133	C0PRNT	4430	D0CMP	3544	DAT9	0766
AERROR	2156	C0RDP5	0606	D0HRU	3542	DATFLG	0157
AGAIN	1346	C0RETD	0614	D0SOP	3543	DATPOT	0151
ALLAGN	0255	C0RETR	0536	D0TM1	3522	DATTRY	0140
AMOUNT	0060	C0SETD	0613	D0TM2	3532	DBREG	0135
APT0	2270	C0SETR	0535	D1CMP	3547	DCLR	6742
ASKNX1	0345	C0SWIT	4431	D1HRU	3545	DGREG	0134
ASKNX2	0406	C0SWST	0745	D1SOP	3546	DISCON	4430
ASKNX3	0425	C0THP1	1021	D1TM1	3523	DISKGO	4441
ASKNX5	0464	C0TTYI	4426	D1TM2	3533	DLAG	6743
ASKSUR	0513	C0TYPE	4435	D2CMP	3552	DLCA	6744
ASREG	0131	CAF	0007	D2HRU	3550	DLOC	6746
AUTO10	0010	CAREG	0126	D2SOP	3551	DLSC	6740
AUTO11	0011	CCNTR1	2757	D2TM1	3524	DOCNT	0247
AUTO12	0012	CHANG	2750	D2TM2	3534	DOHEAD	1262
BADHLT	1707	CHANGR	2742	D3CMP	3555	DONEA	0426
BGHLT	0176	CHEK22	0523	D3HRU	3553	DOPACK	0212
BGN	0200	CHK22	4424	D3SOP	3554	DOSST	0251
BGNBUP	0146	CHKCLA	1200	D3TM1	3525	DRST	6745
BIGHLT	3131	CHKPOT	2000	D3TM2	3535	D8K00	3512
BIGSTP	3136	CHKRES	3012	D4CMP	3560	D8K10	3513
BRKRET	0363	CHKSAV	0141	D4HRU	3556	D8K20	3514
BUPPNT	1400	CHKYN	2035	D4SOP	3557	D8K30	3515
BUPTAL	0117	CHNCDF	2254	D4TM1	3526	D8K40	3516
BYRETR	0506	CHNHLT	2755	D4TM2	3536	D8K50	3517
C0BY1	0230	CHNPOT	2760	D5CMP	3563	D8K60	3520
C0BY2	1300	CKCOUT	0232	D5HRU	3561	D8K70	3521
C0BY3	1061	CKTIM	2450	D5SOP	3562	D8KEX	2302
C0BY4	0515	CLAFLO	1404	D5TM1	3527	D8KGO	2200
C0BY5	1116	CLASIC	4407	D5TM2	3537	D8KP	6741
C0CHAR	1075	CLASIK	1514	D6CMP	3566	D8KSKP	4450
C0CKP	1022	CLOR	0500	D6HRU	3564	DCHK	1600
C0CKPA	4440	CLKCNT	0150	D6SOP	3565	UTR1	1636
C0CKSW	4425	CLRALL	4454	D6TM1	3550	DUMP	2637
C0CNTR	4427	CLRBK	0175	D6TM2	3540	ENDIT	0742
C0CNT7	1145	CLRTN	1315	D7CMP	3571	ERFLG	0165
C0CRLF	4433	CMPPOT	2562	D7HRU	3567	ERHLT0	0556
C0D01	0310	CMREG	0123	D7SOP	3570	ERHLT2	0563
C0D010	1262	CNTRLC	0551	D7TM1	3551	ERHLT3	2561
C0D011	0607	CNTRLD	0600	D7TM2	3541	ERHLT5	2544
C0D02	1033	CNTHLE	0545	DARE6	0125	ERHLT6	0547
C0D03	0350	CNTRLL	0537	DAT1	0756	ERR1	0736
C0D04	1006	CNTRLQ	0500	DAT10	0767	ERRMIS	1320
C0D07	0527	CNTRLR	0511	DAT11	0770	ERR0	1200
C0ECNO	4434	CNTRL5	0521	DAT12	0771	ERRDEX	1355
C0ERR	4436	CNTVAL	0252	DAT2	0757	ERROR	4446

ERTX1	3247	K0001	0001	M12	0105	NTWRKS	1710
ERTX2	3255	K0003	0002	M4	0106	UCT1	2400
ERTX3	3264	K0004	0013	MANUAL	0322	UCT4	2430
ERTX4	3276	K0006	0063	MAXFLD	0143	CTEL	4456
ESAVE	1371	K0007	0064	MAXTIM	0144	ONEIN	4432
EXAMPT8	2151	K0010	0065	MAXTHK	0145	OP1	0021
EXIT	1504	K0017	0066	MEMSET	2142	OP2	0022
EXITA	0440	K0070	0014	MES0	3303	OPRTAL	0116
EXREG	0122	K0077	0102	MES1	3307	PASCNT	0250
EXTICK	1166	K0100	0015	MES10	3414	PCLF	0662
F1OP1	0021	K0200	0016	MES11	3424	PCNTR1	1373
F1OP2	0022	K0212	1425	MES13	3433	PCNTR2	1374
F1SHR	0020	K0215	1424	MES14	3436	PCNTR3	1375
FILCNT	1040	K0240	1512	MES15	3445	PCREG	0120
FILLER	1037	K0260	0067	MES16	3455	PCSAVE	1344
FILLUP	0717	K0277	0070	MES17	3477	PNTBUF	1120
FIRTIM	0170	K0316	2057	MES18	3502	POLOSK	0115
FLDFLG	3372	K0331	2056	MES19	3174	POLNEX	1000
FLDHLT	0206	K0400	0017	MES2	3325	PREXIT	2635
FLSAVE	1347	K1000	0075	MES3	3332	PRINT	2620
FND3UM	0142	K177	2034	MES4	3335	PRN	1450
FORIN	4433	K1777	0076	MES5	3346	PRNDAT	0173
FROCT	1426	K3740	1513	MES6	3363	PRINTER	4455
FWREG	0130	K4000	0073	MES7	3372	PSTE	0665
GENDAT	4426	K4100	0074	MES8	3404	P8KE	0663
GETCH1	0703	K5405	0360	MES9	3410	P8KF	0661
GETDAT	0456	K6500	0541	MES1A	0747	PSTB	0664
GNDAT	1737	K6520	2172	MESAC	1533	PTSTOR	0336
GOBAK	2230	K7000	2155	MESFL	1541	RAD1	1773
GOCDF	1646	K7377	2154	MESMAN	1146	RAD2	1774
GOCMK	2265	K7400	0104	MESMW	1336	RAD3	1775
GOCLE	2260	K7700	0077	MESPAS	0253	RAN1	1767
GOITA	0443	K7760	0100	MESPC	1530	RAN2	1770
GOTIT	1010	K7773	1377	MODSKS	2125	RANDAT	4427
GOTOA	0454	K7775	0110	MOA	7501	RANDOM	1715
GTF	0004	K7777	0101	MUL	7421	RANGEN	4440
HEOTAD	1376	KAERRO	0023	MOSAVE	1346	RANJMS	0522
HLFFLG	3375	KCDF	0103	MRPRN	1456	RDLWRL	2324
INDEXA	0455	KROD	0250	MSKEK	1714	ROST	2541
INERR	2426	K8KP	0546	MYAC	1317	ROSTA	1105
INMODE	1076	KTICK	4425	MYLAS	1546	ROSTAT	4447
INTCM	0136	KTIME	0530	NEWRU	0727	ROTRY	1073
INTOA	0124	LAS	4406	NEXT	0201	REALPC	1316
INTER1	3130	LOAD	2554	NODSKP	2353	RECAL	4443
INTER2	2362	LOADD	4453	NODSKS	2702	RECUF	2605
IOT0	0554	LOCA	2550	NOERR	1672	RECEIV	4444
IOT1	0752	LDCM	0542	NOSET	0242	REDOA	0415
IOT2	0561	LDCMO	4451	NOTEX	1363	REFILL	0723
IOT3	2557	LDCUR	4452	NOTSTA	3041	REREAD	1063
IOT4	2551	LNKDCA	0361	NTCLAS	1270	RERUN	1124
IOT5	2542	LDDGO	2330	NTERN	1255	RESERR	3111
IOT6	0545	M10	0107	NTSOFT	1251	RESRAN	4442

RESTA	3077	TABLB	0471	XC8CNL	1023
RESTOR	3052	TEXAD	3241	XC8ECH	1063
RETRN	2331	TEXAS	3235	XC8ENR	1247
RETURN	2304	TEXCA	3227	XC8ING	0635
REWRT	1047	TEXCM	3221	XC8OCT	1000
RNFLD	0630	TEXDA	3225	XC8PAS	0240
RNWRO	2600	TEXDB	3245	XC8PAU	0337
ROUINS	1302	TEXDG	3243	XC8PNT	0303
ROUTMP	1544	TEXEX	3217	XC8P8W	0656
RSRAN	1761	TEXFW	3233	XC8SW	0262
RUN	0600	TEXIA	3223	XC8TTY	0272
RUNPOT	0154	TEXPC	3213	XC8TYP	1077
SAV1	1771	TEXST	3215	XCHK22	0044
SAV2	1772	TEXWA	3237	XCHKYN	0037
SAVAC	0166	TEXWC	3231	XCKPUT	0036
SAVCM	0174	TIME	3123	XCLAS	0007
SAVE1	0400	TIMER1	3140	XCLUK	0054
SAVEAC	1545	TIMER2	3141	XCLRL	0057
SOKP	0751	TIMER3	2374	XDOLMT	1112
SECFLG	3574	TIMPOY	0152	XDO8W	0520
SELCHK	4436	TIMPCNT	0746	XDSKW0	0041
SETFLD	4435	TPSTA	3000	XDUMP	0030
SETGEN	4434	TRASH1	0111	XERRU	0046
SETREG	1553	TRASH2	0112	XFROCT	0036
SETUP1	1233	TRASH3	0113	XGNDAT	0026
SETUP2	0225	TRDNE	2365	XKTCK	0025
SKPNOP	0237	TRKFLG	3573	XKTICK	1154
SOFT	1245	TRYCNT	0171	XLAS	0006
SPAC	1506	TRYTIM	1060	XLOAD	0053
SPACE	4431	TSAVE1	3045	XLOCA	0052
SPBLK	0164	TSAVE2	3046	XLOCM	0051
SPFLD	0160	TSAVE3	3047	XOCT1	0032
SPSEC	0163	TSAVE4	3050	XUCT4	0033
SPTRK1	0161	TSAVE5	3051	XPRINT	0045
SPTRK2	0162	TSTCHA	0715	XPRN	0055
STAPOT	0153	TYLPT	1121	XRDST	0047
STATER	2337	TYPOAT	3142	XREG	1372
STATRY	0137	TYPE	4445	XWESTR	0043
STATUS	2313	UPAROW	0615	XRNDOU	0040
STFLD	2703	UPDATE	0114	XRNWMD	0027
STGEN	1753	UPONE	1414	XRSRAN	0042
STPHLT	0603	UPTRY	1120	XSDKP	0050
STRAUT	1325	WAIT	2000	XSPAC	0031
STRBUF	3600	WAIT1	2026	XSTFLU	0035
STREG	0121	WAREG	0132	XSTGEN	0034
STRREL	1151	WABRD	2264	XTAOLA	0057
STRTEX	0224	WATME3	0651	XTABLB	0060
STRWRK	2241	WCREG	0127	XTEAT	0172
SVLNK	0167	WRDCHK	1612	XWAIT	0044
SWDAT	2601	WRKDDN	2274	YESNU	4457
SWR	0020	XC8CKP	1041		
TABLA	0461	XC8CNT	0400		

ERRORS DETECTED: 0

LINKS GENERATED: 163

RUN-TIME: 6 SECONDS

3K CORE USED